

Functional characterisation of neuropeptides that act as ligands for both calcitonin-type and pigment-dispersing factor-type receptors in a deuterostome

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eLife Assessment

This **valuable** study characterises receptors for calcitonin-related peptides from a deuterostomian animal, the echinoderm *Apostichopus japonicus*, by a combination of heterologous expression, pharmacological experiments, and the quantification of gene-expression levels. The authors provide **solid** evidence for a functional calcitonin-related peptide system in the sea cucumber, but further work will be needed to confirm the proposed phylogenetic relationships and physiological functions of PDF receptor system in this species. This work should be of interest to scientists studying the signaling pathways, functions, and evolution of neuropeptides, and could be of relevance to improving the culture conditions of this economically key species.

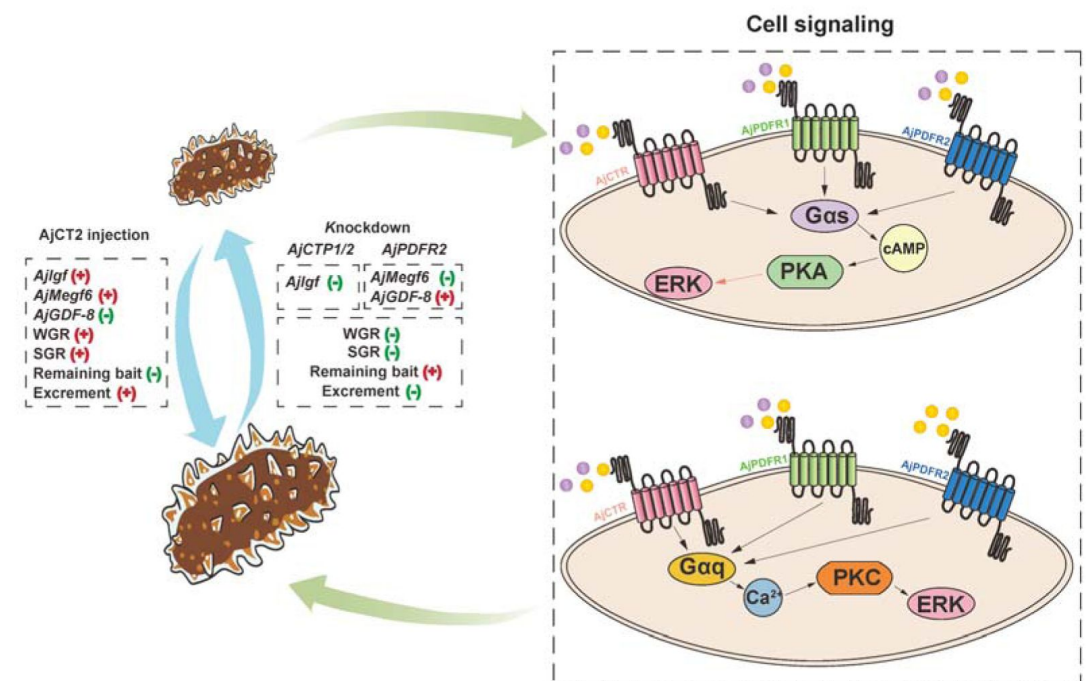
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Abstract

The calcitonin (CT) family of related peptides exerts diverse physiological effects in mammals via two G-protein coupled receptors: CTR and the CTR-like receptor CLR. Phylogenetic analysis of CT-type signaling has revealed the presence of CT-type peptides and CTR/CLR-type proteins in both deuterostome and protostome invertebrates. Furthermore, experimental studies have demonstrated that in the protostome *Drosophila melanogaster* the CT-like peptide DH₃₁ can act as a ligand for a CTR/CLR-type receptor and a pigment-dispersing factor (PDF) receptor. Here we investigated the signaling mechanisms and functions of CT-type neuropeptides in a deuterostome invertebrate, the sea cucumber *Apostichopus japonicus* (phylum Echinodermata). In *A. japonicus*, a single gene encodes two CT-type peptides (AjCT1 and AjCT2) and both peptides act as ligands for a CTR/CLR-type receptor (AjCTR) and two PDF-type receptors (AjPDFR1, AjPDFR2), but with differential activation of downstream cAMP/PKA, Gαq/Ca²⁺/PKC and ERK1/2 signaling pathways. Transcriptic analysis of the

AjCT1/AjCT2-encoding gene revealed expression in the central nervous system and a variety of organ systems, which was visualized immunohistochemically using an antiserum to a starfish CT-type peptide (ArCT). *In vitro* pharmacological experiments demonstrated that AjCT1 and/or AjCT2 cause dose-dependent relaxation of longitudinal muscle and intestine preparations. Furthermore, *in vivo* pharmacological experiments, combined with gain- and loss-of-function experiments, revealed a potential physiological role for AjCT2/AjPDFR2 signaling in promoting feeding and growth in *A. japonicus*. To our knowledge, this is the first study to obtain evidence that CT-type peptides can act as ligands for both CTR/CLR-type and PDF-type receptors in a deuterostome. Moreover, it provides the first evidence for appetite-stimulating and growth-promoting effects of CT-type neuropeptides in bilaterians. Given the economic importance of *A. japonicus* as a foodstuff, the discovery of CT-type peptides as potential regulators of feeding and growth in this species may offer novel strategies for aquaculture applications.

Graphical abstract



A schematic showing proposed molecular mechanisms by which CT-type signaling regulates feeding and growth in *A. japonicus*. AjCT1 and AjCT2 are represented by purple and yellow circles, respectively. For cell signaling, black arrows indicate that this process is activated, whilst the pink arrow indicates that ERK1/2 is activated in the AjCT1/AjPDFR1, AjCT2/AjPDFR1 and AjCT2/AjPDFR2 signaling pathways.

1. Introduction

The peptide calcitonin (CT) was first identified as a hormone that regulates blood calcium levels and subsequently a family of CT-related neuropeptides was discovered in mammals, including calcitonin gene related peptides (CGRPs), amylin (AMY), adrenomedullin (AM), and CT receptor-stimulating peptide (CRSP) (Copp et al., 1962 [\[1\]](#); Sekiguchi et al., 2009 [\[2\]](#); Sekiguchi et al., 2016 [\[3\]](#)). Members of the CT-type family of peptides typically have a disulfide bond in their N-terminal region (Cai et al., 2018 [\[4\]](#)) and exert their physiological effects in mammals by binding to two

related G protein-coupled receptors: CTR (CT receptor) and CLR (CTR-like receptor) (Martins et al., 2014). However, receptor activity-modifying proteins (RAMPs) acting as accessory proteins are key determinants of ligand selectivity for CTR and CLR (Mclatchie et al., 1998). Thus, formation of a complex with a RAMP is essential for CLR function, whereas RAMPs are not required for CTR ligand recognition (Kuwasako et al., 2011; Bomberger et al., 2012). Furthermore, binding of CT to CTR in the absence of RAMPs (Katafuchi et al., 2009) can activate signaling via several downstream pathways, including cAMP accumulation, Ca^{2+} mobilization, and ERK activation (Chen et al., 1998; Pondel, 2000; Walker et al., 2010; Andreassen et al., 2014).

Transcriptome/genome sequencing has enabled analysis of the phylogenetic distribution of CT-type signaling systems, indicating that the evolutionary origin of CT-type peptides and CTR/CLR-related receptors can be traced back to the urbilaterian common ancestor of deuterostomes and protostomes (Furuya et al., 2000; Johnson et al., 2005; Jékely, 2013; Mirabeau and Joly, 2013). Furthermore, in protostomes two different types of CT-related peptides have been identified. Firstly, peptides that are structurally similar to vertebrate CT-type peptides in having an N-terminal disulphide bridge and secondly, diuretic hormone 31 (DH₃₁)-type peptides that lack an N-terminal disulphide bridge (Furuya et al., 2000; Conzelmann et al., 2013). Furthermore, phylogenetic analysis indicates that gene duplication in a common ancestor of the protostomes gave rise to genes encoding CT-like peptides with a disulphide bridge and DH₃₁-type peptides, which were then differentially retained in protostome taxa (Conzelmann et al., 2013; Cai et al., 2018). Receptors for CT-like peptides with a disulphide bridge have been characterised in a molluscan species, the oyster *Crassostrea gigas*, which has seven CTR/CLR-type receptors. Two of these receptors, named Cg-CT-R and Cragi-CTR2, were shown to be specifically activated by the CT-type neuropeptides Cragi-CT1b and Cragi-CT2, respectively, thereby mediating intracellular Ca^{2+} mobilization and cAMP accumulation (Schwartz et al., 2019). CT-like peptides with a disulphide bridge are not present in the insect *Drosophila melanogaster*. However, receptors that are activated by DH₃₁ have been identified in this species and, interestingly, DH₃₁ can act as a ligand for two types of receptors. Thus, DH₃₁ acts as a ligand for a CTR/CLR-related protein (CG17415), but it also acts as a ligand for a protein (CG13758) that is the receptor for pigment-dispersing factor (PDF) (Johnson et al., 2005; Mertens et al., 2005; Shafer et al., 2008). Furthermore, the functional significance of the existence of dual receptor signaling pathways for DH₃₁ in *Drosophila* has been investigated extensively (Johnson et al., 2005; Goda et al., 2019).

Currently, relatively little is known about CT-type signaling systems in deuterostome invertebrates. Analysis of transcriptome/genome sequence data has enabled identification of genes/transcripts encoding CT-type peptides and candidate CTR/CLR-type receptors for these peptides in several deuterostome invertebrate taxa, including urochordates (e.g. *Ciona intestinalis*), cephalochordates (e.g. *Branchiostoma floridae*), hemichordates (e.g. *Saccoglossus kowalevskii*) and echinoderms (e.g. *Strongylocentrotus purpuratus*) (Burke et al., 2006; Sekiguchi et al., 2009; Sekiguchi et al., 2016; Rowe and Elphick, 2012; Martins et al., 2014). Furthermore, analysis of the expression and action of CT-type peptides identified in the starfish *Asterias rubens* and *Patiria pectinifera* has revealed that they act as muscle relaxants (Cai et al., 2018; Semmens et al., 2016). However, experimental confirmation of ligand-receptor partners has thus far only been reported for the cephalochordate *B. floridae*, where three CT-type peptides were shown to act as ligands for a CTR/CLR-type receptor when it is co-expressed with RAMP-type proteins (Sekiguchi et al., 2016). To gain further insights into the mechanisms of CT-type signaling in invertebrate deuterostomes, the aim of this study was to functionally characterise CT-type peptides and their receptors in an echinoderm – the sea cucumber *Apostichopus japonicus*.

A. japonicus is an economically important species in Asian aquaculture (Pangestuti and Arifin, 2018; Guo et al., 2020) and a species representative of the phylum Echinodermata. As a phylum of deuterostome invertebrates, echinoderms form a sister clade to the chordates together with hemichordates and xenoturbellids (Bromham and Degnan, 1999; Bourlat et al., 2006; Dunn et al., 2008) and as such they occupy a key “intermediate” phylogenetic position relative to

the protostome invertebrates and vertebrates (Zandawala et al., 2017 [↗](#)), which make them attractive as experimental systems to address evolutionary questions. In our previous studies, we have established this species as a model system to obtain insights into the evolutionary history and comparative physiology of neuropeptide signaling systems in bilaterians (Chen et al., 2019 [↗](#); Li et al., 2022 [↗](#); Zheng et al., 2022 [↗](#); Cong et al., 2023 [↗](#); Zheng et al., 2024 [↗](#)). A gene encoding two CT-type peptides (AjCT1 and AjCT2) has been identified in *A. japonicus* (Chen et al., 2019 [↗](#)) and a candidate receptor (AjCTR) for these peptides has also been predicted (Huang et al., 2021 [↗](#)). Furthermore, relevant to the discovery that the CT-like peptide DH₃₁ can act as a ligand for both a CTR/CLR-type receptor and a PDF-type receptor in *Drosophila*, the sequences of two PDF-type receptors have been identified in *A. japonicus* (AjPDFR1 and AjPDFR2) (Huang et al., 2021 [↗](#)).

The first aim of this study was to investigate if AjCT1 and/or AjCT2 act as ligands for AjCTR, AjPDFR1 and AjPDFR2 and to characterise the downstream signaling pathways that are activated when AjCT1 and/or AjCT2 bind to their receptor(s). The second aim was to investigate the physiological roles of AjCT1 and AjCT2 in *A. japonicus* by analysing the expression of the gene encoding these peptides, investigating CT-type neuropeptide expression using immunohistochemistry, examining the *in vitro* and *in vivo* pharmacological effects of these peptides and performing loss-of-function tests. The functional characterisation of CT-type signaling in *A. japonicus* reported in this study provides new insights into the evolution and comparative physiology of neuropeptides and a basis for application of neuropeptide-based strategies to improve aquaculture methods for this economically important species.

2. Results

2.1 Characterization and phylogenetic analysis of AjCT1/2

The complete cDNA sequences of the CT-type neuropeptide precursors *AjCTP1* and *AjCTP2* were confirmed. The coding sequence of *AjCTP2* is shorter than *AjCTP1* (Figure 1A [↗](#)) but both transcripts encode precursor proteins that comprise the same predicted 23-residue N-terminal signal peptide (Figure 1-figure supplement 1A [↗](#)). *AjCTP1* encodes both putative CT-type neuropeptides (AjCT1 and AjCT2), while *AjCTP2* only encodes AjCT2 (Figure 1A [↗](#), Figure 1-figure supplement 1A [↗](#) and 1B [↗](#)). Exon and intron structure analysis showed that *AjCTP1* comprises four exons, with AjCT1 and AjCT2 encoded in the third and fourth exons, respectively. *AjCTP2* comprises three exons, excluding the third exon that encodes AjCT1 (Figure 1A [↗](#)). Phylogenetic analysis revealed that *AjCTP1* and *AjCTP2* are closely related to CT-type precursors from other sea cucumbers (*Holothuria glaberrima* and *Holothuria scabra*), and are positioned within a clade that also contains CT-type precursors from other echinoderms (sea urchin *S. purpuratus*, starfish *A. rubens*). Furthermore, the echinoderm CT-type precursors are positioned within a clade that also comprises CT-type precursors from chordates, whilst DH₃₁/CT-type precursors from protostomes are positioned in a separate clade (Figure 1B [↗](#)). Sequence alignment of CT-related neuropeptides revealed that, with the exception of DH₃₁-type peptides, the N-terminal region of CT-related peptides has a pair of conserved cysteine residues that are known or predicted to form a disulfide bond. Another conserved feature is a C-terminal proline residue, with the exception of CGRPs (Figure 1-figure supplement 2 [↗](#)). Accordingly, both AjCT1 and AjCT2 have these conserved characteristics.

2.2 Characterization and phylogenetic analysis of three candidate receptors for AjCT1/2

The sequences of three candidate receptors for AjCT1 and AjCT2 were obtained by RACE and phosphorylation sites were predicted, as shown in Figure 1-figure supplement 3A [↗](#), 3B [↗](#), and 3C [↗](#). Phylogenetic analysis revealed that CTR/CLR-type receptors divided into two clades and an

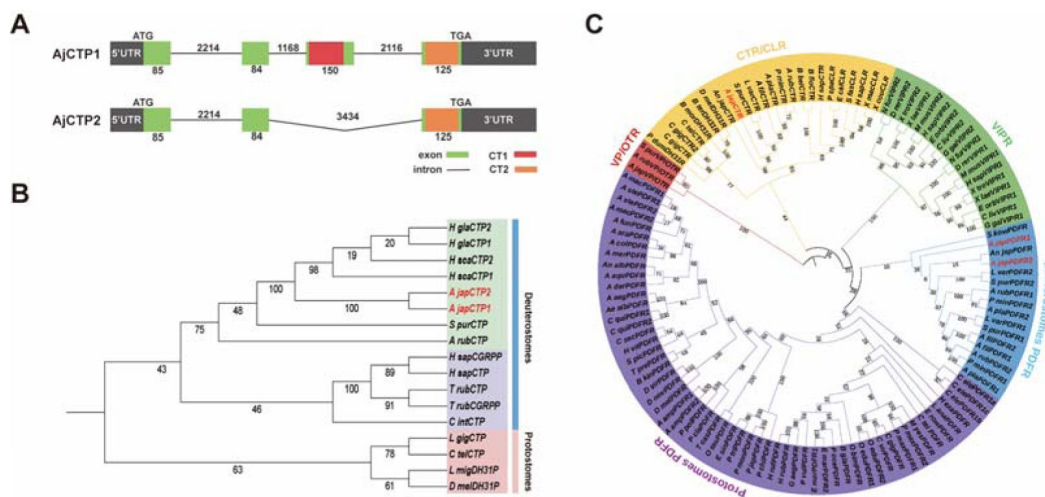


Figure 1.

CT-type neuropeptide precursors and candidate receptors for CT-type peptides in *A. japonicus*.

(A) Alternative splicing schematic of *AjCTP1* and *AjCTP2*. Exons are represented by green rectangles, introns are represented by lines, *AjCTP1* and *AjCTP2* are represented by red and orange rectangles respectively, upstream and downstream non-coding regions are represented by gray rectangles. The numbers represent the length of exons or introns. (B) Phylogenetic analysis of *AjCTP1* and *AjCTP2* and CT-related neuropeptide precursors from other bilaterians. CT-type precursors from deuterostomes are shown in green (echinoderms) and purple (chordates), CT-type precursors from protostomes are shown in pink. (C) Phylogenetic analysis of relationships of *AjCTR*, *AjPDFR1* and *AjPDFR2* (red letters) with CTR/CLR (calcitonin receptor/calcitonin receptor-like receptor), DH31R (diuretic hormone 31 receptor) and PDFR (pigment-dispersing factor receptor) in other taxa (black letters), using VP/OT-type (vasopressin/oxytocin) receptors as the outgroup. Considering that VIP (vasoactive intestinal peptide) and PDF both have roles in regulation of circadian and gut activity, the phylogenetic tree also includes VIPRs. Full species names and accession numbers are listed in [Supplementary table 2](#).

A. japonicus CTR/CLR-type receptor (AjCTR) was positioned in a clade with other deuterostome CTR/CLR-type receptors, whilst the other clade comprised CTR/CLR-type receptors from protostomes. Similarly, two *A. japonicus* PDF-type receptors (AjPDFR1 and AjPDFR2) were positioned in a clade comprising PDF-type receptors from other deuterostomes, whilst another clade comprised PDF-type receptors from protostomes (**Figure 1C**). Domain prediction of the candidate receptors showed that AjCTR had HRM (Hormone receptor) and 7tm_GPCRs (seven-transmembrane G protein-coupled receptor) superfamily domains (**Figure 1-figure supplement 4A**), whereas AjPDFR1 and AjPDFR2 only presented the 7tm_GPCRs superfamily domain (**Figure 1-figure supplement 4B**). To further investigate an orthologous relationship of the candidate receptors with homologous receptors in other echinoderms, we conducted a gene synteny analysis in *A. japonicus*, *A. rubens* and *L. variegatus*. The neighboring genes of *AjCTR* included *GOLGA4*, *NeuroD*, *CRF2*, *ppGalNAc-T*, *URB3*, *SLC40A1*, *CMAH* and *MAGP*. The relative positions of these genes were different from those in *A. rubens* and *L. variegatus* (**Figure 1-figure supplement 4C**). The same situation was also observed for the synteny analysis of PDFRs.

2.3 AjCT1 and AjCT2 act as ligands for AjCTR, AjPDFR1 and AjPDFR2

To investigate if AjCT1 and/or AjCT2 act as ligands for AjCTR, AjPDFR1 and AjPDFR2, here, two receptor assay methods were employed. As shown in **Figure 2A**, both AjCT1 and AjCT2 (10^{-6} M) caused a significant increase in CRE luciferase activity in HEK293T cells transfected with AjCTR, AjPDFR1 or AjPDFR2. Furthermore, this effect was inhibited by pre-incubation of cells with the PKA inhibitor H89. The intracellular accumulation of cAMP was also measured, and this showed that AjCT1 and AjCT2 (10^{-9} – 10^{-5} M) caused a dose-dependent increase in cAMP in HEK293T cells transfected with AjCTR, AjPDFR1 or AjPDFR2 (**Figure 2B**). However, when exposed to AjCT1 or AjCT2, the maximum accumulation of intracellular cAMP was higher in cells transfected with AjCTR than in cells transfected with AjPDFR1 and AjPDFR2 (**Figure 2B**). Measurement of SRE luciferase activity revealed that AjCT1 and AjCT2 (10^{-6} M) administration evoked an increase in activity in HEK293T cells transfected with AjCTR and AjPDFR1 (**Figure 3A**). However, in HEK293T cells transfected with AjPDFR2, an increase in SRE luciferase activity was only observed with AjCT2, not with AjCT1. Furthermore, as shown in **Figure 3A**, AjCT1- and AjCT2-induced increases in SRE luciferase activity were not observed in cells that had been pre-incubated with FR900359 (Gaq protein inhibitor) and Gö 6983 (PKC inhibitor). As shown in **Figure 3-figure supplement 1**, receptors localized in the cell membrane, but after exposure to AjCT1 and AjCT2 (10^{-6} M) for 15 min, internalization of pEGFP-N1/*AjCTR*, pEGFP-N1/*AjPDFR1* and pEGFP-N1/*AjPDFR2* was observed in transfected HEK293T cells (**Figure 3B**). Taken together, all these findings indicate that AjCT1 and AjCT2 can act as ligands for the three candidate receptors and then activate cAMP/PKA signaling and Gaq/ Ca^{2+} /PKC signaling in all combinations, with the exception of AjCT1/AjPDFR2 which only activates the cAMP/PKA cascade.

To investigate whether the activated signaling pathways may exert downstream effects via the ERK-MAPK pathway, the phosphorylation levels of ERK1/2 were analysed in transfected HEK293T cells. pERK1/2 levels in cells transfected with AjCTR or AjPDFR1, but not AjPDFR2, were significantly increased after a 5 min incubation with AjCT1 (**Figure 4A**). Furthermore, this effect was blocked by the PKC inhibitor in AjCTR-expressing cells, and this effect was blocked by both PKA and PKC inhibitors in AjPDFR1-expressing cells (**Figure 4-figure supplement 1A**). pERK1/2 levels in cells transfected with AjCTR or AjPDFR1 or AjPDFR2, were significantly increased after a 5 min incubation with AjCT2 (**Figure 4B**). Moreover, this effect persisted until 15 min after exposure to AjCT in both AjCTR-transfected cells and AjPDFR1-transfected cells but not in AjPDFR2-transfected cells (**Figure 4B**, **Table 1**). Furthermore, these effects were blocked by PKA and PKC inhibitors in AjPDFR1 and AjPDFR2 expressing cells, whereas in AjCTR expressing cells, peptide-induced pERK1/2 activity was only blocked by a PKC inhibitor (**Figure 4-figure supplement 1B**). These results indicate that, with the exception of AjCT1/AjPDFR2, both AjCT1 and AjCT2 can activate the ERK1/2 cascade by binding to the candidate receptors.

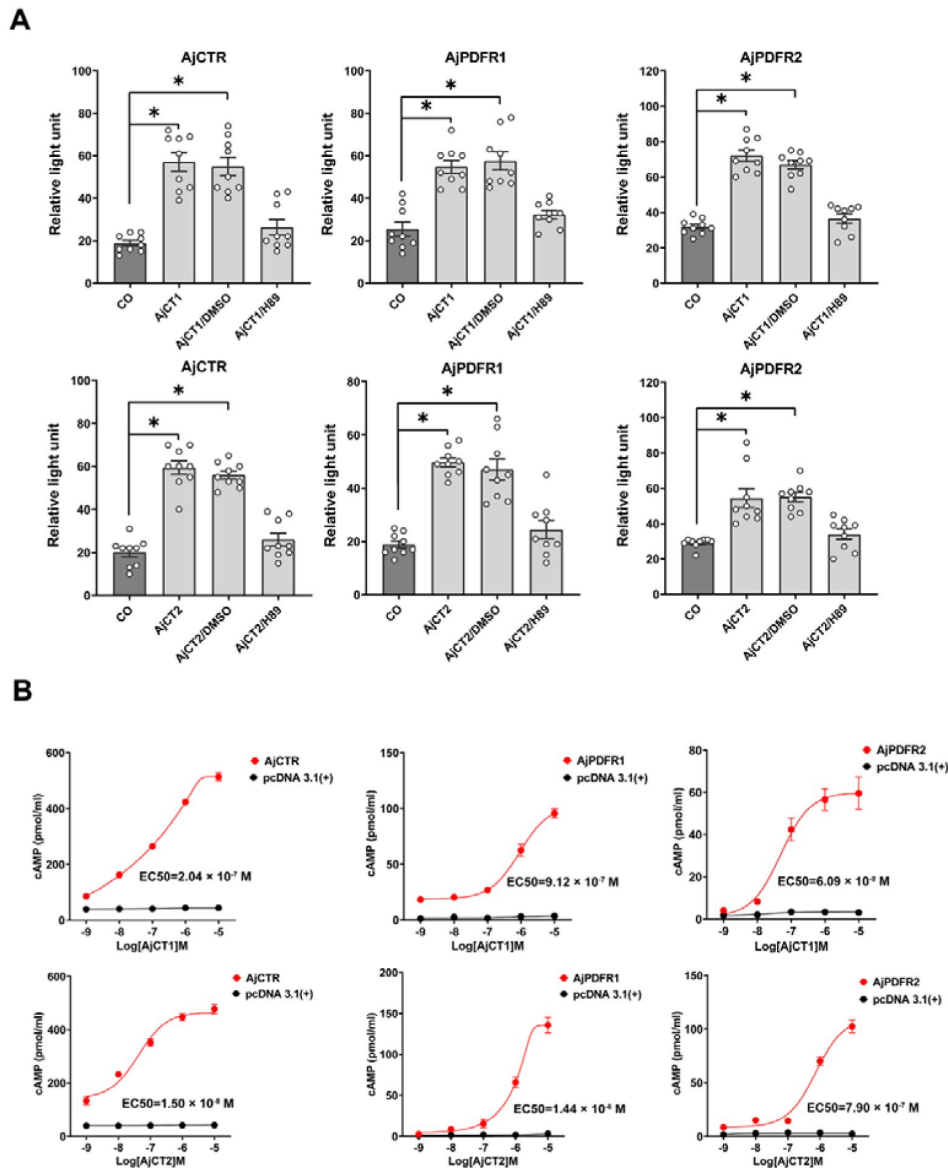


Figure 2.

Pharmacological characterisation of AjCT1 and AjCT2 as ligands for the *A. japonicus* receptors AjCTR, AjPDFR1 and AjPDFR2.

(A) CRE-driven luciferase activity measured in HEK293T cells transfected with one of the three candidate receptors after exposure to AjCT1 or AjCT2 or neuropeptide + DMSO or neuropeptide + H89 (10 μ M) or serum-free DMEM (CO, negative control). The neuropeptide concentration used here was 10^{-6} M. Mean values with standard deviations (n = 9) are shown. * indicates a statistically significant difference with $p < 0.05$. (B) Measurement of cAMP accumulation after exposure to AjCT1 or AjCT2 (10^{-9} – 10^{-5} M). No cAMP elevation is observed in control experiments where HEK293T cells were transfected with empty pcDNA 3.1(+) (black circle). The estimated EC₅₀ of AjCT1 for AjCTR, AjPDFR1 and AjPDFR2 were 2.04×10^{-7} M (95% CI: 1.06×10^{-7} – 8.69×10^{-7} M), 9.12×10^{-7} M (95% CI: 5.56×10^{-7} – 4.45×10^{-6} M) and 6.09×10^{-8} M (95% CI: 2.23×10^{-8} – 2.84×10^{-7} M) respectively. The estimated EC₅₀ of AjCT2 for AjCTR, AjPDFR1 and AjPDFR2 were 1.50×10^{-8} M (95% CI: 1.03×10^{-9} – 3.89×10^{-8} M), 1.44×10^{-6} M (95% CI: 1.04×10^{-6} – 1.99×10^{-6} M) and 7.90×10^{-7} M (95% CI: 5.73×10^{-7} – 1.06×10^{-6} M) respectively. Error bars represent SEM for three independent experiments.

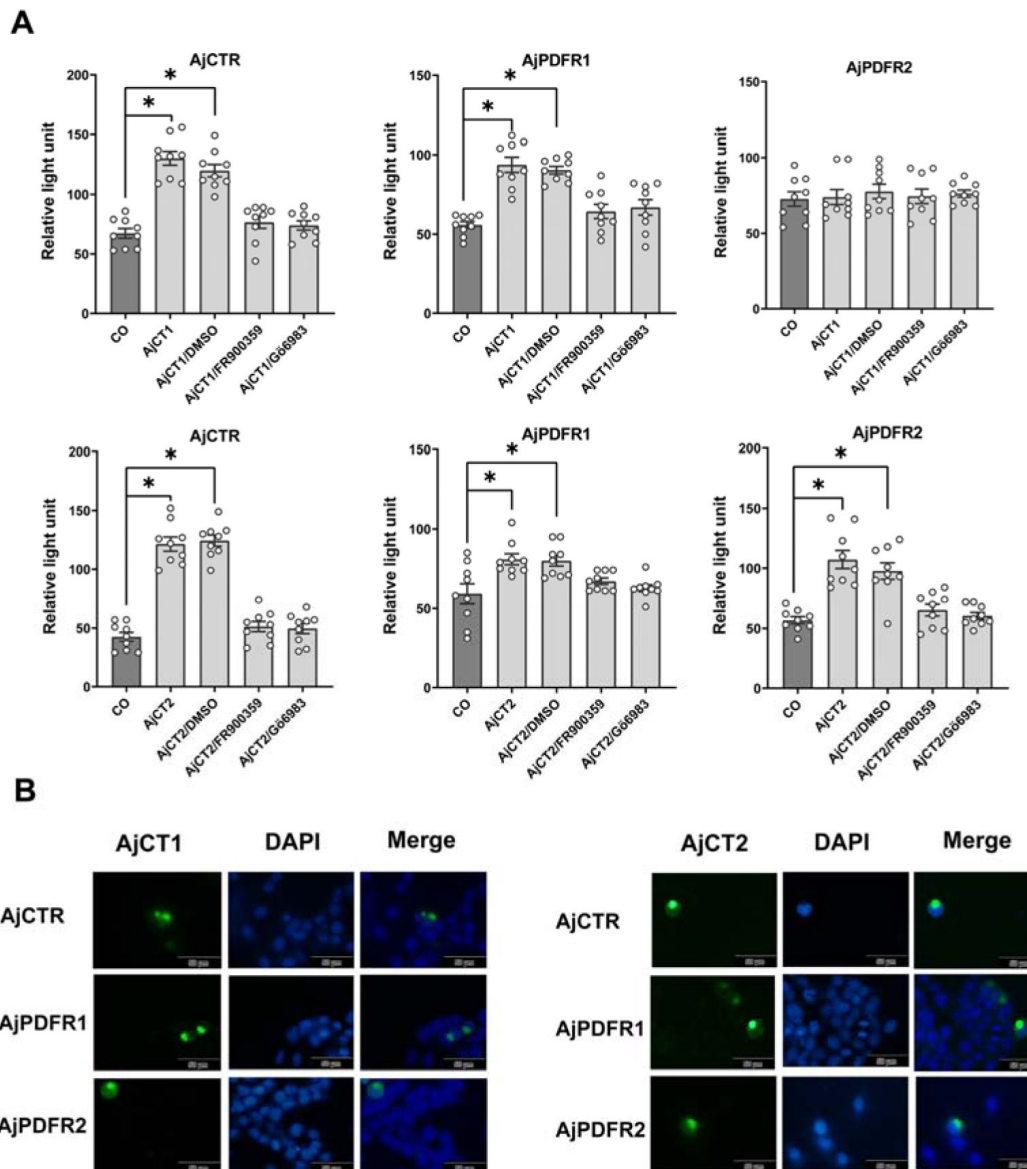


Figure 3.

SRE-driven luciferase activity and receptor internalization in cells transfected with AjCTR, AjPDFR1 or AjPDFR2 and exposed to AjCT1 or AjCT2.

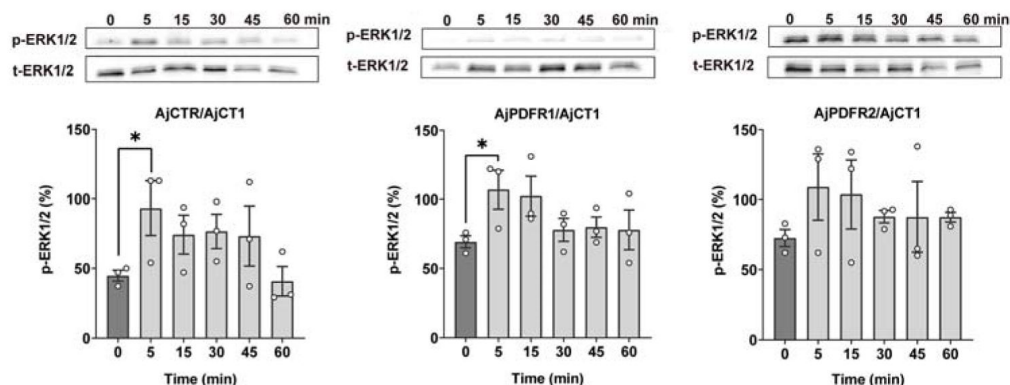
(A) SRE-driven luciferase activity measured after incubation with neuropeptide (10^{-6} M AjCT1 or AjCT2)/DMSO, neuropeptide (10^{-6} M)/FR900359 (1 μ M) and neuropeptide (10^{-6} M)/Gö 6983 (1 μ M). Mean values with standard deviations ($n = 9$) are shown. * indicates a statistically significant difference with $p < 0.05$. (B) Internalization of AjCTR, AjPDFR1 and AjPDFR2 after 15 min treatment with 10^{-6} M AjCT1 or AjCT2. Green fluorescence represents the localization of pEGFP-N1/receptors in cells. The cell nucleus probe (DAPI) was used for cell nucleus staining. Scale bar: 50 μ m. All pictures are representative of three independent experiments.

Figure 4.

ERK1/2 activity in cells transfected with AjCTR, AjPDFR1 or AjPDFR2 and exposed to AjCT1 or AjCT2.

(A) and (B) show the immunoblot intensity of representative phosphorylation bands. The transfected HEK293T cells were incubated with AjCT1 (10^{-6} M) or AjCT2 (10^{-6} M) for 0, 5, 15, 30, 45 and 60 min. The p-ERK1/2 was normalized based on t-ERK1/2. All images are representative of three independent experiments. Error bars represent SEM for three independent experiments. Statistically significant differences are indicated as follows: *: $p < 0.05$, **: $p < 0.01$.

A



B

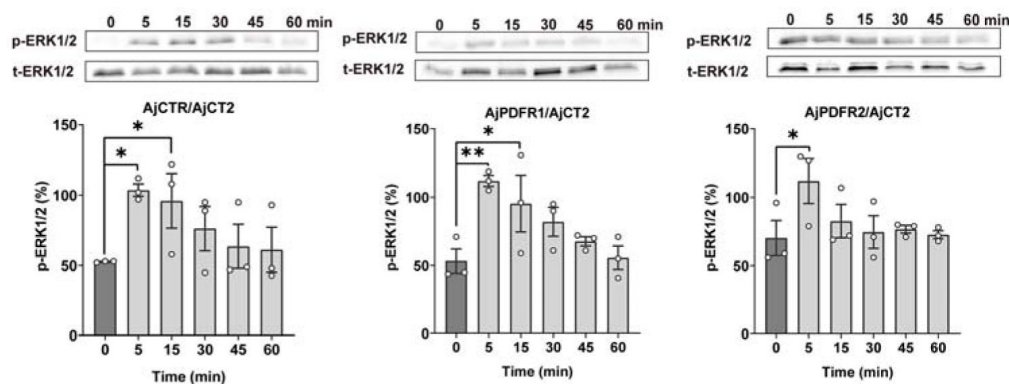


Table 1.

Activation of signaling pathways

Assays Receptors	cAMP/PKA/ERK1/2		Gαq/Ca ²⁺ /PKC/ERK1/2	
	AjCT1	AjCT2	AjCT1	AjCT2
<i>AjCTR</i>	×	×	✓	✓
<i>AjPDFR1</i>	✓	✓	✓	✓
<i>AjPDFR2</i>	×	✓	×	✓

A tick indicates cAMP/PKA/ERK1/2 or Gαq/Ca²⁺/PKC/ERK1/2 activation, the cross indicates no activation, the cross with * indicates that the cAMP/PKA pathway is activated, while the ERK1/2 pathway is not activated.

2.4 Expression profile of AjCTP1/2 and its receptors in *A. japonicus*

qRT-PCR analysis revealed that *AjCTP1/2* had the highest expression level in the circumoral nervous system (CNS) (Figure 5A). The expression levels of three receptors differed from those of *AjCTP1/2*. *AjCTR* (Figure 5B) and *AjPDR1* (Figure 5C) showed the highest expression level in CNS and longitudinal muscle, whereas *AjPDR2* exhibited highest expression in the respiratory tree (Figure 5D).

To investigate the distribution of CT-type neuropeptides in *A. japonicus* using immunohistochemistry we employed an antiserum to the starfish (*Asterias rubens*) CT-type neuropeptide ArCT, which has been used previously for visualization of the expression of ArCT in *A. rubens* (Cai et al., 2018) and a CT-type neuropeptide in the feather star *Antedon mediterranea* (AmCT; Aleotti et al., 2022). The antigen peptide that was used to generate the ArCT antiserum (KYNSPFGASGP-NH₂) shares a C-terminal FGxxGP-NH₂ (where X is variable) motif with both *AjCT1* (FGSAGP-NH₂) and *AjCT2* (FGSGGP-NH₂) and therefore antibodies to ArCT may cross-react with *AjCT1* and/or *AjCT2*. Accordingly, the ArCT antiserum revealed immunostaining in the central nervous system (radial nerve cords and circumoral nerve ring) (Figure 6A–F), which was abolished by pre-incubation of the antiserum with the ArCT antigen peptide (Figure 6-figure supplement 1). More specifically, high magnification images revealed immunostaining was localised in the ectoneural region, but not in the hyponeural region, of the radial nerve cords, with bilaterally symmetrical immunostained cells (neuronal somata) present in the ectoneural epithelium (Figure 6C). Likewise, immunostained cells were revealed in the ectoneural region of the circumoral nerve ring (Figure 6D–F). Furthermore, immunostaining was also revealed in tentacle nerves, which emanate from the circumoral nerve ring (Figure 6D–F), and in several regions of the digestive system, including the pharynx/stomach (Figure 6G), intestine (Figure 6H) and cloaca (Figure 6I). No immunostaining was observed in the longitudinal muscles (Figure 6A, B).

2.5 *AjCT1* and *AjCT2* causes dose-dependent relaxation of longitudinal muscle and intestine preparations from *A. japonicus*

Both *AjCT1* and *AjCT2* caused dose-dependent relaxation of longitudinal muscles from *A. japonicus* at 10^{-10} – 10^{-6} M (Figure 7). The maximum relaxant effects of *AjCT2* (10^{-6} M) on longitudinal muscles are illustrated in Figure 7-figure supplement 1A. At the highest concentration tested (10^{-6} M), *AjCT1* and *AjCT2* caused $30.07 \pm 7.49\%$ and $32.50 \pm 14.49\%$ reversal of acetylcholine (ACh)-induced contraction, respectively (Figure 7A, 7B). *AjCT1* had no effect on the contractile state of intestine preparations, even at concentrations as high as 10^{-5} M (Figure 7C). Unlike *AjCT1*, *AjCT2* caused dose-dependent relaxation of intestine preparations, with $26.33 \pm 8.75\%$ reversal of ACh-induced contraction observed at 10^{-5} M (Figure 7D). Control experiments in which ACh-induced contraction was recorded without addition of CT-type peptides showed that ACh-induced contraction of longitudinal muscles and intestinal preparations persisted for ≥ 15 minutes, confirming that relaxation effects observed could be attributed to addition of *AjCT1* or *AjCT2* (Figure 7-figure supplement 1B, 1C).

2.6 *AjCT2* affects feeding and growth in *A. japonicus*

To investigate the effects of *AjCT1* and *AjCT2* *in vivo*, a long-term injection experiment was carried out at concentrations of 5×10^{-3} mg/mL (low-dose group) and 5×10^{-1} mg/mL (high-dose group), with the injection volume calculated based on the wet weight of sea cucumbers at 1 μ L/g. As shown in Figure 8A, there were no significant changes in weight gain rate (WGR) and specific growth rate (SGR) in the *AjCT1* low concentration group (CT1L), *AjCT1* high concentration group (CT1H) and *AjCT2* low concentration group (CT2L) ($p \geq 0.05$), but in the *AjCT2* high concentration

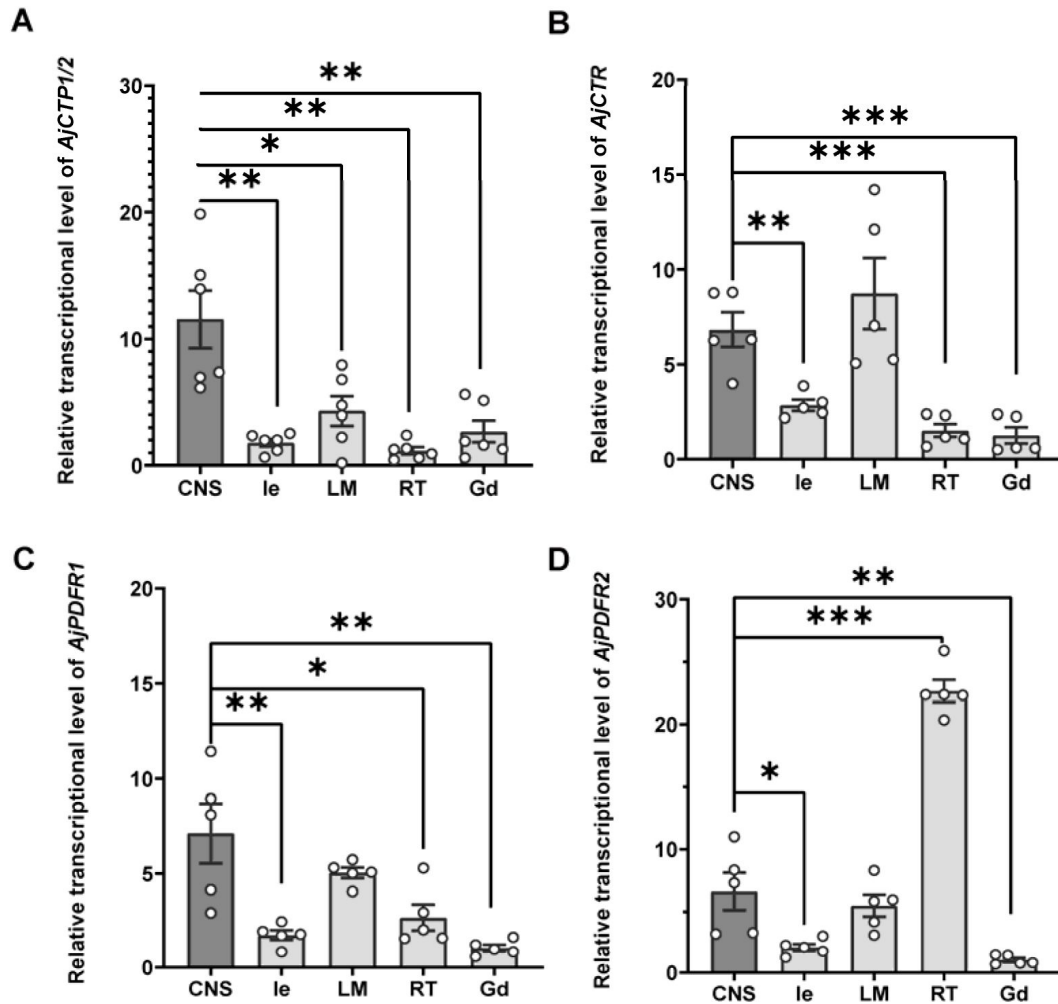


Figure 5.

The relative expression level of *AjCTP1/2* (A), *AjCTR* (B), *AjPDFR1* (C) and *AjPDFR2* (D) in different tissues.

The circumoral nervous system was selected as the control group. Values are means $\pm$ SEM (*AjCTP1/2*, $n = 6$; *AjCTR*, *AjPDFR1* and *AjPDFR2*, $n = 5$). Asterisks represent significant differences as follows: *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$. CNS, circumoral nervous system; Gd, gonad; Ie, intestine; LM, longitudinal muscle; RT, respiratory tree.

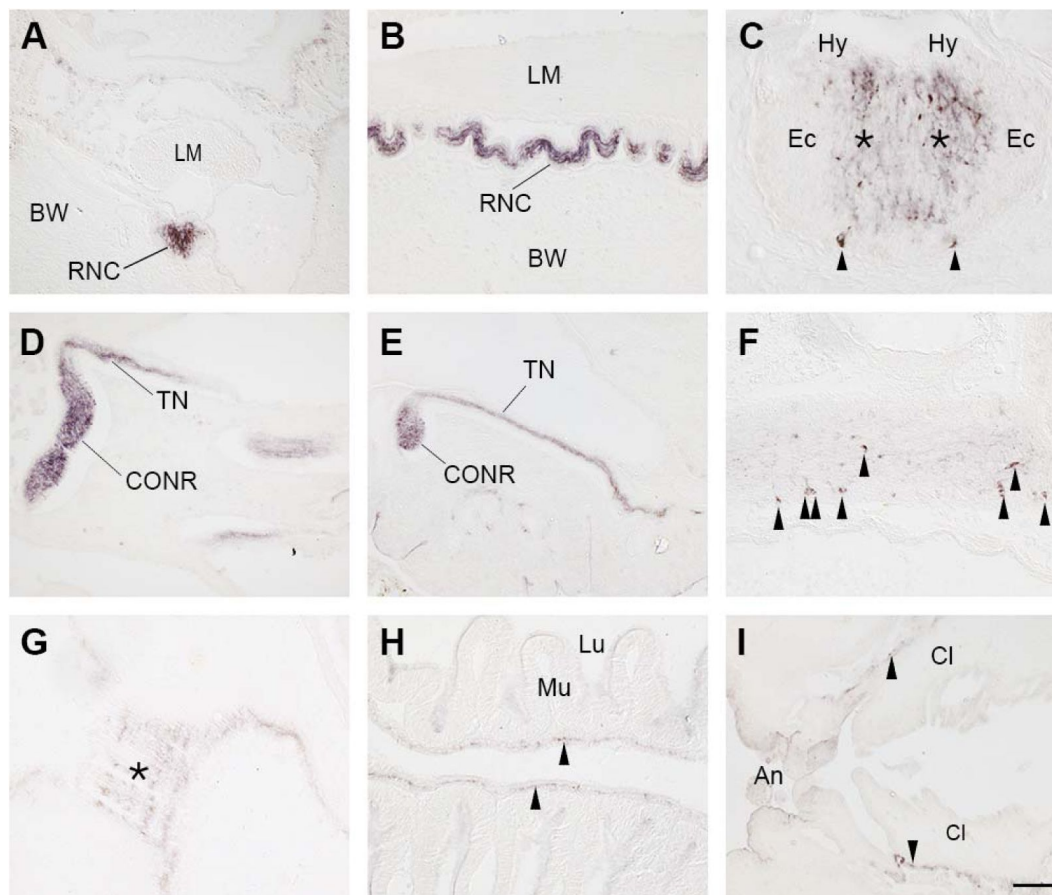


Figure 6.

Immunohistochemical localization of CT-type neuropeptide expression in *A. japonicus* using antibodies to the starfish CT-type neuropeptide ArCT.

A. Transverse section showing immunostaining in the radial nerve cord. B. Longitudinal section showing immunostaining in the radial nerve cord. C. High magnification image of a transverse section showing immunostained cells (arrowheads) and immunostained fibers in the neuropile (asterisks) of the ectoneural region of the radial nerve cord. Note the absence of immunostaining in the hyponeural region. D, E. Longitudinal sections showing immunostaining in the circumoral nerve ring and in tentacle nerves that emanate from the circumoral nerve ring. F. High magnification image of a transverse section showing immunostained cells (arrowheads) in the circumoral nerve ring. G. Longitudinal section showing immunostained fibers (asterisk) at the junction between the pharynx and the stomach. H. Longitudinal section showing immunostained fibers (arrowheads) in the intestine. I. Longitudinal section showing immunostained fibers (arrowheads) in the cloaca. Abbreviations: An, anus; BW, body wall; Cl, cloaca; CONR, circumoral nerve ring; Ec, Ectoneural region of the radial nerve cord; Hy, hyponeural region of the radial nerve cord; LM, longitudinal muscle; Lu, lumen; Mu, mucosa; RNC, radial nerve cord; TN, tentacle nerve. Scale bar, as shown in panel I, is 100 μ m in A, B, D, E, I; 50 μ m in F, G, H; 25 μ m in C.

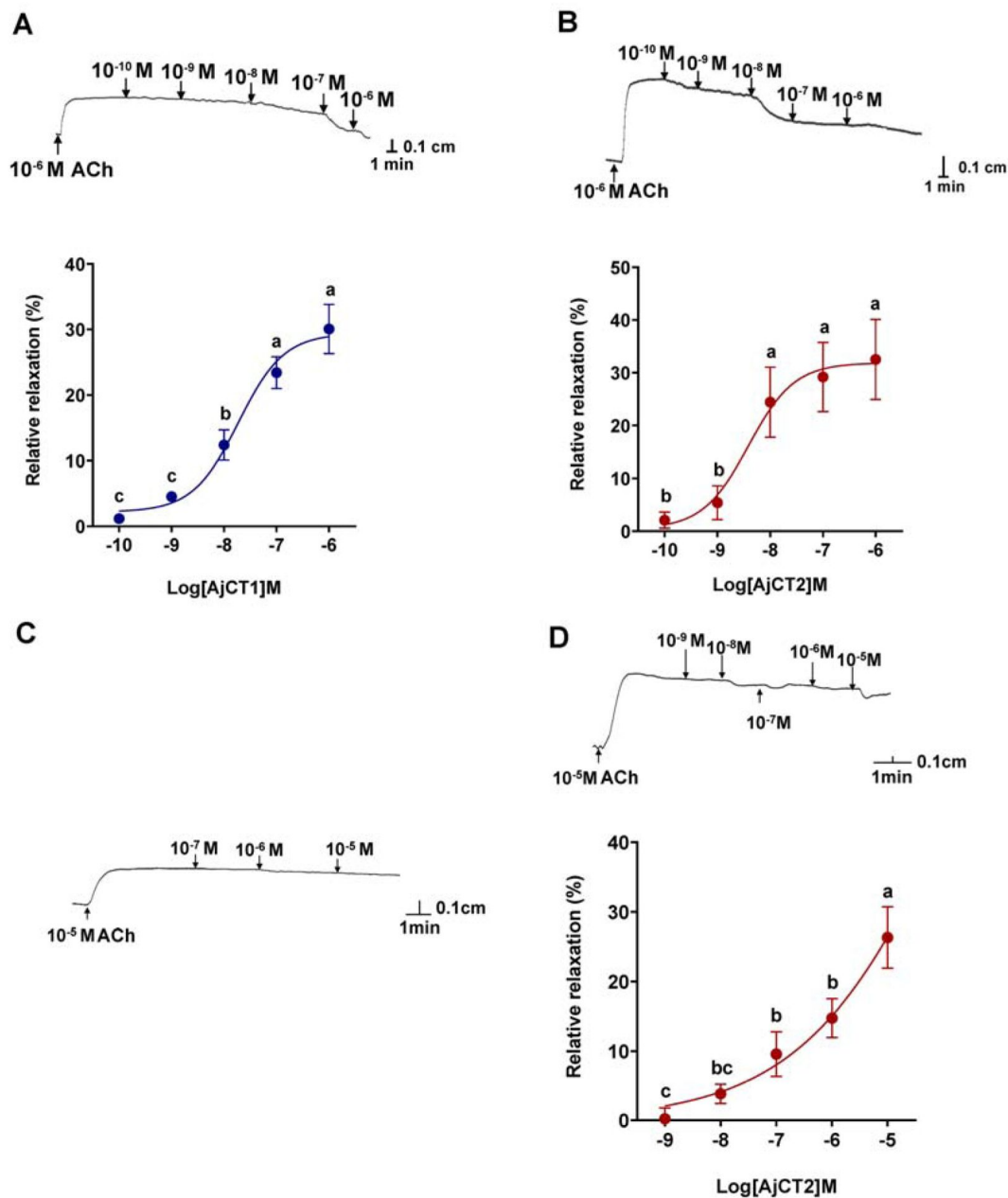


Figure 7.

Pharmacological effects of AjCT1 and AjCT2 on longitudinal muscle and intestine preparations from *A. japonicus*.

(A) and (B) show representative recordings of the relaxing effects of AjCT1 and AjCT2 on longitudinal muscle preparations. The graphs show the dose-dependent relaxing effects of AjCT1 and AjCT2 (10^{-10} – 10^{-6} M), calculated as the percentage reversal of the contracting effect of 10^{-6} M ACh. (C) and (D) show representative recordings of experiments in which AjCT1 (10^{-7} – 10^{-5} M) and AjCT2 (10^{-9} – 10^{-5} M) were tested on intestine preparations. AjCT1 had no effect, but AjCT2 caused relaxation and the graph shows the dose-dependent relaxing effect of AjCT2 (10^{-9} – 10^{-5} M), calculated as the percentage reversal of the contracting effect of 10^{-5} M ACh. All recordings shown are representative of three independent experiments. Mean values $\pm$ SEM were determined from three preparations. Lowercase letters a, b and c above columns indicate the statistical difference at $p < 0.05$, whilst there is no significant difference between bc and c.

group (CT2H) there was a significant increase in WGR and SGR in comparison with the control group (CO) ($p < 0.05$). Correspondingly, the mass of feces increased and the residual bait mass decreased (**Figure 8A-figure supplement 1**). Furthermore, the relative intestinal expression levels of several growth factors were investigated and the results showed that *AjGDF-8*, a growth inhibitory factor (Li et al., 2016), was significantly down-regulated in all the experimental groups. In contrast, the relative expression levels of *AjIgf* and *AjMegf6*, growth promoting factors (Párrizas and LeRoith, 1997; Deribe et al., 2009), were significantly up-regulated in intestines (**Figure 8B**).

Subsequently, the relative intestinal expression levels of *AjCT1* and *AjCT2* receptors were further tested for the CT2H group. The results showed that the relative expression levels of all three receptors were significantly up-regulated, with the ranked fold change being *AjCTR* > *AjPDFR1* > *AjPDFR2* (**Figure 8C**). Conversely, the relative expression levels of the three receptor genes were significantly down-regulated in *siAjCTP1/2-1* treated animals (**Figure 9A**, **9B** and **9C**). The interference rate for *siAjCTP1/2-1* was 66 % (**Figure 9-figure supplement 1**).

In long-term *AjCT2* loss-of-function studies, the *siAjCTP1/2-1* group exhibited significant reductions in WGR and SGR compared to the control group (siNC) ($p < 0.05$, **Figure 10A**). Correspondingly, except for phase I, the *siAjCTP1/2-1* group showed significantly increased remaining bait and decreased excrement across phases II-VI (**Figure 10B**). Furthermore, the growth inhibitory factor *AjGDF-8* was significantly up-regulated and the growth promoting factor *AjMegf6* was significantly down-regulated in the *siAjCTP1/2-1* group (**Figure 10C**).

2.7 *AjPDFR2* mediates CT-type signaling-induced feeding and growth promotion in *A. japonicus*

The differential activation of signaling pathways and associated physiological outcomes between *AjCT2* and *AjCT1* suggested that *AjCT2* may promote feeding and growth via *AjPDFR2*-mediated mechanisms. To investigate this hypothesis, we performed long term knockdown experiments targeting *AjPDFR2*. As shown in **Figure 11A**, *siAjPDFR2-1* effectively knocked down the expression of *AjPDFR2*. In the *siAjPDFR2-1* group, both WGR and SGR were significantly decreased compared to siNC group (**Figure 11B**). Correspondingly, the remaining bait weight was significantly increased in *siAjPDFR2-1* group (except during phase I), and the weight of excrement was significantly decreased in phase V and VI (**Figure 11C**). Additionally, the growth promoting factor *AjIgf* was significantly down-regulated in the *siAjPDFR2-1* group (**Figure 11D**).

3. Discussion

Previous studies have revealed that the CT-type neuropeptide DH_{31} can act as a ligand for a CTR/CLR-type receptor and a pigment-dispersing factor (PDF) receptor in a protostome – the insect *D. melanogaster* (Mertens et al., 2005; Shafer et al., 2008; Schwartz et al., 2019). Here we show that two CT-type peptides (*AjCT1* and *AjCT2*) act as ligands for a CTR/CLR-type receptor and two pigment-dispersing factor (PDF) type receptors in a deuterostome – the sea cucumber *A. japonicus* (phylum Echinodermata). This is an important finding because it indicates that the ability of CT-type neuropeptides to act as ligands for both CTR/CLR-type receptors and PDF-type receptors may be an evolutionarily ancient property that originated in a common ancestor of the Bilateria. Alternatively, the ability of CT-type neuropeptides to act as ligands for PDF-type receptors in *D. melanogaster* and *A. japonicus* may have evolved independently. Further studies on a wider variety of both protostome (e.g. molluscs, annelids) and deuterostome taxa (e.g. other echinoderms, hemichordates) are needed to address this issue. Here, we also investigated the

Figure 8.

***In vivo* pharmacological tests revealed that AjCT1 and AjCT2 affect feeding and growth-related gene expression in *A. japonicus*.**

(A) Weight gain rate (WGR) and specific growth rate (SGR) in experimental and control groups. Values are means $\pm$ SEM ($n = 3$, representing three parallel experiments of each group). (B) The relative intestinal transcript expression levels of *AjGDF-8*, *AjIgf* and *AjMegf6* in different groups after injection of AjCT1 or AjCT2 or sterilized seawater (CO) in *A. japonicus*. Values are means $\pm$ SEM ($n = 5$ biological duplication). (C) The relative intestinal expression levels of AjCT1/2 receptors (*AjCTR*, *AjPDRF1* and *AjPDRF2*) in CT2H and CO groups after 24 days injection experiment. Values are means $\pm$ SEM ($n = 5$ biological duplication). CO: control group (sterilized seawater); CT1L/CT2L: AjCT1/AjCT2 low concentration group (5×10^{-3} mg/mL); CT1H/CT2H: AjCT1/AjCT2 high concentration group (5×10^{-1} mg/mL). * indicates statistically significant differences with $p < 0.05$, ** indicates statistically significant differences with $p < 0.01$.

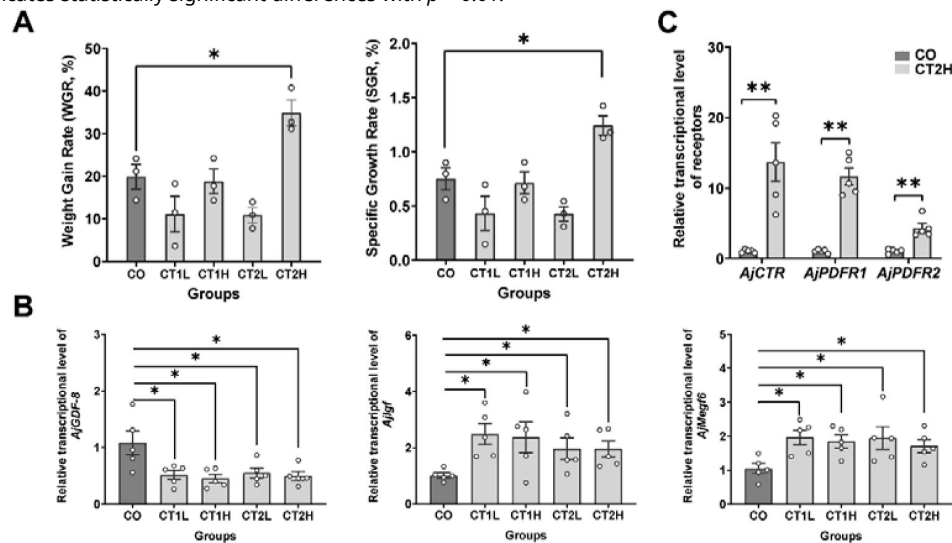
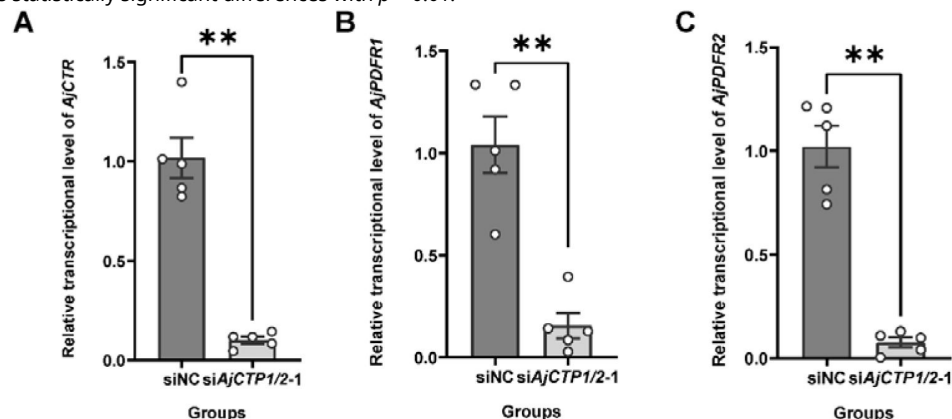


Figure 9.

The relative intestinal expression levels of AjCT1/2 receptors after *AjCTP1/2-1* knockdown.

(A), (B) and (C) show the relative expression levels of *AjCTR*, *AjPDRF1* and *AjPDRF2* respectively. Values are means $\pm$ SEM ($n = 5$). ** indicates statistically significant differences with $p < 0.01$.



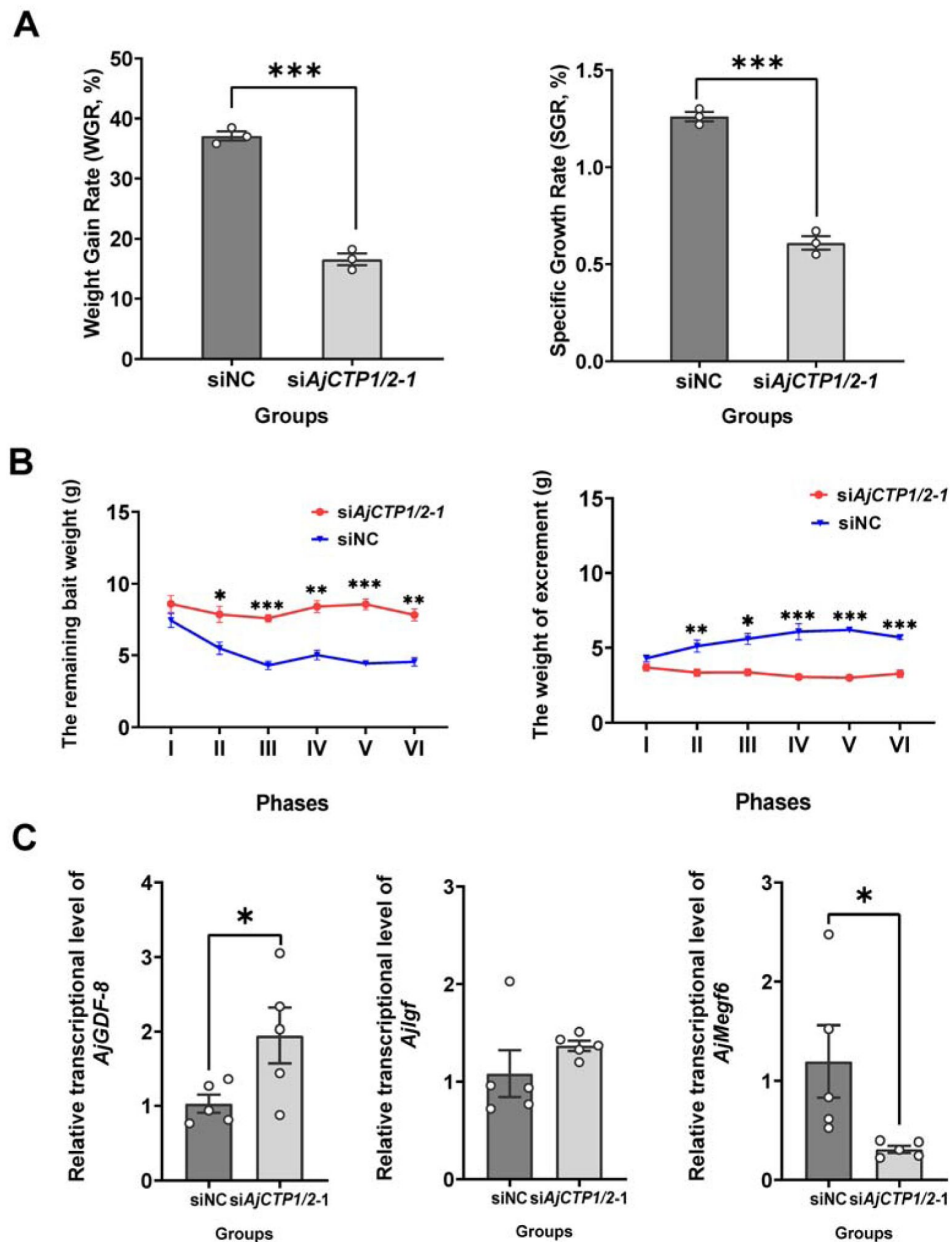


Figure 10.

Effects of 24-day *AjCTP1/2-1* knockdown *in vivo* on feeding and growth in *A. japonicus*.

(A) Weight gain rate (WGR) and specific growth rate (SGR) in siNC and siAjCTP1/2-1 groups. Values are means $\pm$ SEM ($n = 3$, representing three parallel experiments for each group). (B) Calculation of the remaining bait and excrement mass in different phases. The 24-day experiment was divided into six phases (I–VI), with each phase comprising four days. Values are means $\pm$ SEM ($n = 3$, representing three parallel experiments for each group). (C) The relative intestinal transcript expression levels of *AjGDF-8*, *AjIgf* and *AjMegf6* in siNC and siAjCTP1/2-1 groups. Values are means $\pm$ SEM ($n = 5$ biological duplication). siNC: negative control group; siAjCTP1/2-1: effectively experimental groups targeting *AjCTP1/2*. Asterisks represent statistically significant differences as follows: *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$.

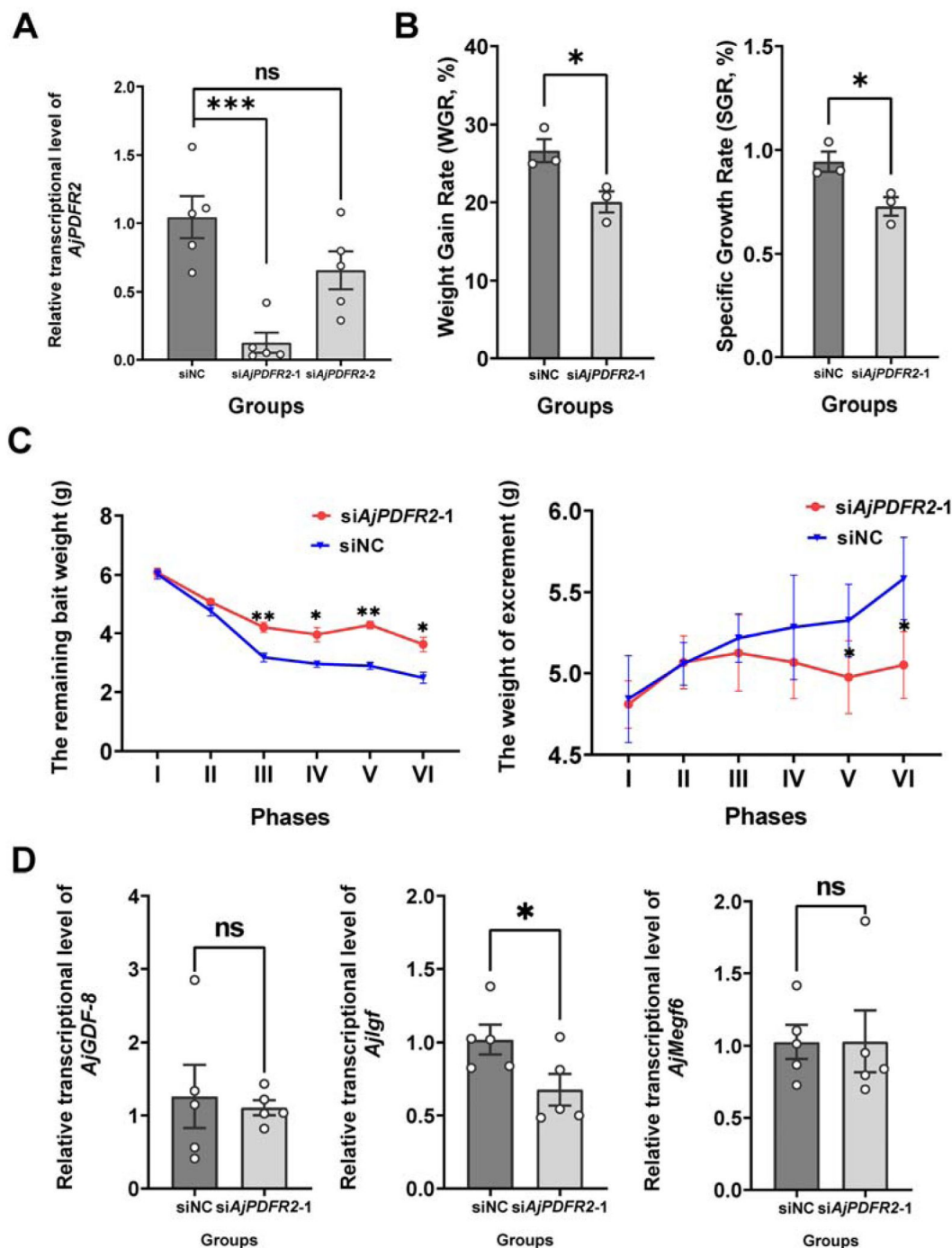


Figure 11.

Effects of *AjPDFR2*-1 knockdown *in vivo* on feeding and growth in *A. japonicus*.

(A) The relative expression level of *AjPDFR2* in siNC, si*AjPDFR2*-1 and si*AjPDFR2*-2. (B) Weight gain rate (WGR) and specific growth rate (SGR) in siNC and si*AjPDFR2*-1 groups. Values are means $\pm$ SEM ($n = 3$, representing three parallel experiments for each group). (C) The remaining bait and excrement weight in different phases. The 24-day experiment was divided into six phases (I–VI), with each phase comprising four days. Values are means $\pm$ SEM ($n = 3$, representing three parallel experiments of each group). (D) The relative intestinal transcript expression levels of *AjGDF-8*, *AjIgf* and *AjMegf6* in siNC and si*AjPDFR2*-1 groups. Values are means $\pm$ SEM ($n = 5$ biological duplication). siNC: negative control group; si*AjPDFR2*-1 and si*AjPDFR2*-2: experimental groups targeting *AjPDFR2*. Asterisks represent statistically significant differences as follows: *: $p < 0.05$, **: $p < 0.01$, ***: $p < 0.001$. ns: not significant.

signal transduction pathways that are activated by AjCT1 and AjCT2 when these peptides bind to their receptors and then proceeded to investigate the expression and actions of these peptides to gain insights into their physiological/behavioral roles in *A. japonicus*, as discussed below.

3.1 Alternative RNA splicing gives rise to transcripts encoding one (AjCT2) or two (AjCT1, AjCT2) CT-type neuropeptides in *A. japonicus*

In vertebrates, alternative RNA splicing gives rise to transcripts encoding CT or CGRP (Amara et al., 1982 [↗](#); Morris et al., 1984 [↗](#)). Analysis of transcriptome/genome sequence data from *A. japonicus* has revealed that in this species alternative RNA splicing gives rise to transcripts encoding one (AjCT2) or two (AjCT1, AjCT2) CT-type neuropeptides (Rowe et al., 2014 [↗](#); Zandawala et al., 2017 [↗](#); Chen et al., 2019 [↗](#); this study). A similar pattern of alternative RNA splicing has been reported for CT-type precursors in other sea cucumbers (*H. scabra* and *H. glaberrima*; class Holothuroidea) and in brittle stars (*Amphiura filiformis* and *Ophiopsila Aranea*; class Ophiuroidea) (Zandawala et al., 2017 [↗](#); Suwansa-Ard et al., 2018 [↗](#)). In contrast, CT-type precursors identified in other echinoderm classes (Asteroidea, Echinoidea and Crinoidea) comprise a single CT-type neuropeptide (Rowe and Elphick, 2012 [↗](#); Mirabeau and Joly, 2013 [↗](#); Rowe et al., 2014 [↗](#); Semmens et al., 2016 [↗](#); Cai et al., 2018 [↗](#); Aleotti et al., 2022 [↗](#)). However, it remains to be determined if this feature of CT-type precursors in sea cucumbers and brittle stars evolved independently or originated in a common ancestor of eleutherozoan echinoderms but with subsequent loss in Asteroidea and Echinoidea (Figure 12 [↗](#)).

Analysis of the sequences of AjCT1 and AjCT2 revealed the presence of two cysteine residues in their N-terminal regions, which is an evolutionarily conserved feature of CT-type peptides. Structural analysis of CT-type peptides in vertebrates and in starfish has revealed that these cysteines form a disulphide bridge in the mature peptides (Cai et al., 2018 [↗](#)) and therefore it is inferred that this post-translational modification also occurs in AjCT1 and AjCT2. Furthermore, the C-terminal residue of AjCT1 and ACT2 is predicted to be an amidated proline, which is a conserved post-translational modification of CT/DH31-type peptides in many taxa, but not vertebrate CGRPs (Cai et al., 2018 [↗](#)).

3.2 Identification and characterisation of receptors for CT-type neuropeptides in *A. japonicus*

Informed by previous genome-wide analysis of GPCRs in *A. japonicus* (Huang et al., 2021 [↗](#)), we identified three candidate receptors for AjCT1 and/or AjCT2. Firstly, AjCTR, which is a homolog of CT/DH₃₁-type receptors that have been identified in other taxa, including vertebrates and insects (Garelja et al., 2022 [↗](#); Johnson et al., 2005 [↗](#); Toribio et al., 2003 [↗](#)). Secondly, AjPDFR1 and AjPDF2, which are homologs of the PDF-type receptor that has been shown to be activated by DH₃₁ in *D. melanogaster* (Mertens et al., 2005 [↗](#); Zandawala, 2012 [↗](#)).

In mammals, CTR and CLR mediate effects of all CT-related neuropeptides, but their ligand specificity relies on the presence and types of receptor activity-modifying proteins (RAMPs) (McLatchie et al., 1998 [↗](#); Purdue et al., 2002 [↗](#); Conner et al., 2004 [↗](#); Hay et al., 2018 [↗](#)). Thus, CTR acts as a functional receptor for CT-type peptides without an associated RAMP (Katafuchi et al., 2009 [↗](#)), whereas activation of downstream signaling via CLR requires an associated RAMP (Hay et al., 2018 [↗](#)). Here, we investigated the occurrence of RAMPs in *A. japonicus* using the tblastn method employed by Sekiguchi et al., (2016) [↗](#), which enabled identification of RAMPs in the cephalochordate *B. floridae*. However, no RAMPs were identified in *A. japonicus*, suggesting that AjCTR may be functionally more similar to CTR than CLR. Accordingly, tblastn analysis of other available echinoderm genomes (*A. rubens*, *S. purpuratus* and *Anneissia japonica*), as well as transcriptome databases for brittle stars (*O. victoriae*, *O. aranea* and *A. filiformis*), also failed to

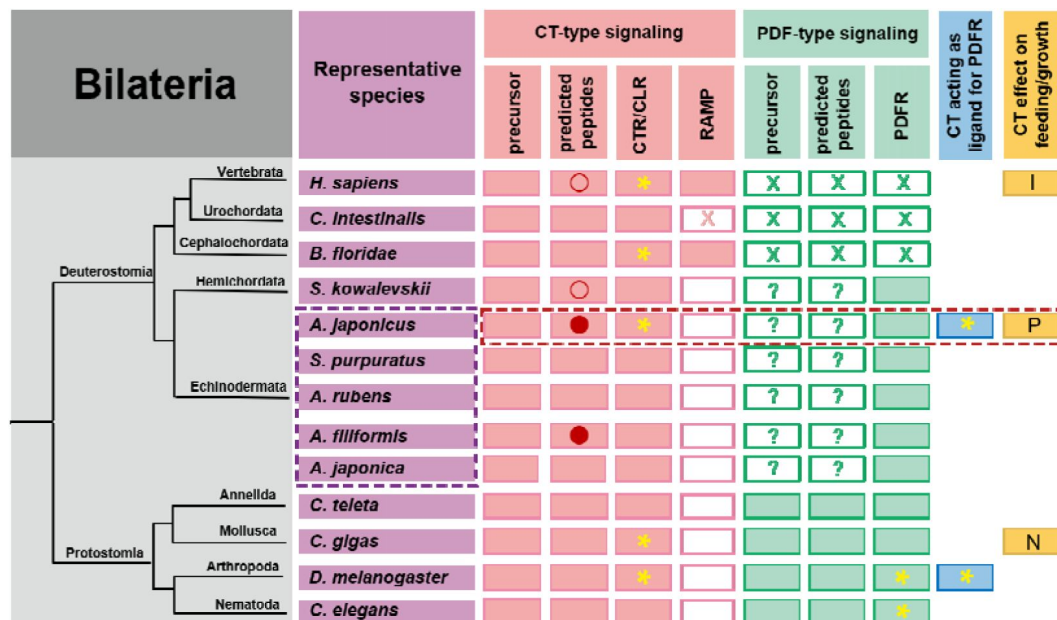


Figure 12.

Phylogenetic summary of the occurrence and properties of CT-type and PDF-type signaling systems and effects of CT-type peptides on feeding/growth regulation in representative species.

Echinoderms are boxed in purple with dashed lines and *A. japonicus*, the focus of this study, is framed in red with dashed lines. Pink filled rectangles indicate the occurrence of CT-type precursors/neuropeptides, CTR/CLR-type receptors and RAMPs. Green filled rectangles indicate the occurrence of PDF-type precursors/neuropeptides and receptors. Blue filled rectangles indicate species in which CT-type peptides have been shown to act as ligands for PDF-type receptors. Yellow filled rectangles indicate species in which effects or lack of effects of CT-type peptides on feeding/growth have been reported (I: Inhibition; P: Promotion; N: No effect). Red circles indicate that neuropeptides are generated by alternative splicing of precursor transcripts, with an unfilled circle indicating that transcripts encode one neuropeptide and with a filled circle indicating that transcripts can encode two neuropeptides in tandem. A yellow asterisk marks species in which neuropeptides have been shown to act as ligands for receptors experimentally. Unfilled rectangles indicate the absence of a protein. Unfilled rectangles containing a cross (X) indicate loss of a protein during evolution. Unfilled rectangles containing a question mark (?) indicate the existence of precursors/peptides that have yet to be identified.

identify RAMPs. Likewise, RAMPs have also not been found in protostomes (Schwartz et al., 2019). Therefore, RAMPs may be unique to chordates, although interestingly RAMPs have also not been identified in the urochordate *C. intestinalis* (Sekiguchi et al., 2009) (Figure 12).

As highlighted above, here we show that the CT-type peptides AjCT1 and AjCT2 both act as ligands for a CTR/CLR-type receptor (AjCTR) and two PDF-type receptors (AjPDFR1, AjPDFR2) in *A. japonicus*. Furthermore, characterization of downstream signaling mechanisms revealed that both AjCT1 and AjCT2 trigger AjCTR-mediated cAMP accumulation, Ca^{2+} mobilization, and ERK signaling. These findings are consistent with previous studies on CT-type receptors in vertebrates and in the mollusc *Crassostrea gigas* (Pondel, 2000; Walker et al., 2010; Andreassen et al., 2014; Schwartz et al., 2019). Both AjCT1 and AjCT2 trigger cAMP accumulation, Ca^{2+} mobilization, and ERK signaling mediated by AjPDFR1. Both AjCT1 and AjCT2 triggered cAMP accumulation mediated by AjPDFR2, but only AjCT2 (not AjCT1) triggered Ca^{2+} mobilization and ERK signaling mediated by AjPDFR2. These differences in the signal transduction mechanisms activated by AjCT1 and AjCT2 may account for differences in the pharmacological activities and physiological roles of these peptides, as discussed below.

3.3 AjCT1 and AjCT2 act as muscle relaxants in *A. japonicus*

Analysis of the expression of the gene encoding the precursors comprising AjCT2 or AjCT1 and AjCT2 revealed the occurrence of transcripts in the circumoral nervous system as well as in peripheral tissues/organs, including the longitudinal muscle, digestive system and gonads. Immunohistochemical analysis using an antiserum to the starfish (*A. rubens*) CT-type neuropeptide ArCT enabled visualisation of CT-type neuropeptide(s) in juvenile specimens of *A. japonicus*. Intense and extensive immunostaining was revealed in the ectoneural region of the radial nerve cords and circumoral nerve ring, consistent with the high levels *AjCTP1/2* transcripts detected in circumoral nervous tissue. Immunostained nerve fibers were also observed in the digestive system (including the intestine), consistent with the detection *AjCTP1/2* transcripts in intestine. However, no immunostaining was observed in longitudinal muscles, which is inconsistent with the detection of *AjCTP1/2* transcripts this tissue. This may reflect differences in the detection sensitivity of the techniques used. In accordance with the expression pattern of *AjCTP1/2* in *A. japonicus*, analysis of CT-type neuropeptide expression in other echinoderms has revealed a widespread pattern of expression in larval and adult starfish (*A. rubens*) and in juvenile and adult featherstars (*Antedon mediterranea*) (Mayorova et al., 2016; Cai et al., 2018; Aleotti et al., 2022).

Several CT-related peptides act as muscle relaxants in vertebrates (Steiner et al., 1991; Pinto et al., 1996; Yoshimoto et al., 1998; Edvinsson et al., 2001; Golpon et al., 2001; Kandilci et al., 2008) and, accordingly, previous studies have revealed that CT-type neuropeptides cause relaxation of muscle preparations in starfish, including locomotory organs (tube feet) and the apical muscle, which is functionally similar to the longitudinal muscle in sea cucumbers (Cai et al., 2018). Therefore, it was of interest to investigate if AjCT1 and/or AjCT2 act as muscle relaxants in *A. japonicus*. Both AjCT1 and AjCT2 caused dose-dependent relaxation of longitudinal muscle preparations, consistent with the relaxing effect of CT-type peptides on starfish apical muscle preparations. It is worth noting that while AjCT2 at 10^{-6} M exhibited the maximum efficacy in reversing ACh-induced contraction, its relaxing effect weakened compared to the 10LL M treatment when applied at progressively higher concentrations. This diminished efficacy likely stems from receptor desensitization, induced by repeated AjCT2 application to longitudinal muscle preparations (Tsurumaki et al., 2003; Arrowsmith and Wray, 2014). Furthermore, AjCT2, but not AjCT1, caused dose-dependent relaxation of intestine preparations from *A. japonicus*. Interestingly, the CT-type peptide ArCT does not cause relaxation of cardiac stomach preparations from the starfish *A. rubens* (Cai et al., 2018). Therefore, from a functional perspective ArCT is more like AjCT1 than AjCT2 and we speculate that this may reflect loss of an exon encoding a AjCT2-like peptide in the Asteroidea lineage. Further insights into this issue and the evolution of CT-type neuropeptide function echinoderms could be obtained by investigation of

the actions of CT-type peptides in brittle stars and sea urchins. Differences in the effects of AjCT1 and AjCT2 on the intestine may be explained by differences in their signal transduction mechanisms. Thus, only AjCT2, not AjCT1, triggered AjPDFR2-mediated activation of Ca^{2+} mobilization and ERK signaling and therefore we speculate that the relaxing effect of AjCT2 on the intestine may be mediated by activation of AjPDFR2.

3.4 AjCT2/AjPDFR2 stimulates feeding and growth in *A. japonicus*

In vivo pharmacological assays demonstrated that only high concentrations of AjCT2 significantly enhanced feeding and growth rates in *A. japonicus*. In contrast, neither a low concentration of AjCT2 nor any concentration of AjCT1 (low or high) induced detectable effects. Furthermore, long-term knockdown of *AjCTP1/2* further validated the essential role of AjCT2 in regulating feeding and growth in this species. To elucidate the receptor mediating AjCT2's feeding- and growth-promoting effects, we selected AjPDFR2 based on its distinct activation profile: AjCT2 selectively activated AjPDFR2, inducing downstream ERK1/2 phosphorylation, whereas AjCT1 exhibited no activity toward this receptor. Given this receptor specificity, we performed *AjPDFR2* knockdown experiments, which revealed phenotypic changes consistent with those in *AjCTP1/2* knockdown animals, including significantly reduced WGR and SGR, alongside increased remaining bait accumulation and diminished excrement output compared to control. Collectively, these results support a model wherein AjCT2 promotes feeding and growth in *A. japonicus* via AjPDFR2-dependent activation of the cAMP/PKA/ERK1/2 and Gαq/Ca²L/PKC/ERK1/2 cascades. Considering the inherent complexity of neuropeptide signaling systems, which involve multiple GPCR subtypes coupled to diverse signaling cascades, ligands bound to the same receptor may activate distinct G protein subforms within a single cell (Møller et al., 2003 [DOI](#); Mendel et al., 2020 [DOI](#)). Receptor activation modes may be modulated by structural polymorphisms or binding site diversity (Wong et al., 2000 [DOI](#); Changeux, 2010 [DOI](#)), as well as by the differential efficacy of peptides in activating receptors *in vivo*.

Collectively, these findings indicate that AjCT2 has a physiological role in stimulating feeding and growth in *A. japonicus*. To the best of our knowledge, this is the first study to report orexigenic and growth-promoting effects of a CT-type neuropeptide in bilaterian animals. Interestingly, previous studies have reported the expression of CT-type neuropeptides in the digestive system of invertebrate chordates (Sekiguchi et al., 2009 [DOI](#); Sekiguchi et al., 2016 [DOI](#)). However, in contrast to the stimulatory effect of AjCT2 on feeding/growth in *A. japonicus*, CT-type neuropeptides have been found to inhibit feeding and growth in vertebrates (Del et al., 2002 [DOI](#); Hay et al., 2018 [DOI](#); Nakamura et al., 2018 [DOI](#)). Thus, it appears that opposing roles of CT-type neuropeptide signaling systems as regulators for feeding/growth have evolved in deuterostomes, with a stimulatory role reported here in an echinoderm and an inhibitory role reported previously in vertebrates (Figure 12 [DOI](#)). Further research is now needed to investigate if the stimulatory effect of AjCT2 on feeding/growth observed in here in *A. japonicus* also applies to other sea cucumber species. Moreover, our discovery of this effect of AjCT2 on *A. japonicus* may provide a basis for development of novel approaches to improve aquaculture of this economically important species.

4. Materials and methods

Key Resources Table				
Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
Cell line (<i>Homo sapiens</i>)	HEK293 cell line	Procell	CL-0005	
Antibody	phosphorylated ERK1/2 kinases (Thr202/Tyr204) (goat)	Cell Signaling Technology	CAT#4930	IF (1:2000)
Antibody	ERK1/2 kinases (137F5) (goat)	Cell Signaling Technology	CAT#4695T	IF (1:1000)
Antibody	HRP-conjugated goat anti-rabbit IgG (goat)	Absin	Abs20040ss	IF (1:2000)
sequence-based reagent	Primers for pcDNA3.1(+) plasmid construction-- AjCTR	This paper	PCR PRIMER	FORWARD CGCGGATCCAAAATGCAGGAGA ACGATTC BamH ☐
sequence-based reagent	Primers for pcDNA3.1(+) plasmid construction-- AjCTR	This paper	PCR PRIMER	REVERSE CCGGAATTCTTAACTACCGTT GTTTTAT EcoR ☐
sequence-based reagent	Primers for pcDNA3.1(+) plasmid construction— AjPDFR1	This paper	PCR PRIMER	FORWARD CGCGGATCCACAATGAGTGTCTG GGAACGT BamH ☐
sequence-based reagent	Primers for pcDNA3.1(+) plasmid	This paper	PCR PRIMER	REVERSE CCGGAATTCTCATTTCCCACCCC ACTCAC

	construction— AjPDFR1			EcoR □
sequence- based reagent	Primers for pcDNA3.1(+) plasmid construction— AjPDFR2	This paper	PCR PRIMER	FORWARD CGCGGATCCAAGATGGCGAACA ATGAAAC BamH □
sequence- based reagent	Primers for pcDNA3.1(+) plasmid construction— AjPDFR2	This paper	PCR PRIMER	REVERSE CCGGAATTCTCATTCAAATT TGCCGT EcoR □
sequence- based reagent	Plasmid construction AjCTR-EGFP	This paper	PCR PRIMER	FORWARD CCGGAATTCTGAAAATGCAGGA GAACGATTC EcoR □
sequence- based reagent	Plasmid construction AjCTR-EGFP	This paper	PCR PRIMER	REVERSE CGCGGATCCCGAACTACCGTTG TTTATCTG BamH □
sequence- based reagent	Plasmid construction AjPDFR1-EGFP			CCGGAATTCTGACAATGAGTGT CGGGAACGT EcoR □
sequence- based reagent	Plasmid construction AjPDFR1-EGFP			CGCGGATCCCGTTTCCACCCC ACTACTGT BamH □
sequence- based reagent	Plasmid construction AjPDFR2-EGFP			CCGGAATTCTGAAGATGGCGAA CAATGAAAC EcoR □
sequence- based reagent	Plasmid construction AjPDFR2-EGFP			CGCGGATCCCGTTCATTAAATTT GCCGTTCA BamH □
sequence- based reagent	Primers for qPCR assays β -Actin	This paper	PCR PRIMER	FORWARD AAGGTTATGCTCTTCCTCACGCT
sequence- based reagent	Primers for qPCR assays β -Actin	This paper	PCR PRIMER	REVERSE GATGTCACGGACGATTTCACG

sequence-based reagent	Primers for qPCR assays <i>β-Tubulin</i>	This paper	PCR PRIMER	FORWARD GAAAGCCTTACGACGGAACA
sequence-based reagent	Primers for qPCR assays <i>β-Tubulin</i>	This paper	PCR PRIMER	REVERSE CACCACGTGGACTCAAATG
sequence-based reagent	Primers for qPCR assays <i>AjCTP1/2</i>	This paper	PCR PRIMER	FORWARD GTTTGGCGGTACCCATTAC
sequence-based reagent	Primers for qPCR assays <i>AjCTP1/2</i>	This paper	PCR PRIMER	REVERSE GGCGTTCGGTCTGTTAATAG
sequence-based reagent	Primers for qPCR assays <i>AjMegf6</i>	This paper	PCR PRIMER	FORWARD CGGTCCTTACTGTGCGGAGA
sequence-based reagent	Primers for qPCR assays <i>AjMegf6</i>	This paper	PCR PRIMER	REVERSE CGTCCTCCCAGCTGGACATC
sequence-based reagent	Primers for qPCR assays <i>AjIgf</i>	This paper	PCR PRIMER	FORWARD TGGAAGGCAGTAAGGCTA
sequence-based reagent	Primers for qPCR assays <i>AjIgf</i>	This paper	PCR PRIMER	REVERSE TAGGCTGATGTGGTTGGA
sequence-based reagent	Primers for qPCR assays <i>AjGDF-8</i>	This paper	PCR PRIMER	FORWARD ATACGCCCATACTCACATCC
sequence-based reagent	Primers for qPCR assays <i>AjGDF-8</i>	This paper	PCR PRIMER	REVERSE TGGCAGGAGTCAACCTTCAT
sequence-based reagent	Primers for qPCR assays <i>AjCTR</i>	This paper	PCR PRIMER	FORWARD GACGAATGGAGGAGAAGG
sequence-based reagent	Primers for qPCR assays <i>AjCTR</i>	This paper	PCR PRIMER	REVERSE CGCCGCAATTAGAGTAGT

sequence-based reagent	Primers for qPCR assays <i>AjPDFR1</i>	This paper	PCR PRIMER	FORWARD TTATCAACCGCCGTCTTC
sequence-based reagent	Primers for qPCR assays <i>AjPDFR1</i>	This paper	PCR PRIMER	REVERSE CAGGAGGAGGAACACATTC
sequence-based reagent	Primers for qPCR assays <i>AjPDFR2</i>	This paper	PCR PRIMER	FORWARD TGTCTCTCTGCCTCTTCT
sequence-based reagent	Primers for qPCR assays <i>AjPDFR2</i>	This paper	PCR PRIMER	REVERSE GTCCTTCTCATCGGTGTG
sequence-based reagent	Primers for 3'RACE <i>AjCTP1</i> <i>AjCTP2</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTCGAGCTG TGTCGATAACATTACGCATGG
sequence-based reagent	Primers for 3'RACE <i>AjPDFR1</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTGAGGGAA AGCCAAACCAGCGAGACG
sequence-based reagent	Primers for 3'RACE <i>AjPDFR2</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTGTGCAAC CGGACGAAGATCCACGTG
sequence-based reagent	Primers for 5'RACE <i>AjPDFR1</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTCGTCTCG CTGGTTTGGCTTTCCCTC
sequence-based reagent	Primers for 5'RACE <i>AjPDFR2</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTGCCTTCC ACAAACATCCATGCGAACGC
sequence-based reagent	Primers for 5'RACE <i>AjCTR</i>	This paper	PCR PRIMER	GATTACGCCAAGCTTACGCCGG GCTCTCAAGAAGTCATTC
sequence	RNAi	This paper	si <i>AjCTP1</i> -1	GGAAUUAUCAAGACAGUAU (dT)(dT) AUACUGUCUUGAAUUAUCC (dT)(dT)
sequence	RNAi	This paper	si <i>AjCTP1</i> -2	GAAAGUCGCUAUGCAGUA (dT)(dT) UACUGCAUUAAGCGACUUUC

				(dT)(dT)
sequence	RNAi	This paper	si <i>AjpDfR2</i> -1	GUAGCGAUUGGUCGAAUAA (dT)(dT) UUAUUCGACCAAUCGCUAC (dT)(dT)
sequence	RNAi	This paper	si <i>AjpDfR2</i> -2	CAGUCAUUGUUAUCUACCA (dT)(dT) UGGUAGAUAAACAAUGACUG (dT)(dT)
sequence	RNAi	This paper	siNC	UUCUCCGAACGUGUCACGU (dT)(dT) ACGUGACACGUUCGGAGAA (dT)(dT)
commercial assay or kit	SMARTer RACE 5'/3' Kit	Takara	Cat No. 634858	
commercial assay or kit	FastPure Gel DNA Extraction Mini Kit	Vazyme	DC301-01	
commercial assay or kit	Hifair® □ 1st Strand cDNA Synthesis SuperMix	YEASEN	Cat No. 11141ES60	
commercial assay or kit	Hieff UNICON Universal Blue qPCR SYBR Green Master Mix	YEASEN	Cat No. 11184ES08	
commercial assay or kit	Firefly luciferase reporter gene assay kit	MK	MF4001	
commercial assay or kit	cAMP Assay kit	R&D systems	Cat No. KGE002B	

4.1 Animals

Adult sea cucumbers were collected from an aquaculture breeding farm in Weihai, Shandong, China. All the animals were acclimated in breeding aquaria (18 L, 32 ppt) for at least seven days and fed daily with a commercially formulated diet. Tissues used for cDNA cloning and qRT-PCR were obtained from six adult *A. japonicus* (~120 g) and kept at -80 L until use. Juvenile specimens of *A. japonicus* (~15 mm in length) were used for immunohistochemical analysis of CT-type neuropeptide expression. *In vitro* functional analysis and *in vivo* experiments were carried out using adult sea cucumbers with body masses of ~120 g and ~20 g, respectively.

Ethical approval

All methods were carried out in accordance with the ARRIVE guidelines (<https://arriveguidelines.org>) for animal experiments. All animal care and use procedures were approved by the Institutional Animal Care and Use Committee of Ocean University of China (Permit Number: 20141201). They were performed according to the Chinese Guidelines for the Care and Use of Laboratory Animals (GB/T 35892-2018).

4.2 Molecular structure and phylogenetic analysis of CT-type neuropeptides and precursors

A. japonicus CT-type neuropeptide precursor sequences (*AjCTP1*, GenBank accession number: MF401985; *AjCTP2*, GenBank accession number: MF401986) were identified in our previous study (Chen et al., 2019). Based on the published sequences, transcript-specific primers for *AjCTP1* and *AjCTP2* (Key resources table) were designed using Primer3Plus (<https://www.primer3plus.com/>) and sequence accuracy was confirmed by SMARTer RACE 5'/3' Kit (TaKaRa, Kusatsu, Japan) according to the manufacturer's instruction. A sequence translation tool (<https://web.expasy.org/translate/>) was used to obtain the corresponding amino acid sequences encoded by the cloned cDNAs. To identify the conserved features of CT-type neuropeptides in Bilateria, sequences were aligned using Clustal-X with default settings and then predicted using the online tool Multiple Sequence Comparison Display Tool (<http://www.bio-software.net/sms/index.html>). A neighbor joining tree was constructed to investigate relationships of *AjCTP1*, *AjCTP2* and CT-type precursors from other Bilateria. Amino acid sequences were aligned using MEGA7 (v.7.170509) with default settings and bootstrap values were calculated from 1000 replicate analysis and then modified using iTOL (<https://itol.embl.de/login.cgi>). The accession numbers or citations for all sequences used for analysis are listed in **Supplementary table 1**.

4.3 Cloning, sequence analysis and phylogenetic tree construction for candidate *AjCT1/2* receptors in *A. japonicus*

Five potential receptors for *AjCT1* and *AjCT2* in *A. japonicus* (GenBank accession number: PIK49998.1, PIK49999.1, PIK49604.1, PIK34007.1 and PIK62458.1) were identified based on genome-wide analysis (Huang et al., 2021). Sequence comparison revealed that both PIK49998.1 and PIK62458.1 were partial sequences of PIK34007.1. Thus, three candidate receptors belonging to Class B (Secretin-like) GPCRs named as *AjCTR* (Calcitonin receptor, GenBank accession number: PIK49604.1), *AjPDFR1* (Pigment-dispersing factor receptor 1, GenBank accession number: PIK34007.1), and *AjPDFR2* (Pigment-dispersing factor receptor 2, GenBank accession number: PIK49999.1) were predicted and then analyzed. To confirm the accuracy of these sequences, cDNAs were cloned using the SMARTer RACE 5'/3' Kit (Takara, Cat No. 634858) and then sequenced (Beijing Genomics Institute, China). Receptor sequence analysis and phylogenetic analysis were performed as described in 4.2. In addition, the online tool NetPhos-3.1 (<https://services.healthtech.dtu.dk/service.php?NetPhos-3.1>) was used to predict potential phosphorylation sites. The conserved domains of protein sequences were predicted by the Batch SMART plugin of TBtools (Toolbox for biologists, v.1.098652) software. For analysis of gene synteny, neighboring genes of *AjCTR*, *AjPDFR1* and *AjPDFR2* were found based on an *A. japonicus* chromosome-level genome assembly (Wang et al., 2022) and the corresponding genes in the starfish *Asterias rubens* and the sea urchin *Lytechinus variegatus* were also identified by performing BLASTp in the NCBI database. Specific primers for cloning of cDNAs encoding the candidate receptors are shown in the Key resources table and the details of the sequences used for constructing the phylogenetic tree are listed in **Supplementary table 2**.

4.4 Testing AjCT1 and AjCT2 as ligands for candidate receptors and exploration of downstream signaling pathways

AjCT1 and AjCT2 (both with a disulfide bridge between the two cysteine residues and with C-terminal amidation) were custom synthesized by GL Biochem Ltd. (Shanghai, China) with >95% purity. To test candidate receptors for CT-type neuropeptides and investigate their downstream activation mechanisms the following assays were used to measure cAMP signaling or Ca²⁺ signaling: i). direct measurement of cAMP levels using ELISA ii). indirect measurement of cAMP signaling using a cAMP response element-Luciferase (CRE-Luc) assay that measures cAMP-induced gene expression, iii). serum response element (SRE)-Luc assays that measure Ca²⁺-induced gene expression, iv). receptor translocation assays and v). Western blotting.

AjCTR, *AjPDR1* and *AjPDR2* incorporating enhanced green fluorescent protein (EGFP) were constructed respectively according to the method described by [Li et al. \(2022\)](#). Briefly, the complete ORFs of candidate receptors were cloned and sequenced. Then, the PCR products were digested with EcoRI and BamHI (TaKaRa, Kusatsu, Japan) and then incorporated into the pEGFP-N1 vector (YouBio, Changsha, China) that was subject to the same double-digestion. Finally, the constructed plasmid with correct sequences was transfected into HEK293T cells seeded on glass coverslips in 12-well plates. After incubation with AjCT1 (10⁻⁶ M), AjCT2 (10⁻⁶ M) or serum-free DMEM (Biosharp, Hefei, China) for 15 min, cells were washed with PBS (Biosharp, Hefei, China) and fixed with 4% PFA for 10 min. Finally, cell nuclei were stained with DAPI (G-CLONE, Beijing, China) for 5 min and an Olympus bright field light microscope was used for image capture. The pcDNA 3.1(+) vectors (YouBio, Changsha, China) comprising the complete ORF of the three candidate receptors were also constructed as described by [Li et al. \(2022\)](#). Then, they were co-transfected with the cAMP reporter pCRE-Luc (YouBio, Changsha, China) or Ca²⁺ signaling reporter pSRE-Luc plasmid including a PKC reporter gene (YouBio, Changsha, China) into HEK293T cells. Incubation was carried out with AjCT1 (10⁻⁶ M), AjCT2 (10⁻⁶ M) or serum-free DMEM (CO, negative control) for 4 h at 37 °C. Finally, luciferase activity was tested using the firefly luciferase assay kit (MK, Shanghai, China). When required, transfected cells described above were co-incubated with H89 (PKA inhibitor) (Absin, Shanghai, China), FR900359 (Gαq protein inhibitor) (Cayman Chemical, Ann Arbor, Michigan) or Gö 6983 (PKC inhibitor) (Cayman Chemical, Ann Arbor, Michigan) before the experiment to explore the potential signaling pathways. To confirm the concentration of intracellular cAMP after stimulation, HEK293T cells expressing pcDNA 3.1(+)/*AjCTR*, pcDNA 3.1(+)/*AjPDR2* or pcDNA 3.1(+)/*AjPDR1* were treated with AjCT1 or AjCT2 (10⁻⁹ – 10⁻⁵ M), respectively, for 15 min after 2 h starvation in serum-free medium. Subsequently, cAMP was measured using an ELISA-based cAMP assay kit (R&D systems, USA&Canada) following the manufacturer's instruction. HEK293T cells transfected with empty pcDNA3.1(+) were used as a negative control. The specific primers used for constructing plasmids are listed in **Supplemental Table 1**.

Western blotting was performed as described previously ([Li et al., 2022](#)). Briefly, HEK293T cells transfected with the constructed pcDNA 3.1(+) plasmid were starved in serum-free medium for 2 h, and then incubated with AjCT1 or AjCT2 (10⁻⁶ M). PVDF membranes containing proteins were blocked with 5% BSA (G-CLONE, Beijing, China) and incubated overnight with antibodies to phosphorylated ERK1/2 kinases (CST, Boston, China). After that, goat anti-rabbit horseradish peroxidase-conjugated secondary antibody (Absin, Shanghai, China) was applied and incubated, followed by the addition of Omni-ECL™ Femto Light Chemiluminescence reagents (EpiZyme, Shanghai, China). Finally, RVL-100-G (ECHO, America) was applied for visualization of the PVDF membrane. When required, transfected cells were co-incubated with PKA or PKC inhibitor before the experiment to identify signaling pathways that activate downstream ERK1/2 cascade.

4.5 Total RNA extraction and Quantitative

Real-Time PCR (qRT-PCR) detection

qRT-PCR was employed to detect *AjCTR*, *AjPDFR1*, *AjPDFR2*, *AjCTP1* and *AjCTP2* transcript expression in five tissues from *A. japonicus*, including the intestine, longitudinal muscle, respiratory tree, gonad and circumoral nervous system, and the circumoral nervous system was selected as the control group. Because it was difficult to design specific primers for *AjCTP1* and *AjCTP2* (Zheng et al., 2022 [DOI](#)), total transcript expression levels for *AjCTP1/2* (*AjCTP1* + *AjCTP2*) were investigated. Total RNA was extracted from the collected tissues using Trizol RNA isolation reagent (Vazyme, Nanjing, China) and Hifair® L 1st Strand cDNA Synthesis SuperMix (YEASEN, Shanghai, China) was used to remove genomic DNA and synthesize cDNA according to manufacturer's instructions. For qRT-PCR, amplification was performed in 20 µL volume reactions based on Hieff UNICON Universal Blue qPCR SYBR Green Master Mix (YEASEN, Shanghai, China) and expression levels were assessed by a Corbett Rotor-Gene Q (Qiagen, Germany). *β*-actin (GenBank accession number: PIK61412.1) and *β*-tubulin (GenBank accession number: PIK51093) were used as house-keeping genes for normalization (Zhao et al., 2014 [DOI](#)). The $2^{-\Delta\Delta CT}$ method was applied to analyze the relative expression levels. Specific sequences of primers used for qRT-PCR are listed in **Supplemental Table 1** [DOI](#).

4.6 Localization of CT-type neuropeptide expression

in *A. japonicus* using immunohistochemistry

Juvenile specimens of *A. japonicus* (~15 mm in length) were fixed for 24 hours at room temperature or for 48 hours at 4 °C in Bouin's fixative (75 mL saturated picric acid in seawater, 25 mL 37% formaldehyde, 5 mL acetic acid) days and then washed in water before being transferred to 50% ethanol followed by 70% ethanol. The fixed specimens were in wrapped in tissue paper soaked in 70% ethanol and then transported from China to the UK. Following dehydration through 90% and 100% ethanol and clearing in xylene, specimens were embedded in paraffin wax and then sectioned (9 or 10 µm) transversely or longitudinally using a Leica RM2145 microtome. Sections were mounted on chrome alum-gelatin coated glass slides and then after incubation in xylene to remove wax followed by 100% ethanol, slides were incubated in 1% hydrogen peroxide in phosphate buffered saline (PBS, pH 7.3) to quench endogenous peroxidases. After using an ethanol series to rehydrate sections, slides were blocked in 5% goat serum in PBS containing 0.1% Tween-20 (PBST). Then slides were incubated overnight at 4 °C with the ArCT antiserum diluted to 1:2000, 1:3000 or 1:4000 in PBST. Following washing in PBST, slides were incubated with Peroxidase-AffiniPure Goat Anti-Rabbit IgG (H + L) conjugated to Horseradish Peroxidase (RRID: AB_2313567; Jackson ImmunoResearch, West Grove, PA) diluted 1:1000 in 2% goat serum/PBST for 3 h. Bound antibodies were revealed using a solution of 0.05% diaminobenzidine (VWR Chemicals, UK), 0.05% nickel chloride (Sigma-Aldrich, Gillingham, UK), 0.015% hydrogen peroxide (VWR Chemicals, UK) diluted in PBS. When staining was observed, slides were washed in water and dehydrated through an ethanol series into xylene and mounted with coverslips over DPX mounting medium (Thermo Fisher Scientific). The specificity of immunostaining was assessed by pre-absorption of the ArCT antiserum (at 1:100 dilution) for 1 h at room temperature with the ArCT antigen peptide (25 µM) prior to testing on sections at 1:3000 dilution. Images of stained sections were taken using a QIClick CCD Colour Camera (Qimaging, UK) linked to a DMRAI light microscope (Leica Microsystems, UK) with Volocity v. 6.3.1 image analysis software (Perkin-Elmer, USA) running on an iMac computer (27 inch with OS Yosemite, v. 10.10). Montages of images were prepared and labelled using GIMP (version 3.0.2) and Adobe Photoshop (version 26.7) running on a MacBook Pro laptop (OS Sequoia 15.5).

4.7 Analysis of the in vitro activity of AjCT1 and AjCT2 on preparations from *A. japonicus*

Longitudinal muscles and intestines (20 mm in length) from *A. japonicus* were dissected and then suspended in a 50 mL glass organ bath containing 20 mL sterilized seawater. Both ends of the preparations were tied with cotton ligatures and then one end was tied to a metal hook and the other end was connected with a High Grade Isotonic Transducer (ADInstruments MLT0015, Oxford, UK). Before testing, the preparation was stabilized for about 20 min, and the resting tension was adjusted to 1.0 g. Informed by previous findings from [Cai et al. \(2018\)](#), we investigated if AjCT1 and AjCT2 caused relaxation of preparations. First, acetylcholine (ACh) was applied to induce the contraction of preparations and then when a stable state of contraction was achieved, 20 μ L AjCT1 or AjCT2 at a range of concentrations was added and relaxation responses were recorded. Output signal (cm) from PowerLab data acquisition hardware (ADInstruments PowerLab 4/26, Oxford, UK) was calculated using LabChart (v8.0.7) software.

4.8 Investigation of the in vivo pharmacological effects of AjCT1 and AjCT2 in *A. japonicus*

A total of 150 adult sea cucumbers were randomly assigned to five groups (30 individuals per group), including two high concentration groups (CT1H and CT2H, 5×10^{-1} mg/mL), two low concentration groups (CT1L and CT2L, 5×10^{-3} mg/mL) and the control group (CO, sterilized seawater), with each group consisting of three parallel experiments comprising ten animals. Before the experiments, sea cucumbers were starved for a day to normalize their physiological status and their wet weights were measured. AjCT1 and AjCT2 were dissolved in sterilized seawater and then injected through the peristome at noon every other day (1 μ L/g wet weight) and the CO group were injected in the same way but with sterilized seawater (1 μ L/g wet weight). Feces and residual bait were collected and weighed after drying every day. After 24 days, wet weights of sea cucumbers were measured again. Then, WGR and SGR were calculated to assess their feeding and growth conditions. Formulas are as follows: $WGR = 100 \times (W_t - W_0) / W_0$, $SGR = 100 \times (\ln W_t - \ln W_0) / D$. In the formula, W_0 represents the initial wet body weight (g), W_t represents the final wet body weight (g), D represents the feeding days. Subsequently, animals were dissected and intestines were collected for detecting the expression levels of the three receptor genes (*AjCTR*, *AjPDFR1*, *AjPDFR2*) and several growth factors, including multiple epidermal growth factor 6 (*AjMegf6*, GenBank accession number: MG018199.1), insulin-like growth factor (*AjIgf*, GenBank accession number: PIK50518.1) and growth and differentiation factor-8 (*AjGDF-8*, GenBank accession number: KP100064.1) ([Li et al., 2016](#); [Wang et al., 2018](#)). The specific primers for qRT-PCR analysis are listed in the Key resources table.

4.9 Loss-of-function experiments by RNA interference (RNAi)

To investigate how receptor expression is affected by knock-down of AjCTP expression, an RNAi experiment was carried out. It was difficult to design *AjCTP1*- and *AjCTP2*- specific interference oligonucleotides and therefore, we designed and customized generic interference oligonucleotides targeting both *AjCTP1* and *AjCTP2* (si*AjCTP1/2*-1 and si*AjCTP1/2*-2, containing the extra 2' ome modification) (Beijing Tsingke Biotech Co., Ltd. China). Meanwhile, a negative control siRNA (siNC) was also customized for the experiment (Key resources table), which does not target genes in *A. japonicus*. The si*AjCTP1/2*-1, si*AjCTP1/2*-2 or siNC dry powder was diluted to 20 μ M with DEPC water following the manufacturer's instruction, then 60 μ L sterilized seawater was mixed with 20 μ L si*AjCTP1/2*-1, si*AjCTP1/2*-2 or siNC, and 20 μ L Lipo6000 transfection reagent (Beyotime, Shanghai, China) as the transfection solution. Eighteen sea cucumbers were randomly divided into three groups and injected with three different transfection solutions respectively. The mixed solutions were injected through the peristome once a day (1 μ L/g wet weight) and intestines were dissected and frozen in liquid nitrogen 3 days after injection. The expression level of *AjCTP1/2* was

investigated by qRT-PCR and the interference rate was calculated using $(1 - 2^{-\Delta\Delta CT})$ to determine the inhibition efficiency of siRNAs. Finally, the relative expression levels of *AjCTR*, *AjPDR1* and *AjPDR2* were investigated using the selected *A. japonicus* intestines.

To assess the effects of CT-type signaling inhibition on feeding and growth in *A. japonicus*, a 24-day RNAi experiment was conducted following the methodology outlined in [Section 4.8](#). Sea cucumbers were randomly allocated into two groups: the negative control group received siNC injections, while the experimental group was injected with si*AjCTP1/2*-1 (effectively targeting *AjCTP1/2*). Throughout the experiment, the dry weight of remaining bait and excrement, along with the wet weight of individual sea cucumbers, were recorded. Subsequently, WGR and SGR, as indices of feeding efficiency and growth performance, were calculated. In addition, the expression levels of the three growth factors (*AjMegf6*, *AjIgf* and *AjGDF-8*) in the intestinal tissue were tested by qPCR.

To investigate the potential mediation of CT-type signaling effects on feeding and growth by *AjPDR2* in *A. japonicus*, an RNAi experiment targeting *AjPDR2* was performed following the same knockdown protocol established for *AjCTP*. Briefly, the effective siRNA targeting *AjPDR2* (si*AjPDR2*-1 or si*AjPDR2*-2) was selected by testing *AjPDR2* expression in the intestines of injected sea cucumbers using qRT-PCR and calculating the interference rate efficiency. Subsequently, the validated siRNA was then employed in a 24-day RNAi experiment. The dry weight of remaining bait and excrement, as well as the wet weight of the sea cucumbers were measured, enabling calculation of WGR and SGR. Furthermore, the expression levels of three key growth factors (*AjMegf6*, *AjIgf* and *AjGDF-8*) in the intestinal tissue were analyzed via qPCR.

4.10 Statistical analysis

Statistical analyses were carried out using GraphPad Prism 8.0. Significant differences between groups were determined by one-way analysis of variance (ANOVA) followed by a Tukey's multiple comparisons test. Statistical significance thresholds were defined as follows: $p < 0.05$ (*) for significant differences and $p < 0.01$ (**) for highly significant differences, with annotations directly marked on the figures. All data were presented as mean $\pm$ SEM from at least three independent biological replicates. Non-linear regression analysis was performed using a four-parameter logistic equation with automatic outlier exclusion to fit the dose-response curve. The half maximal effective concentration (EC_{50}) was derived from the fitted curves through built-in algorithms in GraphPad Prism 8.0.

Supplementary figures



Figure 1-figure supplement 1.

Calcitonin-type (CT-type) neuropeptide precursors in *A. japonicus*.

(A) *AjCTP1* comprises two putative CT-type neuropeptides, AjCT1 and AjCT2, whereas *AjCTP1* generated by alternative splicing comprises only AjCT2. The predicted signal peptide is shown in blue, the predicted cleavage sites (KR or KK) are shown in green, CT-type neuropeptides are shown in red and C-terminal glycines that are potential substrates for amidation are shown in orange. Predicted disulfide bridges in AjCT1 and AjCT2 are formed between the two underlined cysteines. (B) Nucleotide sequences encoding CT-type neuropeptide precursors and the deduced amino acid sequences in *A. japonicus*. 3'UTR and 5'UTR are shown in lowercase and the ORF is shown in uppercase. Blue uppercase labels the predicted signal peptide, the green uppercase shows the predicted cleavage site, AjCT1 and AjCT2 are shown in red and the putative substrate for amidation is shown in orange.

```

A_japCT1  - - - - SCSNKFAG - CAHMKV ANAV L KQNSRGQQQFKF - - GSAG - - Pa
A_japCT2  - - RVGGCGD - FSG - CASLKAGRDLVRAML R - - PSKF - - GSGG - - Pa
H_scaCT1  - - - - SCSDRFSG - CAHLKVAKALLDQ - ARRENSRF - - GISG - - Pa
H_scaCT2  - - - MGGCGD - FSG - CASLKAGRDLVRAML RQ - - PSKF - - GSGG - - Pa
H_glaCT1  - - - - SCSDRFAG - CTHLKLAKVL I DQ - SRRDGNIRF - - GSSG - - Pa
H_glaCT2  - - - MGGCGDKFSG - CASLKAGRDLVRAML RQ - - PSKF - - GSGG - - Pa
S_purCT   - - - SKGCGS - FSG - CMQMEVAKNRVAAL LRNSNAHL F - - GLNG - - Pa
A_rubCT   NGESRGCSG - FGG - CGVLTIGHNAAMRMLAES - NSPF - - GASG - - Pa
A_plaCT   - - SSSGCAEYFGG - CAQLKLQDQDALSRMLADS - NSRF - - GSGG - - Pa
P_pecCT   - - SGTGCTQ - FSG - CAQLKVGQDALSRVLADS - NSRF - - GSGG - - Pa
O_vicCT1  - - SGNGGCAG - FTG - CAQLAAGQNALRNFMHSNRASLFT - GASG - - Pa
O_vicCT2  - - NGNGGCAG - FTG - CAQLAAGQSALQAMIHSGRASLF - - GSGG - - Pa
O_araCT1  - - GTEKGC SG - FSG - CAQLAAGQSALQAMIHGNRASLF - - GSGG - - Pa
O_araCT2  - - GGGGCKS - FSG - CAQLVIGQNAVRNMMHSNRASIFS - GPRG - - Pa
A_filCT1  - - GTEKGCAG - FSG - CAQLAAGQSALQAMIHGNRASLF - - GSGG - - Pa
A_filCT2  - - GGGCRG - FSA - CAQLAIGQDAFRNMMHNNRASIFT - SARG - - Pa
A_medCT   - - - G - C - DVFGG - CAQLKVGRELA I DTLQKGS GGMF - - GSSG - - Pa
An_japCT  - - - GDC - DSFGG - CAQLTASQQLAYRTRHRDLSNLF - - GTGG - - Pa
S_kowCT1  - - - GEGCKG - FAA - CGNLDA GRKHWKANMGGGRTAMESSASGT - - Pa
S_kowCT2  GRGTSACGG - FAT - CKQLEYGRKYATS - - KADLSHF - - GATS - - Pa
B_floCT1  - - GK IACKT - - AW - CMNRLSHNLS SLDNPTDTGVG - - - - A - - Pa
B_floCT2  - - - - DCST - - LT - CFNQKL AHELAMDNRQRTDTANPY - - - - S - - Pa
B_floCT3  - - - - KCES - - GT - CVQMHLADRLRLGLGHNMFTNT - - - GPES - - Pa
C_intCT   - - - - - CDG - VST - CWLHEL GNSVHATAGGKQ - NVGF - - - - G - - Pa
H_sapCGRP - - - - - ACDT - - AT - CVTHRLAGLLSRSGGVVKNNFVPT - NVGSKA Fa
H_sapCT   - - - - - CGN - LST - CMLGTYTQDFNKFHTFP - - QTA I - - GVGA - - Pa
L_gigCT   - - - - - SCSLRLGGMC L TENLNAAANQY - - - - - EYLS - SGLS - - Pa
D_melDH31 - - - - - TVDFGLARGYSG - - - TQEAKHRMGLA - - - - - AANF - - AGG - - Pa
C_telDH31 - - - - - RFDAGYGS - - - RYGV AQSVGSKLMALKQAADW - - - NG - - Pa
C_telCT   - - - - - TCQFNLGGHCA TESAA SVADHW - - - - - HYLN - SPLS - - Pa

```

Figure 1-figure supplement 2.

Comparison of the sequences CT-type neuropeptides from Bilateria: deuterostomes-ambulacraria (green), deuterostomes-chordates (purple) and protostomes (pink).

The conserved residues are shown with white lettering highlighted in gray and black. The accession numbers or citations are shown in the **Supplementary table 1**.

A

AjCTR

```

1 cagatccctttcacoggacctctgcagctgaatcaaaagtgaattcaggagtggtgat
1 M Q E N D
61 gtcaaaacggacactggcagctacgaagaagatcgaaaATCGAGGAGAACGATTCATATT
9 F S F G L L N D P C E E F E D F D
121 ATTTGAGCTTTGTTCCCTTTACACGACCGAGTTCCTGAAGAATTTGAAGCTTG
29 P L A A D Q I D E W R R R R D C C E
181 ACCGCTTCTGACGCTGACCAAAATAGACGAATGGAGGAGAGGCGAGATTCTGTGCG
49 R I E N E P P D E P H C L V M D T W
241 AAAGAATTGAAAGGAGAGCCCTCTGACGAGCGGCATTCGACGCTTTGATGGATACGT
69 D C W N M T P G D L V Q P C P M W V
301 GGGATTGTTGGAAGTGAAGTACACCGGAGATACCGTGGTCAACATGCCATGGTGGG
89 P Q V N P N K D A M K C M P D G W F
361 TTCACAGGTTAATCCGAACAAAGTGCATGAACATGCATGCTGATGGAGCGGT
109 K H P D T N T W N N N C G A Q S
421 TCAAAATCCCGACACTAATTATACCTGGACCAACTACTCTAANTGGCGGCATCACAGA
129 I H V A I F Y T G Y V V I S L C L
481 GCACATCATGTGGCATATTCTACACCGGATACAGTGTTCAGTTATATCACTCTGCT
149 A L F I F Y F Q S L G C P R V I H K
541 TAGCCCTCTCTCATATTCACATCTTTGAGAGCTTGGATGCCGAGGGTACCATCACCA
169 N L F I S F I L C I S N I T W H I T I
601 AAAATTGCTGTTCTCTCTCTCATCTGAGCGTCAATCAACATCACTGCTATCACCA
189 I A S A E K Q E T A C R I H I I A Q
661 TCATTCGACGAGGAGCAACAGGAGCGGCATGCGAATACTCATATCATTCGGC
209 F F V L C N Y F W M L E G L L H V
721 AGTTTTTGTCTTTCGCACTACTTTGGATGCTAGTGAGGATTATACCTACATCGG
229 I V V A V F S E N H N L L L Y I V G M
781 TGATATGTGGTTCGAGTCTCTCGGAAATCAATACTCTCTCTATATATGTGGAT
249 V I P L V P A V F L S L L L T Q D G G
841 GGGTGATTCTATGGTGGCAGCAATGTTTCTCTCACTGTGTATTAACACAGAGTGTG
269 R C W N N A N E I E M V V G G I I A V
901 GAAGTGTGGAGCAACGCTAATGAGATCGAATGGTGTGGTGGCACTATATTTCGAG
289 L L T N A A L L L N I V R V L V K L R
961 TATTATTGACAAACGCTGCTCTCTCTTAACATAGTACGCTTTTATGACGAAGCTTA

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309 A T P S Q G A R T Y V R A V R A T I I L
1021 GGGCACCACCAAGGTGCAAGAACATACGTAAAGCTGTTCGACGCTCATATTTC
329 L P L M G L H I I I V P V R F D N H V
1081 TGCTTCTCTGATGGGCTCCATATATCATCGTTCAGTGAGGCCAAGTATAACCATG
349 A E V I D C I V A F L I S F Q G L F V
1141 TAGCGGAAGTTATTACGATGTATCGTCGCTTCTTAATTTCAATTCAGGGCCTTTTCG
369 A C I F C F F N G E V K M Q I R R K W L
1201 TAGCCTGCATATTTGTTTCTCAATGGCGAGGTAAAGATGCAGATCAGGAGAAATGGC
389 N W Q F R R G D F R S R H I V
1261 TATCAACTGGCAGTTCCAGAAAGTGGTGACTTCGGTCACGACACACCACTACTACTG
409 E A V T P P L V N L N E K D L
1321 TAACCGAAGCTGTACCACTACCAACCTCTGAGTGTCAACTCTCATCGGAGAAAGATT
429 D N I N H P I K K N G I N G D N G K
1381 TGGATACTACATCAATCATCTATCAAGCAAGAAATGGGATTAACGGGCAACATGGGA
449 K K V G F T E A T I A N D M E R R K
1441 AGAGTAAAAATCGTGGATTACAGAGCGGACGATGCTAACGACATGGAAGTCGAA
469 P L M G K D A A E D P D K T V V *
1501 AACATTGTGACGGGAAAGATCGCGCGAAGATCCAGATAAAACACGGTATTTAAa
1561 atgggaggaagaagatgaatgaatgacttcttgagagccggcgtttcatctctctggga
1621 tgtaaaaggggacggatgaatccccatatatctacatctatgtgactctcaaacctt
1681 gcttcggagacactcttgacactagacatcggttactactatgtgactcaaatcttg
1741 ccaaatgacgcagaagaagaacatcatcagagacttttagaaatgatttccatc
1801 tgaacatgttgacatcaagtacagatctgtgtgtgactgtgactgtgactgtgacta
1861 accaagtttgtaaacagctctttagtactgtgtgtgtgactgactgtgtgtaaaat
1921 agttatcacgtctccacagctcttaacaaatgaacgttgtagaaatgtagaacagac
1981 ttacatttgaaatctataaaatctactgtgtcactgaatgactgtgtgttccatttg
2041 gtttaaggggtattctgttaactacggcagatcatccaaacgtagttgtggaaga
2101 ctacgataactcgagtagatgaagcgatgtactgtactgtgtgtgtgactgaactaga
2161 gatttgagaagaaaaaaacatctataatgtgtgtatcgtgaatgtgtagaagcaagctt
2221 caattacattataaactctcagatgactgactgactgactgaagaagagagtagaaga
2281 agtttagagatgaagattttagtactgtatatttagtgaagataaaggaagaa
2341 ctgagtgaaagtgtgttccaaaggtgactgagatttctgtgtactcaagtagaagaa
2401 tgcataaactctggaatgctcttggggcgaactgtgatgatgaggtgcgaactcagc
2461 gtgcctaaagtattgttgaatcgaatgttgaatattatgaagtagaactatcatat
2521 atatgttttgtatatatatatacagcgtttatatatatatatata

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B

AjPDFR1

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1 agagagagaggtggagagagagagagagattcttatcagccaatcggtgtgaatttcaat
61 cgttttaaaagtgcgcgtgtcttccgattctctatagaagtcagttgtgtgatggga
121 acataccaggtgatgtgtctatgagcaacatgacaaagatctgtaaaagcactagctog
181 gtggtctcgtgaatgtttacagcgatgatatactctatataaactcatagatgat
241 agtcattgggtgactgcattacaagtaactgaacacaaatttgcgtcagttggttgg
301 tatttttcatattctttcaggagcattctcttgaagttgtaaaagcgaacgtgttggg
361 ttgtgctttgcagtgagcttttaagttgtctcttactagatgatcatattctctacc
1 H V G W V P C P D N
421 gtaactcaacaatatataaactggcgcaacATGAGTGTGGGAGCTACCTCGCTCGACA
12 S R I I P Y D V T G M T L F C P E T
481 ATTCACGAGGAGTATTATTCTTATGAGCTCAGAGTATGACTTTGTTCTGTCGCGAGA
32 D T L C W P D P P N T T V H I D C
541 CCTATGACTCTACTCTCTGCTGGCGAGTACACCAACGACGACGATGATATAGATT
52 P I F P W N A G I D I D A Q A H R R C G
601 GCCGATTTTTCATGGAACGCTGGAATGACATTGATGCCAGGACATCGTGGTGTG
72 A D G F W E Q H V E D C Q I G P P G
661 GGGCTATGATTCTGGGACAGCATGCTACGAAGATTGCTGCGAGATGAGAACCCCTG
92 T V P P G E D D L K F Y K I N G
781 GAACGTGCTCTCTCGATATGACAGGAGTATCTAAGTTTATATGAAAGTAAATAGT
112 A K I L E V I G I T H L M A V L V A L
781 GTGCGAAGATCTCGAGGTGATGGGATTAATGCTCTTATGGCGGTGCTGGTACAT
132 Y I F H F R L R N H R R I R I Q H L
841 TGTACATATCCATTCTTTCAGAGTTCGCGAATCATGCTATCGAATACATCAACATC
152 F L A F L I R L L D V I F M I N R F K
901 TGTTCTAGCTTCTCTTATCCGACTACTCTCGATGTTATATCATGATTACAGGTCA
172 K K A D L E P V G I N K F P P L V K
961 AAAAGAAAGCTCAGATTCTTTGGAACAGTTGGCATCAACAAATTCACCATGTGTGA
192 V M E L C R E Y A R L C F T W M F V E
1021 AAGTAATGGAACCTTCCGAGAAATAGCCGATTTGTGACGTTACGTGGATGTTTGTGG
212 G I L N L L A V F R K P N F L W
1081 AAGGCATATACCTGACAGTCTTTTATCAACCGCGCTCTCAGAAACCAAGTCTCTCT
232 Y Y L I G W A S P I F V L A F C I A M
1141 GGTATATCTCATTTGGCTGGGCTCACGATACCATTTGTGCTAGCATTTTGCATTGCGA
252 Q K T S N G W H I T E S N Y V F
1201 TGCAAAAGACATCTTCAATGGCTATTGGCATATATACACTGAATCAAAATATTATGATAT

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272 F I E G P R N I I V V M H V F L L L N I
1261 TTTTCATTGAGGACCAAGAACATTATGCTGATGATGATGTTCTCTCTCGTCAACA
292 I R V L I K L R E Q T E K Q V R
1321 TTATCAGAGTCTTGATACAAACTGAGGAAAGCCAAACGAGCAGACGAACCAATCA
312 K A K A C I V L L P L L G T A N L V W
1381 GGAAGGCACAGGCGCTGTATGATTAATGCTGCTGCTGGTACTGCTAATCTGTTT
332 I P K V P E P G K R F Y F A S Y Y
1441 GGATTCCGAAAGTCCAGAGCTCGCAATGCTCTCGGTTTACTTGTGATGACTACT
352 V V L F L D A Y Q G F F L L L Y C F L
1501 AGTGGTCTTATCTCGGATGCTTACCAAGTCTCTCTCGGATTTATGATGCTCT
372 N I D V R V I K R K W Q G W M N A R N
1561 TAAACATTGATGTCGAGTTCAGATAAAAGAAATGGCAAGTGGATGAGCCCGAGAA
392 P Y R A H G V V T T D V R M L
1621 ATCCTTATCGAGCTCATGGATCTGTGTTACTAGCGAACCGATGTACGAGTATCATCT
412 D N N D E W G G K *
1681 TGTATACAAATGACAGTGTGGGTGGGAAATGAagtcaaatatgatggcgctattgaa
1741 ctaattttatcttggcaattggtatccaggtgtgtgttaactagaagcgtgtatgactctta
1801 acaaatgtgtcggcaactcgtgtgcaactgtgtgtgacagcaaacactaactcaac
1861 ccaatgcatcccaatagagtttcaatacctatatatccatgagtgagcagatttgt
1921 acgaatgtgataagtgcaatggtgtgcaaatgaactttgactactactcgttttaaga
1981 cgtcgaattgtactgtgtgtgttaacaagtaactcattgtctatcagcagattgtgaat
2041 ttctcaaaagtatttggataaaatgatcacatgtatgtagcagaaaaagataaaaaa
2101 aagtactctggtgtgatac

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Figure 1-figure supplement 3.

Amino acid sequences and phosphorylation sites of AjCTR, AjPDFR1 and AjPDFR2 in *A. japonicus*.

(A), (B) and (C) show the cDNA sequences encoding the receptors and their deduced amino acid sequences. ORFs are shown in uppercase; 3'UTR and 5'UTR are shown in lowercase. The stop codon is shown by the asterisk (*). Predicted potential phosphorylation sites are shown with squares.

AjPDFR2

0 gaggagcgtctctctttcgcagggtttcttaaaacggtttgtgtgtgtgtgtgcagctc
 61 aaagttgtgttgagcgcaagctttctgattgcacaaacctatgtatgtacggtccgaac
 121 ttgtgtgtgtctctcaaacatggttgattgtttgtgtgtgtttaccgtgaacctttagag
 181 caatcgagatctcttcagggggaaagcttaaacggagagtccagagagcgctactacag
 241 gggagagagatatacttaacttaacgtgcagatcaacaaacggtgtgtatgtgatctcat
 301 tagcagagatgcccctttttgtcaatgagagcagctgaagtagtaggaagcaacgta
 361 tatgattatcaccaagcttgagtcgttcgaagagctccatgtacgtaacgtatcgaaag
 421 acgtatacgaagctgtatcatctgggatagttttgtctctttctgtcatacagtaatac
 481 ggaagagagagcgctcatctttagtcgagccgtgtgtatgtatgaagagtgactgactct
 541 tcccaacgaagactataggatttaactgttcgttggaagacagcgcttcaatgaagcta
 601 ggaagctctgtggaagctcgatgtcgtcacactcgtgtttcaacagaggttttactaca
 661 gagttttgagtaacgctcgggtctactataaagttctcttcagatcgtcgaagTGGGACG
 4 N E T S V L C L E I K T D T D Y D V C
 721 AATGAACAGAGTGATGTGTGTGCGAAATATGTAAAGATTGATTATGATGTATGTGTGT
 781 W P E T A N T A N G I A C C T G C C P F D P F T T
 841 TGGCGAACCAACGCAAGAACCGGAAGTATCGAATCAGCGCTGTCTGCACAGCCCTTCA
 901 C A G C C A G C G A C T T T G A G A G G T T G T G C A G C C T G G A A G A G C C T T C
 961 G A C C C A C G C A C T T T G A G A G G T T G T G C A G C C T G G A A G A G C C T T C
 1021 T G T A T G A T C T G C G T G A G T T A C T T C G C C C T G T A C T T T G C T T C G A G C T T T G
 1081 A A G T G C A A C C G G A G A A G A T C C A C G T G C A T T A T T C A C A G G C T G T T G A T T G C G C T A C C
 1141 A T A G A T G T C G T G T T G A T A T A C A A C A T C A A A T T A A A C C G G A A A T G A T T C G A
 1201 G C A C T G A C A N A T T C G G T T A T T G T G A A G C C T G C A C A T T G A T G A T C A C A G A G
 1261 L C A F A W M F V E L L L N T L I
 1321 A V F G R N F Y Y I A A W V L P I
 1381 C G A G A T T T G G A A G C A A A T T A T C G T T A C T A C A T T G C A T G C T T C C C G A C

224 P F V L A S I A N F A T Y S R C W F L
1381 CGGTTGTGGTGTAGCTGGTGATGCACGAATTTCGCAACATATTCAGAAGTCGTGGCCCTCT

244 H D P Y Y L A L I E I P R D I I F I
1441 CATACTGACTGCCCTCATCTAGCATTAGCAATAATAGAAAGATCTAGAGATATCATATTAAT

264 V V F L H I Y V L V K L N N
1501 CTAAATAGCGTGTCTCTCTTCACATCATTTATGTCTTGGTGACCAAATAAAAAATAC

284 N G E T Q V I R K A M V K A L F V L L P
1561 AAATCTGTGGGAACAACAGCTACATGAAGGAAGCAGTAGAAGCGCTGTGTCTCTGCGCT

304 L G I H L L W G L L P H Q A P A
1621 CTTCTGGGATCAGCCATCTCTTGGTGCTCTTCCATCGCAGCTACGACCACCA

324 V I V I Y H V G I P F Y S F Q G F A
1681 ACAGCATGTGTATCTACCAAGCGGCATATCTGTCTCTATTCGTCTCCAAGATTCGCT

344 V A L L C F L N K E V R I L K R M
1741 GTGCTCTCTCTCTCTGTCTTAAACAGAGGAGTCTGCTGATTAATTAAGAGGAGATGG

364 T V V V N F M D P M R R M I T G T G
1801 AGAATCTGGGATGATTCATGGACACACCGATGAAGAAGATCTCAATGATCGATGCACA

384 D V R I N L M G A A A A A A A *
1861 TCAGATGTGAAGGATTAACCTGTCATMGTAAGGCGAAATTAATGAATGA

Figure 1-figure supplement 3. (continued)

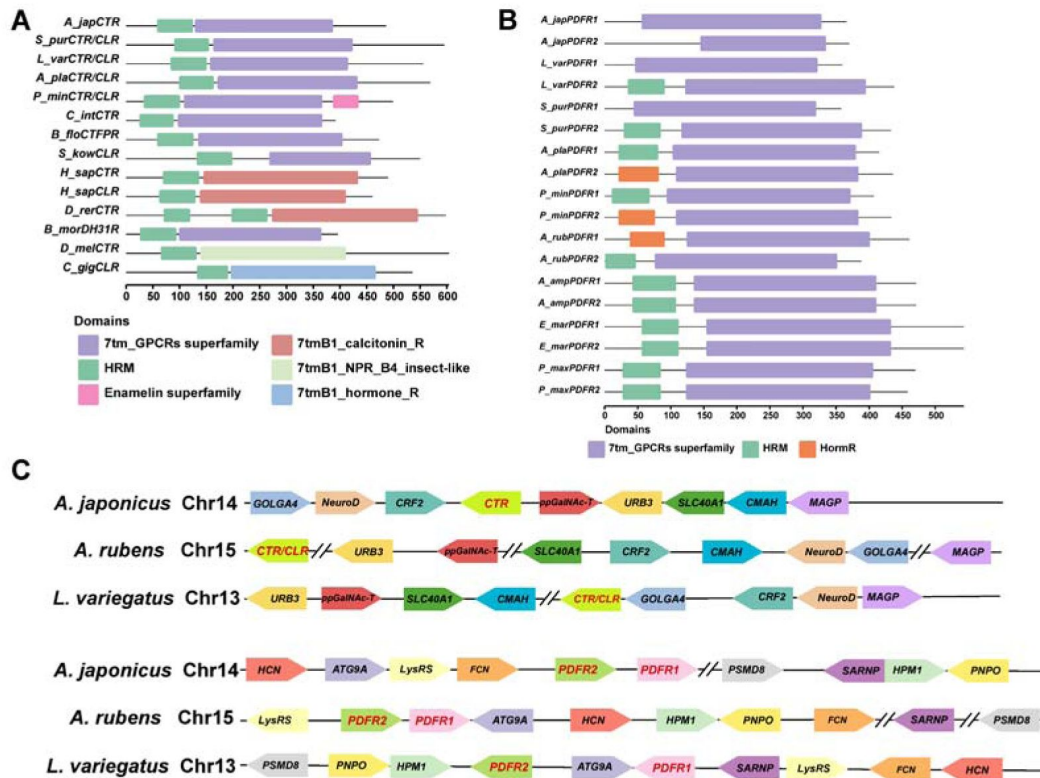


Figure 1-figure supplement 4.

Characterization of AjCT1 and AjCT2 receptors (AjCTR, AjPDFR1 and AjPDFR2).

(A) Conserved domains of CTR/CLR-type protein sequences. Black line represents the length of proteins and different domains are represented by rectangles. (B) Conserved domains of PDFRs in Bilateria. Black line represents the length of proteins and different domains are represented by rectangles. (C) Syntenic analysis of AjCT1 and AjCT2 receptors (AjCTR, AjPDFR1 and AjPDFR2). The chromosomal locations of AjCT1 and AjCT2 receptor genes from *A. rubens*, *L. variegatus* and *A. japonicus* are shown, with different colors used to mark neighboring genes and with their transcriptional direction shown by arrows. *GOLGA4*, golgin subfamily A member 4; *NeuroD*, neurogenic differentiation factor; *CRF2*, corticotropin-releasing factor receptor 2; *CTR*, calcitonin type receptor; *CLR*, calcitonin gene-related peptide type 1 receptor; *ppGalNAc-T*, polypeptide N-acetylgalactosaminyltransferase; *URB3*, E3 ubiquitin-protein ligase ubr3; *SLC40A1*, solute carrier family 40 member 1; *CMAH*, cytidine monophosphate-N-acetylneuraminic acid hydroxylase; *MAGP*, microfibril-associated glycoprotein; *HCN*, hyperpolarization-gated and cyclic nucleotide regulated channel; *ATG9A*, autophagy-related protein 9A-like; *LysRS*, lysine--tRNA ligase; *FCN*, ficolin; *PDFR1*, pigment-dispersing factor receptor 1; *PDFR2*, pigment-dispersing factor receptor 2; *PSMD8*, 26S proteasome regulatory subunit 8; *SARNP*, SAP domain-containing ribonucleoprotein; *HPM1*, histidine protein methyltransferase 1; *PNPO*, pyridoxine-5'-phosphate oxidase.

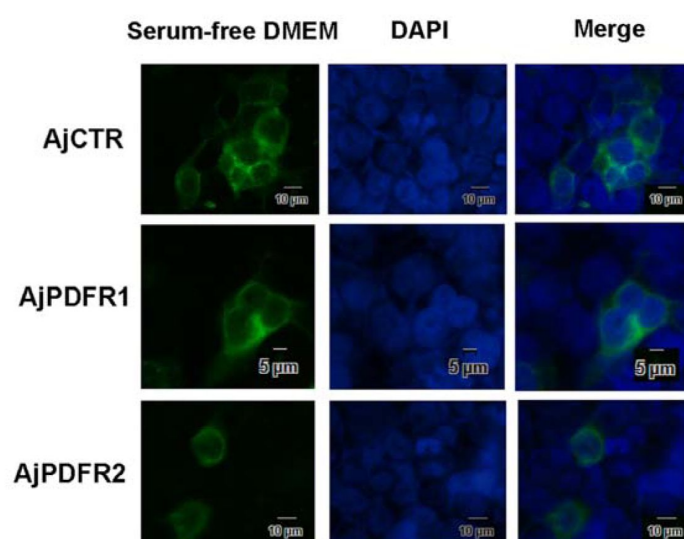
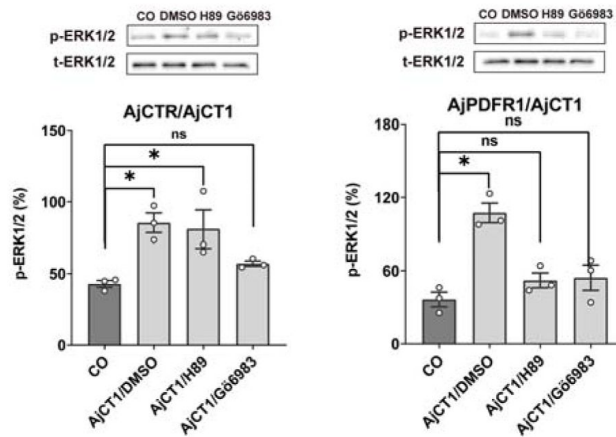


Figure 3-figure supplement 1.

Negative control for analysis of AjCTR, AjPDR1 and AjPDR2 internalization.

The images show the localization of AjCTR, AjPDR1 and AjPDR2 in HEK293T cells without AjCT1 or AjCT2 stimulation. Green fluorescence represents the localization of pEGFP-N1/receptors in cell membrane. Cell nucleus probe (DAPI) was applied for cell nucleus staining. All pictures are representative of three independent experiments.

A



B

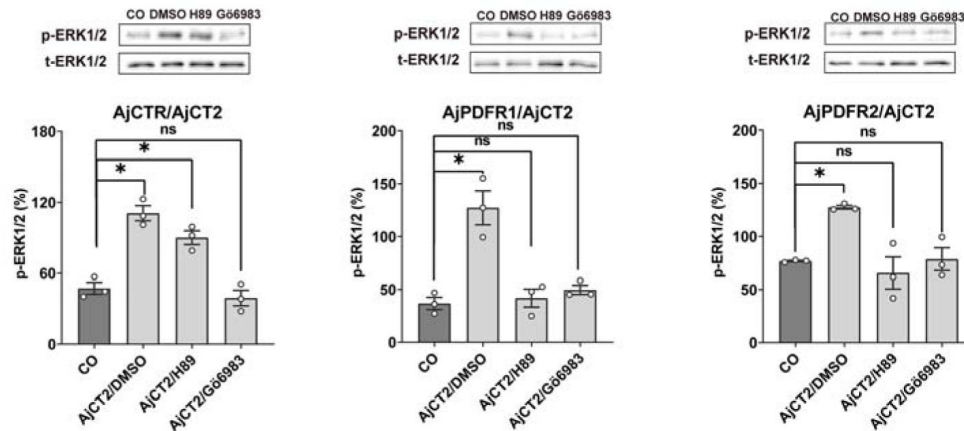


Figure 4-figure supplement 1.

Effects of PKA or PKC inhibitor on AjCT1 or AjCT2 stimulated ERK1/2 phosphorylation in AjCTR, AjPDR1 or AjPDR2 expressing HEK293T cells.

(A) ERK1/2 phosphorylation, activated by AjCT1, is blocked by PKC inhibitor Gö 6983 in AjCTR expressing HEK293T cells and is blocked by PKA inhibitor H89 and PKC inhibitor in AjPDR1 expressing HEK293T cells. (B) ERK1/2 phosphorylation, activated by AjCT2, is blocked by PKC inhibitor in AjCTR expressing HEK293T cells and is blocked by PKA inhibitor and PKC inhibitor in AjPDR1 and AjPDR2 expressing HEK293T cells. Serum-starved HEK293 cells were pre-treated with DMSO, PKA inhibitor, or PKC inhibitor before AjCT1 or AjCT2 stimulation. All images are representative of three independent experiments. Error bars represent SEM for three independent experiments. Statistically significant differences are indicated as follows: *: $p < 0.05$, ns: no significant.

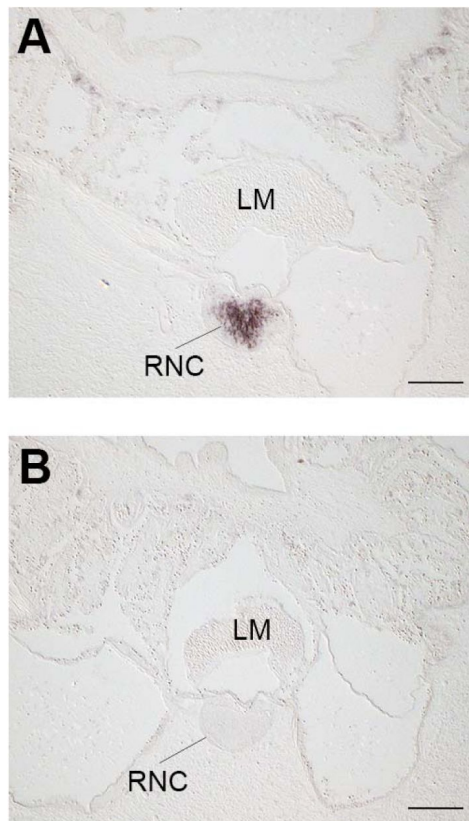


Figure 6-figure supplement 1.

Immunostaining in the radial nerve cord of *A. japonicus* is abolished by pre-absorption of ArCT antiserum with the ArCT antigen peptide.

(A) Transverse section of a juvenile specimen of *A. japonicus* showing immunostaining in the radial nerve cord revealed by an antiserum to the starfish calcitonin-type neuropeptide ArCT, as also shown in [Figure 6A](#). (B) A transverse section neighbouring the section shown in A was incubated with ArCT antiserum that had been pre-absorbed with the ArCT antigen peptide; note the absence of immunostaining in the radial nerve cord, demonstrating the specificity of the immunostaining in A. Abbreviations: LM, longitudinal muscle; RNC, radial nerve cord. Scale is 100 μm .

Figure 7-figure supplement 1.

Positive control on experimental tissues and pharmacological effects of 10^{-6} M AjCT2 on longitudinal muscle from *A. japonicus*.

(A) shows a recording of the relaxing effect of AjCT2 (10^{-6} M) on a longitudinal muscle preparation. (B) and (C) respectively are recordings of the contracting effects of ACh on longitudinal muscle and intestine preparations, showing that the effect of ACh persists for at least 15 minutes. All recordings shown are representative of three independent experiments.

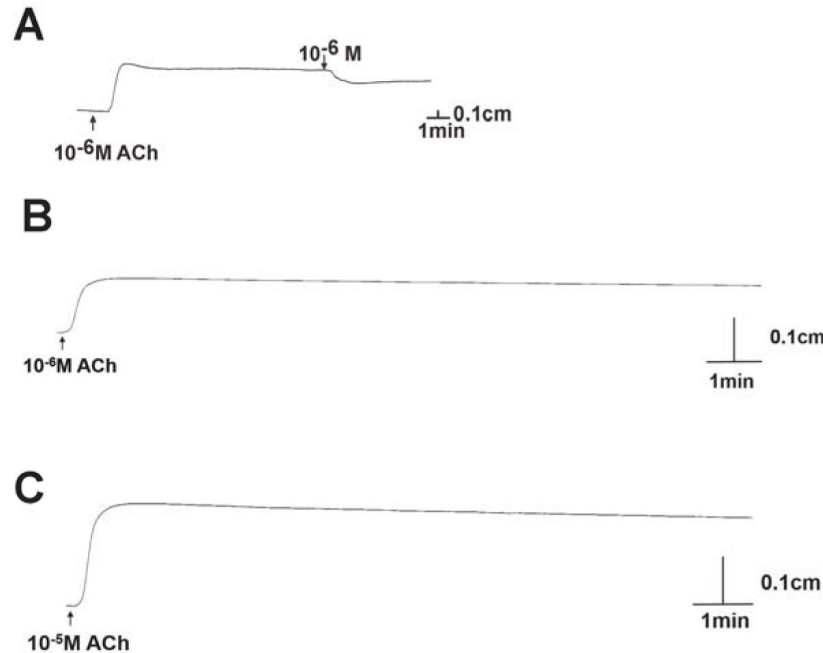
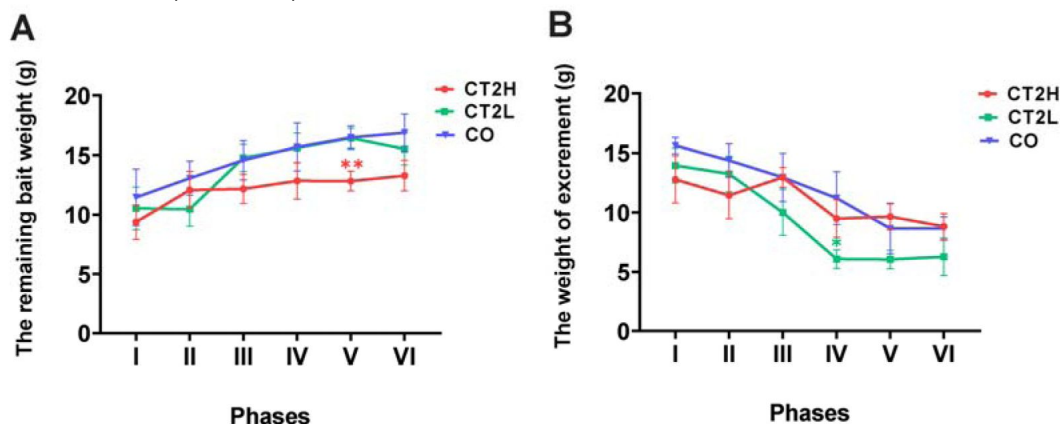


Figure 8A-figure supplement 1.

Measurement of the mass of remaining bait and excrement for AjCT2-treated and control animals.

(A) Remaining bait mass in different phases. (B) The mass of excrement in different phases. The 24-day experiment was divided into six phases (I-VI), with each phase comprising four days for statistical analysis. CO: control group (sterilized seawater); CT2L: AjCT2 low concentration group (5×10^{-3} mg/mL); CT2H: AjCT2 high concentration group (5×10^{-1} mg/mL). Values are means $\pm$ SEM ($n = 3$, representing three parallel experiments of each group). Asterisks represent the significant difference as follows: *: $p < 0.05$, **: $p < 0.01$.



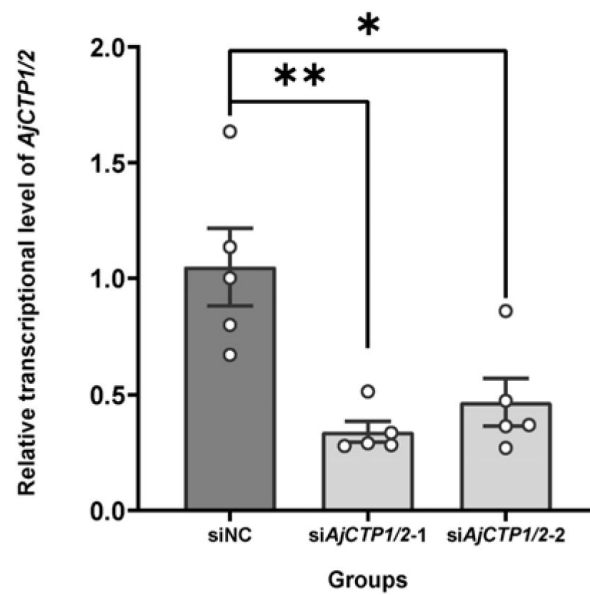


Figure 9-figure supplement 1.

The relative expression level of *AjCTP1/2* in siNC, si*AjCTP1/2-1* and si*AjCTP1/2-2*.

siNC: negative control group; si*AjCTP1/2-1* and si*AjCTP1/2-2*: experimental groups targeting *AjCTP1/2*. Values are means ± SEM (n = 5). Asterisks represent the significant differences as follows: *: $p < 0.05$, **: $p < 0.01$.

Supplementary tables

Abbreviation	Species Names	Accession Numbers And/or Citations
<i>H_sapCTP</i>	<i>Homo sapiens</i>	GI:179819
<i>H_sapCGRPP</i>	<i>Homo sapiens</i>	NP_001029125.1
<i>T_rubCTP</i>	<i>Takifugu rubripes</i>	GenBank: CAC81278.1
<i>T_rubCGRPP</i>	<i>Takifugu rubripes</i>	GenBank: CAC81277.1
<i>C_intCTP</i>	<i>Ciona intestinalis</i>	GenBank: BAI63095.1
<i>B_floCTP1</i>	<i>Branchiostome floridae</i>	GI:260826569
<i>B_floCTP2</i>	<i>Branchiostome floridae</i>	GI:260826567
<i>B_floCTP3</i>	<i>Branchiostome floridae</i>	GI:260826573
<i>S_kowCTP</i>	<i>Saccoglossus kowalevskii</i>	GI:187217193
<i>A_japCTP1</i>	<i>Apostichopus japonicus</i>	GenBank: AWU78769.1
<i>A_japCTP2</i>	<i>Apostichopus japonicus</i>	GenBank: AWU78770.1
<i>A_rubCTP</i>	<i>Asterias rubens</i>	GenBank: KT601715.1
<i>P_pecCTP</i>	<i>Patiria pectinifera</i>	GenBank: AWW63418.1
<i>A_plaCTP</i>	<i>Acanthaster planci</i>	Smith et al. (2017)
<i>S_purCTP</i>	<i>Strongylocentrotus purpuratus</i>	GI:115767208
<i>O_vicCTP1</i>	<i>Ophionotus. victoriae</i>	GenBank: ASK86238.1
<i>O_vicCTP2</i>	<i>Ophionotus. victoriae</i>	GenBank: ASK86238.1
<i>O_araCTP1</i>	<i>Ophiopsila aranea</i>	Zandawala et al. (2017)
<i>O_araCTP2</i>	<i>Ophiopsila aranea</i>	
<i>A_filCTP1</i>	<i>Amphiura filiformis</i>	
<i>A_filCTP2</i>	<i>Amphiura filiformis</i>	
<i>H_scaCTP1</i>	<i>Holothuria scabra</i>	Suwansa-Ard et al. (2018)
<i>H_scaCTP2</i>	<i>Holothuria scabra</i>	
<i>H_glaCTP1</i>	<i>Holothuria glaberrima</i>	
<i>H_glaCTP2</i>	<i>Holothuria glaberrima</i>	
<i>A_medCTP</i>	<i>Antedon mediterranea</i>	Aleotti et al., (2022)
<i>An_japCTP</i>	<i>Anneissia japonica</i>	
<i>L_gigCTP</i>	<i>Lottia gigantea</i>	GI:163526287
<i>D_melDH31P</i>	<i>Drosophila melanogaster</i>	NP_523514.1
<i>L_migDH31P</i>	<i>Locusta migratoria</i>	GenBank: AKN21237.1
<i>C_telCTP</i>	<i>Capitella teleta</i>	GI: 161220966
<i>C.telDH31P</i>	<i>Capitella teleta</i>	Rowe et al. (2014)

Supplementary table 1.

Accession numbers or citations of calcitonin-type family peptides or precursors in Bilateria

Classification	Abbreviation	Species Names	Accession Numbers And/or Citations
CTR/CLR	<i>B_floCTR</i>	<i>Branchiostoma floridae</i>	XP_035687900.1
	<i>S_purCTR</i>	<i>Strongylocentrotus purpuratus</i>	XP_003725171.1
	<i>A_japCTR</i>	<i>Apostichopus japonicus</i>	PIK49604.1
	<i>A_rubCTR</i>	<i>Asterias rubens</i>	XP_033639437.1
	<i>P_minCTR</i>	<i>Patiria miniata</i>	XP_038079052.1
	<i>A_plaCTR</i>	<i>Acanthaster planci</i>	XP_022079821.1
	<i>A_filCTR</i>	<i>Amphiura filiformis</i>	XP_072028982.1
	<i>L_varCTR</i>	<i>Lytechinus variegatus</i>	XP_041478637.1
	<i>H_sapCTR</i>	<i>Homo sapiens</i>	NP_001158209
	<i>H_sapCLR</i>	<i>Homo sapiens</i>	NP_005786
	<i>B_belCTR</i>	<i>Branchiostoma belcheri</i>	XP_019645944.1
	<i>P_spaCLR</i>	<i>Polyodon spathula</i>	XP_041102166.1
	<i>E_calCLR</i>	<i>Erpetoichthys calabaricus</i>	XP_028672925.1
	<i>S_fasCLR</i>	<i>Stegostoma fasciatum</i>	XP_048390160.1
	<i>X_couCLR</i>	<i>Xiphophorus couchianus</i>	XP_027880277.1
	<i>X_macCLR</i>	<i>Xiphophorus maculatus</i>	XP_005798205.1
	<i>An_japCTR</i>	<i>Anneissia japonica</i>	XP_033104142.1
	<i>B_terDH31R</i>	<i>Bombus terrestris</i>	XP_003398998.1
	<i>B_morDH31R</i>	<i>Bombyx mori</i>	NP_001127732
	<i>C_telCTR</i>	<i>Capitella teleta</i>	ELU10120.1
	<i>C_gigCTR</i>	<i>Crassostrea gigas</i>	AJ551182
	<i>C_gigCTR2</i>	<i>Crassostrea gigas</i>	MK290745
	<i>P_dumDH31R</i>	<i>Platynereis dumerilii</i>	KP293952
	<i>D_melDH31R</i>	<i>Drosophila melanogaster</i>	NP_725278.1
Deuterostomes PDFR	<i>A_japPDFR1</i>	<i>Apostichopus japonicus</i>	PIK34007.1
	<i>A_japPDFR2</i>	<i>Apostichopus japonicus</i>	PIK49999.1
	<i>A_plaPDFR1</i>	<i>Acanthaster planci</i>	XP_022080055.1
	<i>S_purPDFR1</i>	<i>Strongylocentrotus purpuratus</i>	XP_030845952.1
	<i>L_varPDFR1</i>	<i>Lytechinus variegatus</i>	XP_041477899.1
	<i>P_minPDFR1</i>	<i>Patiria miniata</i>	XP_038077382.1
	<i>A_rubPDFR1</i>	<i>Asterias rubens</i>	XP_033639833.1
	<i>S_purPDFR2</i>	<i>Strongylocentrotus purpuratus</i>	XP_030845954.1
	<i>L_varPDFR2</i>	<i>Lytechinus variegatus</i>	XP_041478256.1
	<i>A_plaPDFR2</i>	<i>Acanthaster planci</i>	XP_022080156.1
	<i>P_minPDFR2</i>	<i>Patiria miniata</i>	XP_038077934.1
	<i>A_rubPDFR2</i>	<i>Asterias rubens</i>	XP_033640032.1

Supplementary table 2.

Accession numbers or citations of CTR/CLR, PDF receptors, VP/OT-type receptors and VIP receptors in Bilateria

	<i>An_jap</i> PDFR	<i>Anneissia japonica</i>	XP_033104115.1
	<i>S_kow</i> PDFR	<i>Saccoglossus kowalevskii</i>	XP_006825104.1
	<i>A_fil</i> PDFR1	<i>Amphiura filiformis</i>	XP_072029557.1
	<i>A_fil</i> PDFR2	<i>Amphiura filiformis</i>	XP_072029560.1
VP/OTR	<i>A_jap</i> VP/OTR	<i>Apostichopus japonicus</i>	PIK60987
	<i>A_rub</i> VP/OTR	<i>Asterias rubens</i>	MK279533
	<i>S_pur</i> VP/OTR	<i>Strongylocentrotus purpuratus</i>	XM_779280.4
VIPR	<i>H_sap</i> VIPR1	<i>Homo sapiens</i>	KAI4029133.1
	<i>H_sap</i> VIPR2	<i>Homo sapiens</i>	KAI4016600.1
	<i>M_mus</i> VIPR1	<i>Mus musculus</i>	EDL09145.1
	<i>M_mus</i> VIPR2	<i>Mus musculus</i>	AAI38573.1
	<i>G_gal</i> VIPR1	<i>Gallus gallus</i>	NP_001090992.2
	<i>G_gal</i> VIPR1	<i>Gallus gallus</i>	NP_001014970.1
	<i>C_liv</i> VIPR1	<i>Columba livia</i>	PKK32249.1
	<i>C_liv</i> VIPR2	<i>Columba livia</i>	PKK32065.1
	<i>X_lae</i> VIPR1	<i>Xenopus laevis</i>	XP_018122563.1
	<i>X_lae</i> VIPR2	<i>Xenopus laevis</i>	XP_041423294.1
	<i>X_tro</i> VIPR1	<i>Xenopus tropicalis</i>	XP_031759879.1
	<i>X_tro</i> VIPR2	<i>Xenopus tropicalis</i>	NP_001120616.1
	<i>E_orb</i> VIPR1	<i>Emys orbicularis</i>	XP_065254637.1
	<i>E_orb</i> VIPR2	<i>Emys orbicularis</i>	XP_065255130.1
	<i>D_rer</i> VIPR1	<i>Danio rerio</i>	AAI62979.1
	<i>D_rer</i> VIPR2	<i>Danio rerio</i>	AAI62944.1
	<i>N_fur</i> VIPR1	<i>Nothobranchius furzeri</i>	KAF7213846.1
	<i>N_fur</i> VIPR2	<i>Nothobranchius furzeri</i>	SBP44365.1
Protostomes PDFR	<i>P_chi</i> PDFR	<i>Penaeus chinensis</i>	XP_047472721.1
	<i>O_edu</i> PDF1R	<i>Ostrea edulis</i>	XP_048754972.1
	<i>C_vir</i> PDFR	<i>Crassostrea virginica</i>	XP_022310010.1
	<i>C_gig</i> PDFR	<i>Crassostrea gigas</i>	XP_034317823.1
	<i>E_sin</i> PDFR	<i>Eriocheir sinensis</i>	XP_050724174.1
	<i>P_tri</i> PDFR	<i>Portunus trituberculatus</i>	XP_045102386.1
	<i>P_max</i> PDF1R	<i>Pecten maximus</i>	XP_033762294.1
	<i>D_vir</i> PDFR	<i>Drosophila virilis</i>	XP_032288784.1
	<i>A_aeg</i> PDFR	<i>Aedes aegypti</i>	XP_021711391.1
	<i>Ae_alb</i> PDFR	<i>Aedes albopictus</i>	XP_029719191.1
	<i>An_alb</i> PDFR	<i>Anopheles albimanus</i>	XP_035773075.1
	<i>H_ruf</i> PDFR	<i>Haliotis rufescens</i>	XP_048242666.1
	<i>A_amp</i> PDF1R	<i>Amphibalanus amphitrite</i>	XP_043235105.1
	<i>A_mac</i> PDF1R	<i>Anopheles maculipalpis</i>	XP_050069493.1
	<i>A_aqu</i> PDFR	<i>Anopheles aquasalis</i>	XP_050085801.1
	<i>A_dar</i> PDFR	<i>Anopheles darlingi</i>	XP_049535712.1
	<i>D_nov</i> PDFR	<i>Drosophila novamexicana</i>	XP_030566780.1
	<i>A_fun</i> PDFR	<i>Anopheles funestus</i>	XP_049279756.1

Supplementary table 2. (continued)

<i>G_aeg</i> PDFR	<i>Gigantopelta aegis</i>	XP_041366859.1
<i>P_mon</i> PDFR	<i>Penaeus monodon</i>	XP_037798712.1
<i>M_yes</i> PDFR	<i>Mizuhopecten yessoensis</i>	XP_021374228.1
<i>C_sec</i> PDFR	<i>Cryptotermes secundus</i>	XP_023726525.1
<i>A_ste</i> PDF1R	<i>Anopheles stephensi</i>	XP_035902082.1
<i>B_gla</i> PDFR	<i>Biomphalaria glabrata</i>	KAI8755269.1
<i>O_bim</i> PDFR	<i>Octopus bimaculoides</i>	XP_014783031.1
<i>A_mer</i> PDFR	<i>Anopheles merus</i>	XP_041761546.1
<i>A_ara</i> PDFR	<i>Anopheles arabiensis</i>	XP_040175433.1
<i>T_pre</i> PDFR	<i>Trichogramma pretiosum</i>	XP_014225639.1
<i>C_qui</i> PDF1R	<i>Culex quinquefasciatus</i>	XP_038122111.1
<i>E_mar</i> PDF1R	<i>Elysia marginata</i>	GFS10866.1
<i>E_row</i> PDFR	<i>Euperipatoides rowelli</i>	QKL20129.1
<i>T_cas</i> PDFR	<i>Tribolium castaneum</i>	XP_015835556.1
<i>B_kin</i> PDFR	<i>Belonocnema kinseyi</i>	XP_033223584.1
<i>A_col</i> PDFR	<i>Anopheles coluzzii</i>	XP_040220269.2
<i>D_mel</i> PDFR	<i>Drosophila melanogaster</i>	AAT84083.1
<i>S_pic</i> PDFR	<i>Schistocerca piceifrons</i>	XP_047108870.1
<i>P_jap</i> PDFR	<i>Penaeus japonicus</i>	XP_042879729.1
<i>P_max</i> PDF2R	<i>Pecten maximus</i>	XP_033762295.1
<i>P_pol</i> PDFR	<i>Pollicipes pollicipes</i>	XP_037077774.1
<i>A_amp</i> PDF2R	<i>Amphibalanus amphitrite</i>	XP_043235103.1
<i>E_mar</i> PDF2R	<i>Elysia marginata</i>	GFS10866.1
<i>P_oce</i> PDFR	<i>Plakobranthus ocellatus</i>	GFO33290.1
<i>O_edu</i> PDF2R	<i>Ostrea edulis</i>	XP_048755458.1
<i>C_qui</i> PDF2R	<i>Culex quinquefasciatus</i>	XP_038122111.1
<i>H_exe</i> PDFR	<i>Hypsibius exemplaris</i>	OQV13188.1
<i>P_vul</i> PDFR	<i>Patella vulgata</i>	XP_050418767.1
<i>A_ste</i> PDF2R	<i>Anopheles stephensi</i>	XP_035902080.1
<i>A_mac</i> PDF2R	<i>Anopheles maculipalpis</i>	XP_050069491.1
<i>P_cla</i> PDFR	<i>Procambarus clarkii</i>	XP_045622346.1
<i>H_vit</i> PDFR	<i>Homalodisca vitripennis</i>	XP_046669177.1
<i>O_tau</i> PDFR	<i>Onthophagus taurus</i>	XP_022919236.1
<i>L_ana</i> PDFR	<i>Lingula anatina</i>	XP_013413106.1
<i>H_rub</i> PDFR	<i>Haliotis rubra</i>	XP_046569917.1
<i>C_ele</i> PDFR1a	<i>Caenorhabditis elegans</i>	NP_001021170.1
<i>C_ele</i> PDFR1b	<i>Caenorhabditis elegans</i>	EF141317.1
<i>C_ele</i> PDFR1c	<i>Caenorhabditis elegans</i>	EF141318
<i>C_tel</i> PDFR	<i>Capitella teleta</i>	ELU10117.1

Supplementary table 2. (continued)

Additional information

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Author Contributions

Xiao Cong, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing—original draft preparation; Huachen Liu, Investigation, Methodology, Validation; Lihua Liu, Methodology, Validation; Michaela Egertová, Investigation, Methodology, Visualization; Maurice R. Elphick, Formal analysis, Validation, Visualization, Writing—review & editing; Muyan Chen, Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing—review & editing

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Reviewer #1 (Public review):

Summary:

The manuscript characterizes a functional peptidergic system in the echinoderm *Apostichopus japonicus* that is related to the widely conserved family of calcitonin/diuretic hormone 31 (CT/DH31) peptides in bilaterian animals. In vitro analysis of receptor-ligand interactions, using multiple receptor activation assays, identifies three cognate receptors for two CT-like peptides in the sea cucumber, which stimulate cAMP, calcium, and ERK signaling. Only one of these receptors clusters within the family of calcitonin and calcitonin-like receptors (CTR/CLR) in bilaterian animals, whereas two other receptors cluster with invertebrate pigment dispersing factor receptors (PDFRs). In addition, this study sheds light on the expression and in vivo functions of CT-like peptides in *A. japonicus*, by quantitative real-time PCR, immunohistochemistry, pharmacological experiments on body wall muscle and intestine preparations, and peptide injection and RNAi knockdown experiments. This reveals a conserved function of CT-like peptides as muscle relaxants and growth regulators in *A. japonicus*.

Strengths:

This work combines both in vitro and in vivo functional assays to identify a CT-like peptidergic system in an economically relevant echinoderm species, the sea cucumber *A. japonicus*. A major strength of the study is that it identifies three G protein-coupled receptors for AjCT-like peptides, one related to the CTR/CLR family and two related to the PDFR family. A similar finding was previously reported for the CT-related peptide DH31 in *Drosophila melanogaster* that activates both CT-type and PDF-type receptors. Here, the authors expand this observation to a deuterostomian animal, which suggests that receptor promiscuity is a more general feature of the CT/DH31 peptide family and that CT/DH31-like peptides may activate both CT-type and PDF-type receptors in other animals as well.

Besides the identification of receptor-ligand pairs, the downstream signaling pathways of AjCT receptors have been characterized, revealing broad and in some cases receptor-specific effects on cAMP, calcium, and ERK signaling.

Functional characterization of the CT-related peptide system in heterologous cells is complemented with ex vivo and in vivo experiments. First, peptide injection and RNAi knockdown experiments establish transcriptional regulation of all three identified receptors in response to changing AjCT peptide levels. Second, ex vivo experiments reveal a conserved role for the two CT-like peptides as muscle relaxants, which have differential effects on body wall muscle and intestine preparations. Finally, peptide injection and knockdown experiments uncover a growth-promoting role for one CT-like peptide (AjCT2). Injection of AjCT2 at high concentration, or long-term knockdown of the AjCT precursor, affects diverse growth-related parameters including weight gain rate, specific growth rate, and transcript levels of growth-regulating transcription factors. The authors also reveal a growth-promoting

function for the PDFR-like receptor AjPDFR2, suggesting that this receptor mediates the effects of AjCT2 on growth.

Weaknesses:

The authors present a more detailed phylogenetic analysis in the revised version, including a larger number of species. But some clusters in the analysis are not well supported because they have only low bootstrap values. This makes it difficult to interpret the clustering in some parts of the tree.

Expression of CT-like peptides was investigated both at transcript and protein level, but insight into the expression of the three peptide receptors is limited. This makes it difficult to understand the mechanism underlying the (different) functions of the two CT-like peptides in vivo. The authors identify differences in signal transduction cascades activated by each peptide, which might underpin distinct functions, but these differences were established only in heterologous cells.

The authors show overlapping phenotypes for a long-term knockdown of the AjCT precursor and the AjPDFR2 receptor, suggesting that the growth-regulating functions of AjCT2 are mediated by this receptor pathway. However, it remains unclear whether this mechanism underpins the growth-regulating function of AjCT2, until further in vivo evidence for this ligand-receptor interaction is presented. For example, the authors could investigate whether knockdown of AjPDFR2 attenuates the effects of AjCT2 peptide injection. In addition, a functional PDF system in this species remains uncharacterized, and a potential role of PDF-like peptides in growth regulation has not yet been investigated in *A. japonicus*. Therefore, it also remains unclear whether the ability of CT-like peptides to activate PDFRs is an evolutionary ancient property of this peptide family or whether this is an example of convergent evolution in some protostomian (*Drosophila*) and deuterostomian (sea cucumber) species.

<https://doi.org/10.7554/eLife.101799.2.sa2>

Reviewer #2 (Public review):

Summary:

The authors show that *A. japonicus* calcitonins (AjCT1 and AjCT2) activate not only the calcitonin/calcitonin-like receptor, but they also activate the two "PDF receptors", *ex vivo*. They also explore secondary messenger pathways that are recruited following receptor activation. They determine the source of CT1 and CT2 using qPCR and in situ hybridization and finally test the effects of these peptides on tissue contractions, feeding and growth. This study provides solid evidence that CT1 and CT2 act as ligands for calcitonin receptors; however, evidence supporting cross-talk between CT peptides and "PDF receptors" is weak.

Strengths:

This is the first study to report pharmacological characterization of CT receptors in an echinoderm. Multiple lines of evidence in cell culture (receptor internalization and secondary messenger pathways) support this conclusion.

Weaknesses:

The authors claim that *A. japonicus* CTs activate "PDF" receptors and suggest that this cross-talk is evolutionary ancient since similar phenomenon also exists in the fly *Drosophila melanogaster*. These conclusions are not fully supported. The authors perform phylogenetic analysis to show that the two "PDF" receptors form an independent clade. The bootstrap

support is quite low in a lot of instances, especially for the deuterostomian and protostomian PDFR clades which is below 30. With such low support, it is unclear if the clade comprising deuterostomian "PDFR" is in fact PDFRs and not another receptor type whose endogenous ligand (besides CT) remains to be discovered.

<https://doi.org/10.7554/eLife.101799.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Weaknesses:

(1). Analysis of transcript expression is limited to the CT-peptide encoding gene, while no gene expression analysis was attempted for the three identified receptors. Differences in the activation of downstream signaling pathways between the three receptors are also questionable due to unclarities in the statistical analysis and variation in the control and experimental data in heterologous assays. Together, this makes it difficult to propose a mechanism underlying differences in the functions of the two CT-like peptides in muscle control and growth regulation.

We appreciate the reviewer's rigorous critique. The manuscript has been comprehensively revised as follows:

(1) For the expression analysis of the three identified receptors, the updated results are presented in Figure 5, with the detailed descriptions in Results section 2.4 (line 287-290) and Materials and Methods section 4.5 (line 767).

(2) For the statistical tests and methodological clarity, statistical tests were indeed performed for all experiments. However, we acknowledge that the original labeling methods required enhanced methodological clarity, and we apologize for any confusion caused. All figures have been revised to improve the visibility of differences, and statistical test information has been added to both the figure legends and the Materials and methods section "4.10 Statistical Analysis" (line 900-910).

(3) For the variation in the control and experimental data, the minor observed variations in control conditions across experiments primarily arise from two methodological factors: 1) Each experimental set used cells transfected with distinct receptor subtypes (e.g., AjPDFR1 vs. AjPDFR2), inherently introducing baseline variability due to differential receptor expression profiles. 2) Independent cell culture batches were employed for replicate experiments to ensure biological reproducibility. Importantly, these minor variations did not compromise the statistical significance of downstream signaling differences ($p < 0.01$ for all comparative analyses). Therefore, differences in the activation of downstream signaling pathways between the three receptors are reliable.

(2) The authors also suggest a putative orexigenic role for the CT-like peptidergic system in feeding behavior. This effect is not well supported by the experimental data provided, as no detailed analysis of feeding behavior was carried out (only indirect measurements were performed that could be influenced by other peptidergic effects, such as on muscle relaxation) and no statistically significant differences were reported in these assays.

Thank you for the reviewer's valuable comments. Our revised manuscript now includes the following multidimensional analyses to strengthen evidence of the orexigenic role of AjCT2:

Firstly, in sea cucumbers, the mass of remaining bait is a common indicator of feeding condition. After long-term AjCT2 injection, this value was significantly decreased in comparison with control group during phase V (Figure 8A-figure supplement 1), which indicates that AjCT2 promotes feeding in *A. japonicus*. Correspondingly, in long-term loss-of-function experiments (newly added in the revised manuscript), the remaining bait in the siAjCTP1/2-1 group was significantly increased in comparison with siNC group from phase II to IV (Figure 10B). The detailed descriptions of these supplementary experiments have been added to Results Section 2.6 (lines 390-396) and Materials and Methods Section 4.9 (line 879-888).

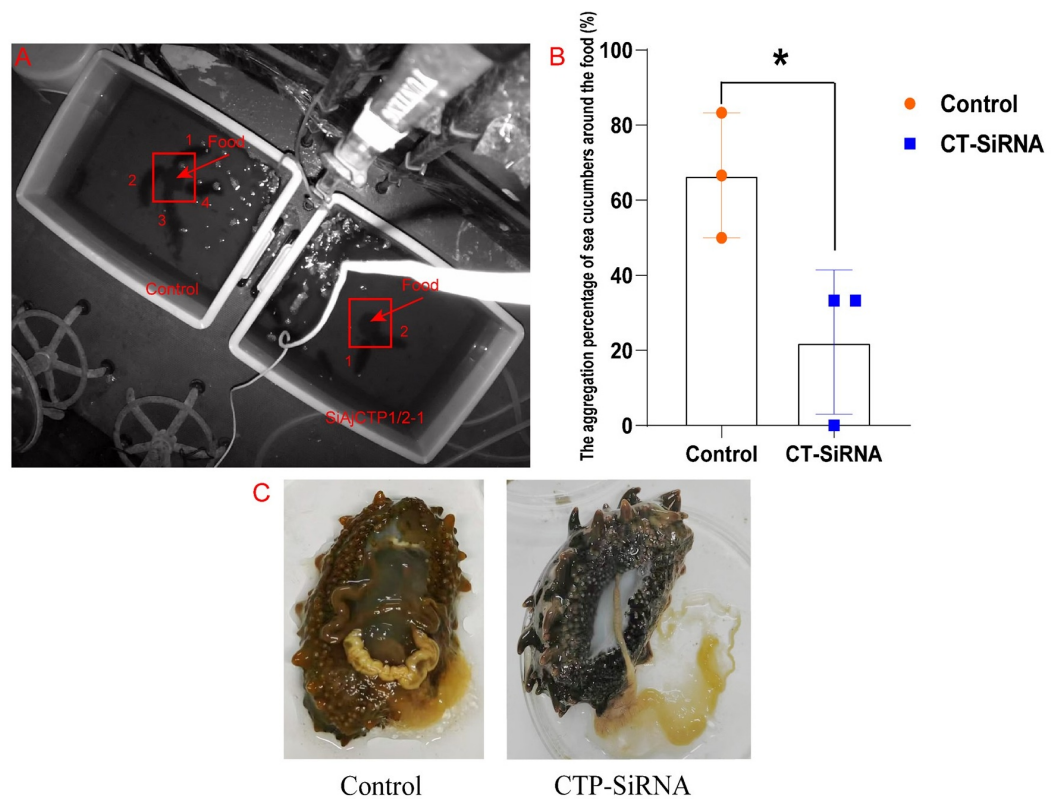
Secondly, after 24 days of continuous injections of siAjCTP1/2-1, we monitored the feeding behavior of these sea cucumbers over three consecutive days. Each day, we removed residual bait and feces, then repositioned fresh food at the tank center. We calculated the aggregation percentage (AP) of sea cucumbers around the food during the feeding peak (2:00-4:00) each day, which is the most reliable indicator of feeding behavior in this species. The results showed that the AP in siAjCTP1/2-1 group was significantly lower than that in control group. Post-dissection observations revealed reduced intestinal food content and significant intestinal degeneration in the siAjCTP1/2-1 group (The figure has been added below). These results indicate that long-term functional loss of AjCT2 reduces food intake and influences the feeding behavior of *A. japonicus*.

In response to the comment regarding “No statistically significant differences were reported in these assays”, we have modified the figures to clearly visualize the differences and added statistical test details in both the figure legends and the Materials and methods section “4.10 Statistical analysis” (lines 900–910).

Author response image 1.

The feeding behavior of *A. japonicus* after long-term loss-of-function of AjCT2. (A) A record of feeding behavior. The red arrow refers to the food and the red box represents the feeding area. The numbers in the figure represent individuals entering into the feeding area. (B) The aggregation percentage (AP) of sea cucumbers around the food during the feeding peak (2:00-4:00) (n=3 days). (C) The degenerated intestine of sea cucumber after 24 days of siAjCTP1/2-1 injection. Data in the graph represent the mean $\pm$ standard deviation. *Significant differences between groups ($p < 0.05$). Control: siNC injection group; CT-SiRNA: siAjCTP1/2 injection

group.



(3) Overall, details regarding statistical analyses are not (clearly) specified in the manuscript, and there are several instances where statements are not supported by literature evidence.

Thank you for the reviewer's comments. Again, we sincerely apologize for the confusion caused. To clarify, statistical tests were performed for all experiments. However, the original labeling may have been somewhat messy. We have revised all figures to enhance the visibility of differences and provided detailed statistical test information in both the figure legends and the Materials and Methods section titled "4.10 Statistical Analysis" (lines 900–910). Additionally, we have supplemented the revised manuscript with further literature evidence to support our statements: (1) citation to Furuya et al. (2000), Johnson et al. (2005), Jékely (2013) and Mirabeau et al. (2013) have been added to clarify the foundation studies on DH31 and DH31 receptors in invertebrates (line 73-74); (2) Conzelmann et al. (2013) and Furuya et al. (2000) were cited to validate the present of two different types of CT-related peptides in protostomes: CT-type peptides (with an N-terminal disulphide bridge) and DH31-type peptides (lacking this feature) (line 78-79); (3) Johnson et al. (2005) was referenced to support the dual ligand-receptor interactions of DH31 in *Drosophila*, specifically its binding to both CG17415 (a CTR/CLR-related protein) and CG13758 (the PDF receptor) (line 94); (4) Johnson et al. (2005) and Goda et al. (2019) were cited to reinforce the functional significance of dual DH31 receptor pathways in *Drosophila*, as extensively studied in prior research (line 95-97).

Reviewer #2 (Public review):

Weaknesses:

*(1) The authors claim that *A. japonicus* CTs activate "PDF" receptors and suggest that this cross-talk is evolutionarily ancient since a similar phenomenon also exists in the fly *Drosophila melanogaster*. These conclusions are not fully supported for several reasons. The authors perform phylogenetic analysis to show that the two "PDF" receptors form an independent clade. This clade is sister to the clade comprising CT receptors. This phylogenetic analysis suffers from several issues. Firstly, the phylogenies lack bootstrap support. Secondly, the resolution of the phylogeny is poor because representative members from diverse phyla have not been included. For instance, insect or other protostomian PDF receptors have not been included so how can the authors distinguish between "PDF" receptors or another group of CT receptors? Thirdly, no in vivo evidence has been presented to support that CT can activate "PDF" receptors in vivo.*

We thank the reviewers for their constructive comments. As suggested, we expanded our taxon sampling to include more representative members across diverse phyla and reanalyzed the phylogenetic relationships (including bootstrap tests) in Figure 1C. The revised analysis revealed two distinct clades: one containing CTR/CLR-type receptors and the other PDF-type receptors. Specifically, AjCTR clustered within the CTR/CLR-type receptor group, while AjPDFR1 and AjPDFR2 were placed in the PDF-type receptor clade. The full species names for all taxa were provided in the Supplementary Table 2.

To provide in vivo evidence supporting CT-mediated activation of "PDF" receptors, we conducted the following experiments: Firstly, we confirmed that AjPDFR1 and AjPDFR2 were the functional receptors of AjCT1 and AjCT2 (Figure 2, 3 and 4). Secondly, injection of AjCT2 and siAjCTP1/2-1 in vivo induced corresponding changes in AjPDFR1 and AjPDFR2 expression levels in the intestine (Figure 8C, 9A, 9B and 9C).

(2) The source of CT which mediates the effects on longitudinal muscles and intestine is unclear. Is it autocrine or paracrine signaling by CT from the same tissue or is it long-range hormonal signaling?

Thank you for this feedback. We have now analysed CT-type neuropeptide expression in *A. japonicus* using immunohistochemistry with the antiserum to the *A. rubens* CT-type peptide ArCT, which has previously been shown to cross-react with CT-type neuropeptides in other echinoderms (Aleotti et al., 2022). We have added related descriptions in the following sections: Results (section 2.4, line 299-336), Discussion (section 3.3, line 545-554) and Materials and methods (section 4.6, line 785-817). Consistent with this previous finding, the ArCT antiserum labelled neuronal cells and fibers in the central and peripheral nervous system and in the digestive system of *A. japonicus* (Figure 6). The specificity of immunostaining was confirmed by performing pre-absorption tests with the ArCT antigen peptide (Figure 6-figure supplement 1). The detection of immunostaining in the innervation of the intestine is consistent with PCR results and the relaxing effect of AjCT2 on intestine preparations. Interestingly, no immunostaining was observed in longitudinal muscle, which is inconsistent with the detection of AjCT1/2 transcripts in this tissue. This may reflect differences in the sensitivity of the methods employed to detect transcripts (PCR) and mature peptide (immunohistochemistry). The absence of ArCT-like immunoreactivity in the longitudinal muscles suggests that AjCT1 and AjCT2 may exert relaxing effects on this tissue in vivo via hormonal signaling mechanisms. However, because AjCT1/2 expression in the longitudinal muscles may be below the detection threshold of the ArCT antibodies, we can't rule out the possibility that AjCT1/2 are released within the longitudinal muscles physiologically.

(3) Pharmacology experiments showing the effects of CT1 and CT2 on ACh-induced contractions were performed. Sample traces have been provided but no traces with ACh alone have been included. How long do ACh-induced contractions persist? These controls are necessary to differentiate between the eventual decay of ACh effects and relaxation

induced by CT1 and CT2. The traces also do not reflect the results portrayed in dose-response curves. For instance, in Figure 6B, maximum relaxation is reported for 10-6M. Yet, the trace hardly shows any difference before and after the addition of 10-6M peptide. The maximum effect in the trace appears to be after the addition of 10-8M peptide.

Thank you for the reviewer's comments. As requested, we have included representative traces of ACh-induced contraction of longitudinal muscle and intestinal preparations (Figure 7—figure supplement 1B and 1C). Notably, the positive control (ACh) maintained contraction effects for at least 15 minutes, consistent with its known pharmacological properties. Regarding Figure 7B (previous Figure 6B), the trace illustrates the cumulative effects of successive neuropeptide treatments at increasing concentrations. A gradual reduction in response amplitude was observed at the highest peptide concentration, likely reflecting receptor desensitization, a phenomenon previously reported for neuropeptide Y and oxytocin (Tsurumaki et al., 2003; Arrowsmith and Wray, 2014). These results are now explicitly described in the Results Section 2.5 (lines 340-345 and 348-352) and discussed in Section 3.3 (lines 569-574). In response to the reviewer's suggestion, we further tested the pharmacological effects of AjCT2 at 10^{-6} M. As shown in Figure 7—figure supplement 1A, this concentration induced maximal relaxation, confirming its dose-dependent efficacy.

(4) I am unsure how differences in wet mass indicate feeding and growth differences since no justification has been provided. Couldn't wet mass also be influenced by differences in osmotic balance, a key function of calcitonin-like peptides in protostomian invertebrates? The statistical comparisons have not been included in Figure 7B.

We appreciate the reviewer's insightful comments. We fully concur that wet mass constitutes an inadequate indicator for evaluating feeding and growth variations. Consequently, we reassessed *A. japonicus* growth parameters using two established metrics: weight gain rate (WGR) and specific growth rate (SGR), to delineate differences between experimental and control groups. Notably, the high-concentration AjCT2 injection group exhibited statistically significant increases in both WGR and SGR relative to controls (Figure 8A). This demonstrates a putative physiological role of AjCT2 signaling in enhancing feeding efficiency and growth performance in *A. japonicus*. Detailed methodologies are provided in the Materials and methods Section 4.8 (lines 847-851), with corresponding results presented in the Results Section 2.6 (lines 370-375). Besides, Cong et al., (2024) reported holotocin-induced osmoregulatory function in *A. japonicus*, manifested by significant wet weight elevation and body bloating. However, our AjCT2 intervention showed no such phenotypic alterations, suggesting that AjCT2 likely does not participate in osmotic balance regulation, at least under these experimental conditions. Crucially, the observed WGR and SGR enhancements following AjCT2 administration was not caused by osmoregulatory effects.

(5) While the authors succeeded in knocking down CT, the physiological effects of reduced CT signaling were not examined.

Thank you for the reviewer's comment. We have supplemented the experiments to investigate the physiological effects of long-term reduced CT signaling following the reviewer's suggestions, including measuring the dry weight of remaining bait and excrement, calculating the weight gain rate and specific growth rate, and testing the expression levels of three growth factors (AjMegf6, AjGDF-8 and AjIgf) to further assess AjCT2's role in feeding and growth. The results demonstrated that weight gain rate and specific growth rate in the siAjCTP1/2-1 group were significantly decreased (As shown in Figure 10A). Correspondingly, except in phase I, the siAjCTP1/2-1 group exhibited a significant increase in remaining bait and a decrease in excrement during phases II-VI (Figure 10B). Furthermore, the growth inhibitory factor AjGDF-8 was significantly up-regulated and the growth promoting factor

AjMegf6 was significantly down-regulated in siAjCTP1/2-1 group (Figure 10C). These findings further support the potential physiological role of AjCT2 signaling in promoting feeding and growth in *A. japonicus*. The added results are presented in Figure 10, with related descriptions in Section 2.6 (Results, lines 390-396), Section 3.4 (Discussion, line 597-603) and Section 4.9 (Materials and Methods, lines 879-888).

Reviewer #1 (Recommendations for the authors):

(1) The abstract states that loss-of-function tests (RNAi knockdown) reveal a potential physiological role for AjCT2 signaling in promoting feeding and growth in A. japonicus. However, RNAi knockdown was only followed by analysis of transcript expression of CT-like receptors and not by the assessment of feeding or growth.

Thank you for this helpful feedback. In the revised manuscript, we have supplemented the experiments to investigate the physiological effects of long-term reduced CT signaling, as suggested by the reviewer. These include measuring the dry weight of remaining bait and excrement, calculating the weight gain rate and specific growth rate, and testing the expression levels of the three growth factors (AjMegf6, AjGDF-8 and AjIgf) to further assess the function of AjCT2 on feeding and growth in *A. japonicus*. The results are as follows:

- (1) The weight gain rate and specific growth rate in the siAjCTP1/2-1 group were significantly decreased (As shown in Figure 10A).
- (2) Correspondingly, except for the phase I, the siAjCTP1/2-1 group had significantly increased remaining bait and decreased excrement during phases II-VI (Figure 10B).
- (3) The growth inhibitory factor AjGDF-8 was significantly up-regulated, while the growth promoting factor AjMegf6 was significantly down-regulated in the siAjCTP1/2-1 group (Figure 10C).

These findings further support the potential physiological role of AjCT2 signaling in promoting feeding and growth in *A. japonicus*. We have incorporated these results into Figure 10 and added related descriptions in the following sections: Results (section 2.6, line 390-396), Discussion (section 3.4, line 597-603) and Materials and methods (section 4.9, line 879-888).

Regarding the original statement in the abstract “Furthermore, in vivo pharmacological experiments and loss-of-function tests revealed a potential physiological role for AjCT2 signaling in promoting feeding and growth in *A. japonicus*.” This sentence effectively summarizes our findings. Therefore, we have retained it in the revised manuscript while supplementing the missing experimental details as requested.

(2) Information on the statistical tests that were performed is lacking for most experiments. It is recommended to include this information in the figure legends, in addition to the methods section. Details on the phylogenetic analysis (parameters and statistics used) and calculation of half maximal effective concentrations (calculation methods and confidence intervals) also need to be included in the manuscript.

Thank you for this constructive feedback. As the reviewer suggested, statistical test information has been incorporated into both the figure legends and the “4.10 Statistical Analysis” subsection of the Materials and methods (lines 900-910). Specifically:

- (1) Phylogenetic analysis details (parameters and statistical approaches) are now provided in the Materials and methods section 4.2 (line 675-682);
- (2) Bootstrap test results supporting the phylogenetic trees have been added to Figure 1B and 1C;

(3) Half-maximal effective concentration (EC_{50}) calculations, including methodologies and confidence intervals, are documented in both the Figure 2B legend and the “4.10 Statistical Analysis” section (lines 900-910).

(3) In some figures (e.g. Figure 5A, 7A), the n number indicated does not match the number of data points shown in the figure panel. It is not clear what n represents here. In Figure 6B, an x-axis label is missing. In some figure legends (e.g. Figure 4 - Figure Supplement 1), the error bars and significance levels are not defined.

We apologize for this error; we have corrected all quantity errors related to “n” in the manuscript’ figure legends. And also, the x-axis label was added in Figure 7B (previous Figure 6B), error bars and significance levels were defined in all figure legends clearly

(4) It would be useful to explain what the difference is between the Cre and SRE luciferase assay and why these two assays were used to study receptor-activated signaling cascades. The source of the synthetic peptides is mentioned, but it is recommended to also state the purity of the synthetic peptides.

Thank you for the valuable comments. As stated in the introduction (line 66-69)- “binding of CT to CTR in the absence of RAMPs can activate signaling via several downstream pathways, including cAMP accumulation, Ca^{2+} mobilization, and ERK activation.” Based on this established mechanism, we selected cAMP and Ca^{2+} signaling pathways as biomarkers for studying receptor-activated cascades, with the following experimental rationale: CRE-Luc Reporter System functions as a cAMP response element detector and SRE-Luc Reporter System serves as an intracellular Ca^{2+} level indicator. In CRE-Luc detection, when the receptor is activated by a ligand, it couples with Gas protein to activate the cAMP/PKA signaling pathway. The accumulation of cAMP can lead to the phosphorylation of PKA, and then enhance the transcription of CRE-containing genes. Therefore, significant increase in CRE-Luc activity directly correlates with cAMP accumulation. Similarly, SRE-Luc activity reflects dynamic changes in intracellular Ca^{2+} levels. We have added the explanation of this part in the materials and methods section 4.4 (line 715-721). The purity of the synthetic peptides was >95%, and we have also added this information in section 4.4 (line 715) according to the reviewer’s suggestion.

(5) In Figure 3B, it is difficult to see receptor internalization in response to the application of synthetic CT-like peptides, and a control condition (without peptide application) is lacking.

Thank you for the reviewer’s comment. The control condition (without peptide application) was added in Figure 3-figure supplement 1, which shows the localization of pEGFP-N1/receptors in the cell membrane. Upon stimulation with synthetic CT-like peptides (Materials and methods section 2.3), the receptors exhibit clear internalization into the cytoplasm, as visualized in Figure 3B through comparative analysis.

(6) Differences in the activation of downstream signaling cascades between the three receptors are questionable because there is substantial variation in the experimental data and control conditions in different experiments (for example, in Figures 3A and 4A). To better represent this variation, it is recommended to plot individual data points onto the bar graphs in all figures and to nuance the interpretation of putative differences in downstream signaling of different receptors. Differences in the physiological roles of CT-like peptides may be explained by various mechanisms, including differences in peptide/receptor expression or in the potency of peptides to activate different receptors in vivo. It would be useful to elaborate on these different explanations in the discussion.

We appreciate the reviewer's critical assessment. The observed variations in control conditions across experiments (e.g., Figures 3A & 4A) primarily arise from two methodological factors: ① Each experimental set used cells transfected with distinct receptor subtypes (e.g., AjPDFR1 vs. AjPDFR2), inherently introducing baseline variability due to differential receptor expression profiles. ② Independent cell culture batches were employed for replicate experiments to ensure biological reproducibility. Importantly, these minor variations did not compromise the statistical significance of downstream signaling differences ($p < 0.01$ for all comparative analyses). And according to the reviewer's suggestion, we have plotted individual data points onto the bar graphs in all figures.

And also, according to the reviewer's suggestion, we have expanded the discussion on receptor-specific signaling cascades in Section 3.4 (lines 589-609). Key findings include: In vivo pharmacological assays demonstrated that only high concentrations of AjCT2 significantly enhanced feeding and growth rates in *A. japonicus*. In contrast, neither a low concentration of AjCT2 nor any concentration of AjCT1 (low or high) induced detectable effects. Furthermore, long-term knockdown of AjCTP1/2 further validated the essential role of AjCT2 in regulating feeding and growth in this species. To elucidate the receptor mediating AjCT2's feeding- and growth-promoting effects, we selected AjPDFR2 based on its distinct activation profile: AjCT2 selectively activated AjPDFR2, inducing downstream ERK1/2 phosphorylation, whereas AjCT1 exhibited no activity toward this receptor. Given this receptor specificity, we performed AjPDFR2 knockdown experiments, which revealed phenotypic changes consistent with those in AjCTP1/2 knockdown animals, including significantly reduced WGR and SGR, alongside increased remaining bait accumulation and diminished excrement output compared to control. Collectively, these results support a model wherein AjCT2 promotes feeding and growth in *A. japonicus* via AjPDFR2-dependent activation of the cAMP/PKA/ERK1/2 and Gαq/Ca²⁺/PKC/ERK1/2 cascades. Considering the inherent complexity of neuropeptide signaling systems, which involve multiple GPCR subtypes coupled to diverse signaling cascades, ligands bound to the same receptor may activate distinct G protein subforms within a single cell (Møller et al., 2003; Mendel et al., 2020). Receptor activation modes may be modulated by structural polymorphisms or binding site diversity (Wong et al., 2000; Changeux, 2010), as well as by the differential efficacy of peptides in activating receptors in vivo.

(7) For the peptide injection experiments, it is recommended to explain the different animal groups in the results section. In addition, injection in the control condition seems to have a small effect on the wet weight. Therefore, it would be useful to compare control-injected and peptide-injected groups after injection.

Thank you for the reviewer's comments. We have provided an expanded explanation of the animal group classifications in Section 2.6 (lines 367–375). We fully agree that a comparative analysis between the experimental and control groups post-injection is essential. However, since wet weight measurement is suboptimal for demonstrating feeding and growth variations, we re-evaluated the data using two validated metrics: weight gain rate (WGR) and specific growth rate (SGR) of *A. japonicus*. The results revealed that the high-concentration AjCT2 injection group exhibited significantly elevated weight gain rate and specific growth rate compared to the control group, suggesting a potential role of AjCT2 signaling in promoting feeding and growth in *A. japonicus*. These results are presented in Figure 8A, with detailed descriptions in Results Section 2.6 (lines 370–375) and methodology in Materials and Methods Section 4.8 (lines 847-851).

(8) Regarding the RNAi knockdown experiments, it is not clear from the methods section what the siNC control exactly is, and how the interference rate is calculated.

Thank you for this comment. The siNC control was siRNA which does not target any genes in *A. japonicus*, with interference rates quantified through the $2^{-\Delta\Delta CT}$ method to assess siRNA inhibition efficiency. These methodological details have been incorporated into Materials and Methods Section 4.9 (lines 866–867 and 874–876) for enhanced clarity.

Reviewer #2 (Recommendations for the authors):

(1) Both the phylogenies are missing bootstrap tests. Please include this analysis. The phylogenetic analyses should also include other Family B ligands and receptors from both vertebrates and invertebrates because it is widely assumed that PDF is related to VIP given their shared roles in circadian clock and gut regulation. Therefore, this analysis needs to be more comprehensive than currently presented. Drosophila melanogaster receptors have also been excluded in spite of the Drosophila PDFR exhibiting ligand promiscuity. The legend should also include the full species names of the various taxa (or modify the figure to include full names) instead of referring to another table. The supplementary table was not available to this reviewer.

Thank you for the reviewer's constructive comments. According to the reviewer's suggestion, we have incorporated the VIPRs and *Drosophila melanogaster* receptors into the comparative analysis and reanalyzed the phylogenies in Figure 1C, and both phylogenies included bootstrap tests (Figure 1B, 1C) in the revised manuscript. The full species names of the various taxa are listed in supplementary tables 1 and 2 in the revised manuscript.

(2) Expression data indicate that AjCTP1/2 is expressed in both the longitudinal muscles and intestine. What are the cell types that express AjCTP1/2? Given that the authors show an effect of CT1 and CT2 on both of these tissues, it would be important to know whether this is local regulation (paracrine or autocrine) vs long-distance hormonal control by the nervous system. This can be addressed by performing in situ hybridization or immunohistochemistry of CT (using Asterias rubens CT antibody: <https://doi.org/10.3389/fnins.2018.00382>) on these tissues.

Thank you for this feedback. We have now analysed CT-type neuropeptide expression in *A. japonicus* using immunohistochemistry with the antiserum to the *A. rubens* CT-type peptide ArCT, which has previously been shown to cross-react with CT-type neuropeptides in other echinoderms (Aleotti et al., 2022). We have added related descriptions in the following sections: Results (section 2.4, line 299–336), Discussion (section 3.3, line 545–554) and Materials and methods (section 4.6, line 785–817). Consistent with this previous finding, the ArCT antiserum labelled neuronal cells and fibers in the central and peripheral nervous system and in the digestive system of *A. japonicus* (Figure 6). The specificity of immunostaining was confirmed by performing pre-absorption tests with the ArCT antigen peptide (Figure 6-figure supplement 1). The detection of immunostaining in the innervation of the intestine is consistent with PCR results and the relaxing effect of AjCT2 on intestine preparations. Interestingly, no immunostaining was observed in longitudinal muscle, which is inconsistent with the detection of AjCT1/2 transcripts in this tissue. This may reflect differences in the sensitivity of the methods employed to detect transcripts (PCR) and mature peptide (immunohistochemistry). The absence of ArCT-like immunoreactivity in the longitudinal muscles suggests that AjCT1 and AjCT2 may exert relaxing effects on this tissue in vivo via hormonal signaling mechanisms. However, because AjCT1/2 expression in the longitudinal muscles may be below the detection threshold of the ArCT antibodies, we can't rule out the possibility that AjCT1/2 are released within the longitudinal muscles physiologically.

(3) While Drosophila DH31 can activate both PDF and DH31 receptors, the EC50 values differ drastically. Importantly, there is an independent gene encoding PDF which is a more sensitive ligand for the PDF receptor. This is in stark contrast to the situation

presented here where the authors have yet to identify the PDF gene in their system. Outside Drosophila this cross signaling between the two systems has not been observed in any species. Based on this, I would argue that the ability of CTs to activate PDFR is not an evolutionary ancient property but rather an example of convergent evolution if supported by more evidence.

We sincerely appreciate the reviewers' insightful comments. We agree that we cannot rule out the possibility that ability of CT-type peptides to activate PDF-type receptors in *Drosophila* and *A. japonicus* has arisen independently. Therefore, we have modified the text in the discussion accordingly so that this alternative explanation for the effects of CT-type peptides on PDF-type receptors is also presented: "Alternatively, the ability of CT-type neuropeptides to act as ligands for PDF-type receptors in *D. melanogaster* and *A. japonicus* may have evolved independently. Further studies on a wider variety of both protostome (e.g. molluscs, annelids) and deuterostome taxa (e.g. other echinoderms, hemichordates) are needed to address this issue."

(4) AjCT1 and CT2 can activate the two PDF receptors ex vivo. However, their EC50 values are larger and the responses are lower compared to those seen for the CT receptor. Similar cross-talk between closely related peptide families is often observed in ex vivo systems (see: <https://doi.org/10.1016/j.bbrc.2010.11.089>, <https://doi.org/10.1073/pnas.162276199>, <https://doi.org/10.1093/molbev/mst269> and others). However, very few signaling systems exhibit this type of cross-talk in vivo. Without any in vivo evidence, I suspect that the more likely possibility is that the bona fide endogenous ligand for PDF receptors remains to be discovered. The authors could, however, perform peptide and receptor knockdown experiments and show overlap in phenotypes following CT knockdown and PDFR knockdown to support their claim.

We sincerely appreciate the reviewers' insightful critique. According to the reviewer's suggestion, we have supplemented CTP and AjPDFR2 knockdown experiments, and measured the dry weight of remaining bait and excrement, as well as calculating the weight gain rate and specific growth rate in response to phenotypic changes. The results showed that weight gain rate and specific growth rate in experimental groups were significantly decreased respectively (As shown in Figure 10A and 11B). Correspondingly, except for the I phase, the siAjCTP1/2-1 group had significantly increased remaining bait and decreased excrement in II-VI phases (Figure 10B), the remaining bait weight was significantly increased in siAjPDFR2-1 group (except during phase I), while the weight of excrement was significantly decreased in phase V and VI (Figure 11C). Therefore, AjCT and AjPDFR2 knockdown experiments showed overlap in phenotypes, providing evidence that AjCT does act as an endogenous ligand for PDFR. These results were added in Figure 10 and Figure 11. The related description was added in the results section 2.6 (line 390-396), section 2.7 (line 427-439) and the materials and methods section 4.9 (line 879-898). We acknowledge, however, that other peptides, in addition AjCT1 and AjCT2, may also act as ligands for AjPDFR1 and AjPDFR2 in vivo and on-going studies in the Chen (OUC) and Elphick (QMUL) labs are attempting to address this issue

(5) Why are receptor transcripts upregulated following peptide injection? Usually, increased ligand levels/signaling result in a compensatory decrease in receptor levels. These negative feedback loops maintain optimum signaling levels. Since the authors have successfully implemented RNAi for this CT precursor, what are the phenotypes on growth and feeding?

We thank the reviewers for raising these critical points. Our responses are structured as follows: Firstly, our findings align with established mechanisms of neuropeptide-induced receptor modulation (Please check the reference Tiptanavattana et al. 2022). Secondly, based on the reviewer's suggestion, we have supplemented the experiments to detect the phenotype

variations on growth and feeding based on long-term reduced CT signaling, including measuring the dry weight of remaining bait and excrement, calculating the weight gain rate and specific growth rate, as well as testing the expression levels of the three growth factors (AjMegf6, AjGDF-8 and AjIgf). The results showed that weight gain rate and specific growth rate in siAjCTP1/2-1 group were significantly decreased (As shown in Figure 10A), Correspondingly, except for the I phase, the siAjCTP1/2-1 group had more remaining bait and less excrement in II-VI phases (Figure 10B). Furthermore, the growth inhibitory factor AjGDF-8 was significantly up-regulated and the growth promoting factors AjMegf6 were significantly down-regulated in siAjCTP1/2-1 group (Figure 10C). We have added these results in Figure 10, with detailed description in the results section 2.6 (line 390-396) and in the materials and methods section 4.9 (line 879-888). And after long-term continuous injections of siAjCTP1/2-1, we further recorded the feeding behavior of these sea cucumbers for three consecutive days. The remaining bait and feces were cleaned and the food was re-placed in the middle of the tank each day. We calculated the aggregation percentage (AP) of sea cucumbers around the food during the peak feeding period (2:00-4:00) each day, which is the best indicator for sea cucumber feeding behavior detecting. The results showed that the AP in siAjCTP1/2-1 group was significantly lower than that in control group. After dissection, we also found the intestines of siAjCTP1/2-1 group had less food and significantly degenerated (see author response image 1). All these results supported that long-term functional loss of AjCT2 negatively influence the feeding and growth of *A. japonicus*.

Other comments:

(6) What criteria do the authors use to classify some proteins as "type", some as "like" and others as "related"? In my opinion, DH31 could be referred to as CT-like or CT-type. Please use one term for clarity unless there is a scientific explanation behind this terminology.

Thank you for the reviewer's comment. If you look at the paper by Cai et al. (2018) you will see in Figure 14 that CT-type peptides and DH31-type peptides are paralogous, probably due to a gene duplication in the common ancestor of the protostomes. The CT-related peptides in protostomes that have a disulphide bridge we would describe as CT-type because they have conserved a feature that is found in CT-type peptides in deuterostomes. Whereas the DH31 peptides we would describe as CT-like. But there is not a formal rule on this. It is possible the duplication event that gave rise to DH31 and CT-type peptides occurred in the common ancestor of the Bilateria but DH31-type signaling was lost in deuterostomes. On the other hand, if the gene duplication that gave rise to DH31-type peptides and CT-type peptides in protostomes did occur in a common ancestor of the protostomes, then DH31 and CT-type peptides in protostomes could be described as co-orthologs of CT-type peptides in deuterostomes. In this case, both CT peptides and DH31 peptides in protostomes could be described as CT-type. Here is a useful link for explanation of terms: <https://omabrowser.org/oma/type/>

(7) Was genomic DNA removal step performed before cDNA synthesis for qRT-PCR?

Thank you for the reviewer's comment. The genomic DNA removal step was performed before cDNA synthesis for qRT-PCR and we have added the information in the section 4.5 (line 774-776).

(8) Line 70: The presence of calcitonin-like peptides (DH31) and DH31 receptors in invertebrates was discovered long before the discoveries by Jekely 2013 and Mirabeau and Joly 2013. Please credit these original studies: <https://pubmed.ncbi.nlm.nih.gov/10841553/> and <https://pubmed.ncbi.nlm.nih.gov/15781884/>.

Thank you for the reviewer's comment. We have credited these original studies in the revised manuscript.

| (9) Lines 72-74: Please cite <https://pubmed.ncbi.nlm.nih.gov/24359412/>.

Thank you for the reviewer's comment. We have cited it in the revised manuscript.

| (10) Line 87: Please cite <https://pubmed.ncbi.nlm.nih.gov/15781884/>.

Thank you for the reviewer's comment. We have cited it in the revised manuscript.

| (11) Lines 89-91: The functional significance of DH31 signalling to PDPR in *Drosophila* is known. See: <https://pubmed.ncbi.nlm.nih.gov/15781884/> and <https://pubmed.ncbi.nlm.nih.gov/30696873/>. There are several studies that have shown the functions of DH31 signalling via DH31R.

Thank you for the reviewer's comment. We have corrected it and added all this studies in the revised manuscript.

| (12) Figure 1 Supplement 1: The tertiary models for CT1 and CT2 look completely different. This prediction is not in line with both ligands activating the same receptor.

Thank you for the reviewer's comment. We have deleted this supplementary figure.

| (13) Figure 1 Supplement 3 legend: Please add panel labels next to the corresponding receptor.

Thank you for the reviewer's comment. We have added panel labels next to the corresponding receptors as you suggested.

| (14) Figure 2: What does CO refer to?

Thank you for the reviewer's comment. CO (Control) refers to the stimulation of HEK293T transfected cells with serum-free DMEM, and we have added the detailed information in Figure 2 legend (line 251-252).

| (15) Figure 3: Due to the low magnification of the cells, it is difficult to see the localization of the receptor. It would also be more appropriate to use a membrane marker rather than DAPI which does not label the cytoplasm or membrane where the receptor can be found.

we appreciate the reviewer's insightful comment regarding the experimental controls. The baseline receptor localization data under non-stimulated conditions are presented in Figure 3—figure supplement 1, demonstrating constitutive membrane distribution of pEGFP-N1-tagged receptors. Upon stimulation with synthetic CT-like peptides, qualitative imaging analysis revealed significant ligand-induced receptor internalization into the cytoplasm (Figure 3B).

| (16) Figure 9: Please include PDF precursor and receptor as separate columns. Also, *Drosophila* CT/DH31 receptors have been characterized.

Thank you for the reviewer's comment. We have added PDF precursor, predicted peptides and receptors as separate columns in the revised manuscript Figure 12. And also, we

corrected the error summary of *Drosophila* CT/DH31 receptors according to your suggestions.

(17) Table 1: It is not very clear why there are multiple columns for ERK1/2 with different outcomes.

Thank you for the reviewer's comment. Although the cAMP/PKA or Gαq/Ca²⁺/PKC signaling is activated after ligand binding to receptors, the downstream ERK1/2 cascade is not necessarily activated. Therefore, we counted the activation status of cAMP/PKA and its downstream ERK1/2 cascade, and Gαq/Ca²⁺/PKC and its downstream cascade in Table 1 respectively. We have optimized Table1 to make it clearer in the revised manuscript.

<https://doi.org/10.7554/eLife.101799.2.sa0>